Warsaw University of Technology

Lecture Notes

Linear Algebra

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1 Complex numbers

1.1 Motivation

The equation

$$x^2 + 1 = 0$$

has no real solutions, since $x^2 \geq 0$ for every $x \in \mathbb{R}$. Nonetheless, we often need solutions of such equations. The remedy is to *extend* the set of real numbers: we introduce a new number

$$i, \qquad \text{required to satisfy} \qquad i^2 = -1.$$

Clearly i is not a real number. As we shall see, adjoining i to $\mathbb{R}$ produces a new number system — the complex numbers — in which *every* polynomial equation has a solution.

Born of a scandal: the cubic. Historically, nobody extended $\mathbb{R}$ because of $x^2 + 1 = 0$. A quadratic equation with no solution was simply considered *unsolvable*, and that was the end of it. The trouble began with *cubic* equations in sixteenth-century Italy. Scipione del Ferro and Niccolò Tartaglia found a formula for solving them, which Gerolamo Cardano published in his *Ars Magna* (1545): the equation $x^3 = px + q$ is solved by

$$x = \sqrt[3]{\frac{q}{2} + \sqrt{\frac{q^2}{4} - \frac{p^3}{27}}} + \sqrt[3]{\frac{q}{2} - \sqrt{\frac{q^2}{4} - \frac{p^3}{27}}}.$$

Warning. Try the formula on $x^3 = 15x + 4$, i.e. $p = 15$, $q = 4$. It gives

$$x = \sqrt[3]{2 + \sqrt{-121}} + \sqrt[3]{2 - \sqrt{-121}},$$

which contains the square root of a *negative* number. Yet $x = 4$ plainly solves the equation, since $64 = 60 + 4$. The formula is right; it just speaks a language nobody understood yet.

So complex numbers were not forced on mathematicians by quadratics without real roots. They were forced on them by cubics whose roots are perfectly real.

Bombelli's "wild thought". In 1572 Rafael Bombelli found a way out. He had no theory of complex numbers: no definition and no i. His idea, which he himself called a "wild thought", was to treat the meaningless symbol $\sqrt{-1}$ as if it were a number, to obey every ordinary rule of algebra, and to add one bookkeeping rule: $\left(\sqrt{-1}\right)^2 = -1$.

Example 1.1. Expanding the cube by the binomial formula and then applying the bookkeeping rule twice gives

$$\left(2 + \sqrt{-1}\right)^3 = 8 + 12\sqrt{-1} + 6\left(\sqrt{-1}\right)^2 + \left(\sqrt{-1}\right)^3 = 8 + 12\sqrt{-1} - 6 - \sqrt{-1} = 2 + 11\sqrt{-1} = 2 + \sqrt{-121}.$$

In the same way, $\left(2 - \sqrt{-1}\right)^3 = 2 - \sqrt{-121}$. So Bombelli could take

$$\sqrt[3]{2 + \sqrt{-121}} = 2 + \sqrt{-1}, \qquad \sqrt[3]{2 - \sqrt{-121}} = 2 - \sqrt{-1},$$

and Cardano's formula yields

$$x = \left(2 + \sqrt{-1}\right) + \left(2 - \sqrt{-1}\right) = 4.$$

The strange symbol entered the computation, cancelled out, and delivered the honest, checkable answer 4, all without any claim about what $\sqrt{-1}$ actually *is*. That claim remained an unpaid debt for almost three centuries.

Descartes dismissed the new numbers as "imaginary" in 1637, and the insult stuck as their name. Only in the nineteenth century, with Gauss's picture of complex numbers as points of the plane and Hamilton's definition of them as pairs of real numbers, was the debt finally paid. This chapter pays it too: our definition of $\mathbb{C}$ in the next section is precisely the licence for Bombelli's game.

So much for the algebra — but why should a computer scientist care about numbers that mathematicians themselves long dismissed as "imaginary"? Because complex numbers turned out to be one of the most practical inventions in the history of mathematics. You have already used them today, probably several times:

- **Multimedia and signal processing.** Every JPEG image, every MP3 file and every video call relies on the *Fast Fourier Transform* (FFT), an algorithm that decomposes a signal into a combination of simple waves. The FFT is formulated entirely in terms of complex numbers — more precisely, in terms of the *roots of unity* that we compute later in this chapter. The reason your phone can stream music in real time is that multiplying complex numbers of modulus 1 is a very cheap way of manipulating waves.
- **Computer graphics and games.** As we shall see, multiplying by a complex number of modulus 1 *rotates* the plane: rotating a 2D sprite is literally one complex multiplication. The same idea, one dimension up, leads to *quaternions* — a four-dimensional cousin of $\mathbb{C}$ used by every serious game engine to rotate 3D objects smoothly and without the numerical artefacts of angle-based approaches.
- **Quantum computing.** The state of a qubit is a pair of complex numbers, and everything a quantum computer does is linear algebra over $\mathbb{C}$. The famous interference effects behind quantum speedups are exactly additions of complex numbers with different arguments — the geometry of this chapter.
- **Fractals.** The Mandelbrot set — perhaps the most famous computer-generated image in history — is produced by iterating the map $z \mapsto z^2 + c$ over complex numbers and colouring each pixel by how fast the modulus $|z|$ escapes to infinity.

There is also an internal reason to start the course here. Together with the real numbers, the complex numbers will provide the scalars for everything that follows: vector spaces, matrices, determinants and eigenvalues. Moreover, some purely real-looking problems of linear algebra secretly live in $\mathbb{C}$: in Chapter 7 we will meet perfectly ordinary real matrices whose eigenvalues simply do not exist over $\mathbb{R}$ and force us into the complex plane whether we like it or not. Learning to compute in $\mathbb{C}$ now is a small investment with a large payoff.

1.2 The complex numbers

Definition 1.2. A *complex number* is an expression $a + bi$, where $a, b \in \mathbb{R}$. The set of all complex numbers is denoted by $\mathbb{C}$.

Remark 1.3. Formally, a complex number is simply an ordered pair $(a, b) \in \mathbb{R} \times \mathbb{R}$, and $a + bi$ is a convenient notation for that pair: a stands for $(a, 0)$, the symbol i denotes the pair $(0, 1)$, and real numbers are identified with the pairs $(a, 0)$. There is nothing to "imagine": $\mathbb{C}$ is the plane $\mathbb{R}^2$ equipped with the multiplication defined below. Two complex numbers are equal exactly when both their real parts and their imaginary parts agree.

Definition 1.4. Addition and multiplication of complex numbers are defined by

$$(a + bi) + (c + di) = (a + c) + (b + d)i,$$
$$(a + bi) \cdot (c + di) = (ac - bd) + (ad + bc)i.$$

The rule for multiplication is exactly what one obtains by expanding $(a+bi)(c+di)$ formally and using $i^2 = -1$.

Example 1.5.

$$(2 + 3i) + (1 - 2i) = 3 + i, \qquad (2 + 3i) \cdot (1 - 2i) = 2 - 4i + 3i - 6i^2 = 2 - i + 6 = 8 - i.$$

Fact 1.6 (Inversion). Let $z = a + bi \neq 0$ and put

$$\frac{1}{z} = \frac{a}{a^2 + b^2} - \frac{b}{a^2 + b^2}\, i.$$

Then $z \cdot \frac{1}{z} = 1$, i.e. $\frac{1}{z}$ is a multiplicative inverse of z.

Proof. Since $z \neq 0$, at least one of the real numbers a, b is non-zero, so $a^2 + b^2 > 0$ and the real numbers $\frac{a}{a^2+b^2}$ and $\frac{b}{a^2+b^2}$ are well defined. By the definition of multiplication,

$$(a+bi)\left(\frac{a}{a^2+b^2} - \frac{b}{a^2+b^2}\,i\right) = \frac{a^2+b^2}{a^2+b^2} + \frac{-ab+ba}{a^2+b^2}\,i = 1 + 0i = 1. \qquad \square$$

Remark 1.7. Where does the formula come from? The key observation is that $(a+bi)(a-bi) = a^2 + b^2$ is a non-zero *real* number. So, informally, if we pretend that fractions of complex numbers behave like fractions of real numbers, we may multiply the numerator and denominator of $\frac{1}{a+bi}$ by the conjugate $a - bi$:

$$\frac{1}{a+bi} = \frac{a-bi}{(a+bi)(a-bi)} = \frac{a-bi}{a^2+b^2}.$$

This is not a proof, since it uses division by complex numbers before we know it makes sense; but it is a convenient way to *find* the inverse, and the fact above confirms that the result is correct.

Example 1.8.

$$\frac{1}{i} = \frac{-i}{i\cdot(-i)} = \frac{-i}{1} = -i, \qquad \frac{1}{1+i} = \frac{1-i}{(1+i)(1-i)} = \frac{1-i}{2} = \tfrac{1}{2} - \tfrac{1}{2}\,i.$$

Theorem 1.9 (Rules of arithmetic in $\mathbb{C}$). *Addition and multiplication of complex numbers satisfy the following rules, for all $z, w, u \in \mathbb{C}$.*

(1) Associativity of addition: $(z+w)+u = z+(w+u)$.

(2) Commutativity of addition: $z+w = w+z$.

(3) Zero: *the number* $0 = 0+0i$ *satisfies* $z+0 = z$.

(4) Opposite numbers: *for* $z = a+bi$ *the number* $-z = (-a)+(-b)i$ *satisfies* $z+(-z) = 0$.

(5) Associativity of multiplication: $(z\cdot w)\cdot u = z\cdot(w\cdot u)$.

(6) Commutativity of multiplication: $z\cdot w = w\cdot z$.

(7) One: *the number* $1 = 1+0i$ *satisfies* $z\cdot 1 = z$.

(8) Inverses: *if* $z \neq 0$, *then the number* $\frac{1}{z}$ *from the Inversion fact satisfies* $z\cdot\frac{1}{z} = 1$.

(9) Distributivity: $z\cdot(w+u) = z\cdot w + z\cdot u$.

These are exactly the rules we use, usually without thinking, when we compute with real numbers. So every familiar algebraic manipulation, such as expanding brackets, adding fractions, cancelling a non-zero factor or solving a linear equation $az = b$ with $a \neq 0$, works for complex numbers just as it does for real numbers.

1.3 Modulus, conjugate and the complex plane

Definition 1.10. Let $z = a + bi$ be a complex number. We define

$$\begin{aligned}
&\mathfrak{Re}(z) = a && \text{(the } \textit{real part} \text{ of } z\text{)},\\
&\mathfrak{Im}(z) = b && \text{(the } \textit{imaginary part} \text{ of } z\text{)},\\
&|z| = \sqrt{a^2+b^2} && \text{(the } \textit{modulus} \text{ of } z\text{)},\\
&\bar{z} = a - bi && \text{(the } \textit{conjugate} \text{ of } z\text{)}.
\end{aligned}$$

Example 1.11. For $z = 3-4i$ we have $\mathfrak{Re}(z) = 3$, $\mathfrak{Im}(z) = -4$ (note that the imaginary part is a *real* number), $|z| = \sqrt{9+16} = 5$ and $\bar{z} = 3+4i$. Moreover $z\cdot\bar{z} = (3-4i)(3+4i) = 9+16 = 25 = |z|^2$.

Fact 1.12 (Properties). For complex numbers z, w,

$$z\cdot\bar{z} = |z|^2, \qquad |z| = \sqrt{z\cdot\bar{z}}, \qquad \mathfrak{Re}(z) = \tfrac{1}{2}(z+\bar{z}), \qquad \mathfrak{Im}(z) = \tfrac{1}{2i}(z-\bar{z}),$$

and $|z-w|$ is the distance between the points z and w.

Fact 1.13 (Algebra of conjugates and moduli). For all $z, w \in \mathbb{C}$:

$$\overline{z+w} = \bar{z} + \bar{w}, \qquad \overline{z \cdot w} = \bar{z} \cdot \bar{w}, \qquad \bar{\bar{z}} = z, \qquad |z \cdot w| = |z|\,|w|, \qquad \left|\frac{z}{w}\right| = \frac{|z|}{|w|} \quad (w \neq 0).$$

Fact 1.14 (Triangle inequality). For all $z, w \in \mathbb{C}$,

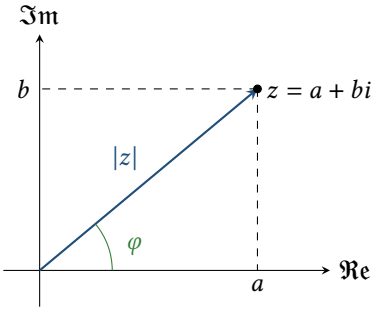
$$|z + w| \;\leq\; |z| + |w|,$$

with equality exactly when one of z, w is a non-negative real multiple of the other. Geometrically: in the triangle with vertices 0, z and $z + w$, no side is longer than the sum of the other two.

Example 1.15 (Geometric loci). Since $|z - z_0|$ is the distance from z to z_0, conditions on moduli describe geometric figures in the complex plane:

- $\{z \in \mathbb{C} \mid |z - z_0| = r\}$ is the circle of radius r centred at z_0; for instance $|z + 3 - i| = 5$ describes the circle of radius 5 centred at $z_0 = -3 + i$ (note the signs: $|z + 3 - i| = |z - (-3 + i)|$).
- $\{z \in \mathbb{C} \mid |z - z_0| \leq r\}$ is the closed disc of radius r centred at z_0, and $\{z \in \mathbb{C} \mid |z - z_0| > r\}$ is the exterior of that disc.
- Conditions on $\mathfrak{Re}$ and $\mathfrak{Im}$ describe half-planes and lines: $\mathfrak{Re}(z) > 0$ is the open right half-plane, while $\mathfrak{Im}(z) = 1$ is a horizontal line.

Since a complex number $z = a + bi$ is a pair of real numbers, it can be drawn as a point (or vector) in the plane — the *complex plane* (or *Argand diagram*). Its modulus $|z|$ is the length of the vector, and its *argument* φ is the angle the vector makes with the positive real axis.



1.4 Polar form and De Moivre's formula

Definition 1.16. Let $z = a + bi$ be a non-zero complex number. An *argument* of z (also called its *phase*), denoted $\arg(z)$, is any angle $\varphi \in \mathbb{R}$ satisfying

$$\cos\varphi = \frac{a}{|z|}, \qquad \sin\varphi = \frac{b}{|z|}.$$

Geometrically, φ is the angle between the positive real axis and the vector z. For $z = 0$ the quotients above are undefined, and we adopt the convention $\arg(0) = 0$. The pair $\big(|z|, \arg(z)\big)$ describes z in *polar coordinates*, as opposed to the Cartesian coordinates (a, b).

Theorem 1.17 (Polar form). *Every complex number $z = a + bi$ can be written as*

$$z = |z|\,(\cos\varphi + i\sin\varphi), \qquad \varphi = \arg(z),$$

since $a = |z|\cos\varphi$ and $b = |z|\sin\varphi$ (for $z = 0$ both sides vanish, whatever the angle). Conversely, $|z| = \sqrt{a^2 + b^2}$, while for $z \neq 0$ the argument φ may be found from the equation $\tan\varphi = \frac{b}{a}$ (for $a \neq 0$) together with the quadrant in which the point (a, b) lies.

Warning. The equation $\tan\varphi = \frac{b}{a}$ alone does *not* determine the argument: two angles in $(-\pi, \pi]$ satisfy it. One must always check in which quadrant the point (a, b) lies. For instance, $z = -1 + i$ lies in the second quadrant, so $\arg(z) = \frac{3\pi}{4}$, even though $\tan\frac{3\pi}{4} = -1 = \tan\left(-\frac{\pi}{4}\right)$.

Fact 1.18. If φ is an argument of a complex number z, then so is $\varphi + 2k\pi$ for any $k \in \mathbb{Z}$.

Definition 1.19. The argument φ of a complex number z for which $\varphi \in (-\pi, \pi]$ is called the *principal argument.*

Example 1.20. Let $z = 1 + i$. Then $|z| = \sqrt{2}$ and, reading the angle off the complex plane, $\arg(z) = \frac{\pi}{4} = 45°$. In polar form,

$$1 + i = \sqrt{2}\left(\cos\tfrac{\pi}{4} + i\sin\tfrac{\pi}{4}\right).$$

Theorem 1.21 (Multiplication in polar form). *Let $z_1 = |z_1|(\cos\varphi_1 + i\sin\varphi_1)$ and $z_2 = |z_2|(\cos\varphi_2 + i\sin\varphi_2)$. Then*

$$z_1 z_2 = |z_1|\,|z_2|\,\big(\cos(\varphi_1 + \varphi_2) + i\sin(\varphi_1 + \varphi_2)\big), \qquad \frac{z_1}{z_2} = \frac{|z_1|}{|z_2|}\,\big(\cos(\varphi_1 - \varphi_2) + i\sin(\varphi_1 - \varphi_2)\big).$$

That is, multiplication multiplies the moduli and adds the arguments.

Theorem 1.22 (Multiplication in polar form). *Let $z_1 = |z_1|(\cos\varphi_1 + i\sin\varphi_1)$ and $z_2 = |z_2|(\cos\varphi_2 + i\sin\varphi_2)$. Then*

$$z_1 z_2 = |z_1|\,|z_2|\,\big(\cos(\varphi_1 + \varphi_2) + i\sin(\varphi_1 + \varphi_2)\big), \qquad \frac{z_1}{z_2} = \frac{|z_1|}{|z_2|}\,\big(\cos(\varphi_1 - \varphi_2) + i\sin(\varphi_1 - \varphi_2)\big).$$

Thus, $|z_1 z_2| = |z_1||z_2|$ and $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$.

Proof. For the product of $z_1 z_2$:

$$\begin{aligned} z_1 z_2 &= |z_1|\,|z_2|(\cos\varphi_1 + i\sin\varphi_1)\cdot(\cos\varphi_2 + i\sin\varphi_2) \\ &= |z_1|\,|z_2|\Big[(\cos\varphi_1\cos\varphi_2 - \sin\varphi_1\sin\varphi_2) + i(\cos\varphi_1\sin\varphi_2 + \sin\varphi_1\cos\varphi_2)\Big]. \end{aligned}$$

Applying the angle addition identities for sine and cosine:

$$\cos(\varphi_1 + \varphi_2) = \cos\varphi_1\cos\varphi_2 - \sin\varphi_1\sin\varphi_2, \qquad \sin(\varphi_1 + \varphi_2) = \sin\varphi_1\cos\varphi_2 + \cos\varphi_1\sin\varphi_2,$$

we obtain

$$z_1 z_2 = |z_1|\,|z_2|\,\big(\cos(\varphi_1 + \varphi_2) + i\sin(\varphi_1 + \varphi_2)\big).$$

□

Remark 1.23 (Multiplication is rotation and scaling). The theorem has a geometric reading worth memorising: multiplying by $z = |z|(\cos\varphi + i\sin\varphi)$ rotates the plane by the angle φ and scales it by the factor $|z|$. In particular, multiplication by a number of modulus 1 is a pure rotation — this is the fact behind the use of complex numbers (and, one dimension up, quaternions) in computer graphics — and multiplication by i is the rotation by 90°: one checks directly that $i\cdot(a + bi) = -b + ai$.

Theorem 1.24 (De Moivre's formula). *Set $z^0 = 1$ and $z^n = z^{n-1}\cdot z$. If $z = |z|(\cos\varphi + i\sin\varphi)$ and $n \in \mathbb{N}$, then*

$$z^n = |z|^n\big(\cos(n\varphi) + i\sin(n\varphi)\big).$$

Example 1.25. We compute $(1 + i)^{100}$ using $1 + i = \sqrt{2}(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4})$:

$$(1 + i)^{100} = \big(\sqrt{2}\big)^{100}\left(\cos\tfrac{100\pi}{4} + i\sin\tfrac{100\pi}{4}\right) = 2^{50}\big(\cos 25\pi + i\sin 25\pi\big) = 2^{50}(\cos\pi + i\sin\pi) = -2^{50}.$$

Example 1.26 (Trigonometric identities from De Moivre's formula). Take $z = \cos\alpha + i\sin\alpha$ (a number of modulus 1). De Moivre's formula gives $\cos 3\alpha + i\sin 3\alpha = (\cos\alpha + i\sin\alpha)^3$, and expanding the cube,

$$(\cos\alpha + i\sin\alpha)^3 = \cos^3\alpha + 3i\cos^2\alpha\sin\alpha - 3\cos\alpha\sin^2\alpha - i\sin^3\alpha.$$

Comparing real and imaginary parts of both sides yields both triple-angle formulas at once:

$$\cos 3\alpha = \cos^3\alpha - 3\cos\alpha\sin^2\alpha, \qquad \sin 3\alpha = 3\cos^2\alpha\sin\alpha - \sin^3\alpha.$$

The same method expresses $\cos n\alpha$ and $\sin n\alpha$ in terms of $\cos\alpha$ and $\sin\alpha$ for any $n \in \mathbb{N}$.

Remark 1.27 (Exponential notation). In mathematical analysis one proves *Euler's formula* $e^{i\varphi} = \cos\varphi + i\sin\varphi$, which turns the polar form into the compact expression $z = |z|\,e^{i\varphi}$. The multiplication rule and De Moivre's formula then become instances of the ordinary laws of exponents:

$$|z_1|e^{i\varphi_1}\cdot|z_2|e^{i\varphi_2} = |z_1||z_2|\,e^{i(\varphi_1+\varphi_2)}, \qquad \big(|z|e^{i\varphi}\big)^n = |z|^n e^{in\varphi}.$$

We will not rely on this notation in these notes, but it is ubiquitous in engineering, physics and signal processing, and it is the form in which the Fast Fourier Transform is usually written.

1.5 Roots of complex numbers

Definition 1.28. For a complex number $z \neq 0$ and $n \in \mathbb{N}$, any number w satisfying $w^n = z$ is called an *n-th root* of z.

Theorem 1.29. *Let $z = |z|(\cos\varphi + i\sin\varphi)$ and $n \in \mathbb{N}$. Then every n-th root of z is of the form*

$$z_k = \sqrt[n]{|z|}\left(\cos\tfrac{\varphi+2k\pi}{n} + i\sin\tfrac{\varphi+2k\pi}{n}\right), \qquad k = 0, 1, \ldots, n-1.$$

Fact 1.30. All of the n-th roots of a complex number z lie on the circle centred at 0 of radius $\sqrt[n]{|z|}$, and they divide its circumference into n equal arcs.

Example 1.31. We find the square roots of $2i$. Writing $2i = 2\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$,

$$z_0 = \sqrt{2}\left(\cos\tfrac{\pi}{4} + i\sin\tfrac{\pi}{4}\right) = 1 + i,$$
$$z_1 = \sqrt{2}\left(\cos\tfrac{5\pi}{4} + i\sin\tfrac{5\pi}{4}\right) = -1 - i.$$

Indeed, one checks directly that $(1+i)^2 = 2i$ and $(-1-i)^2 = 2i$.

Example 1.32. We find the cube roots of -8. Writing $-8 = 8(\cos\pi + i\sin\pi)$, the roots are

$$z_k = 2\left(\cos\tfrac{\pi+2k\pi}{3} + i\sin\tfrac{\pi+2k\pi}{3}\right), \qquad k = 0, 1, 2,$$

that is,

$$z_0 = 2\left(\cos\tfrac{\pi}{3} + i\sin\tfrac{\pi}{3}\right) = 1 + \sqrt{3}\,i, \qquad z_1 = 2(\cos\pi + i\sin\pi) = -2, \qquad z_2 = 2\left(\cos\tfrac{5\pi}{3} + i\sin\tfrac{5\pi}{3}\right) = 1 - \sqrt{3}\,i.$$

Note that z_0 and z_2 are conjugate — as they must be, being the non-real roots of the polynomial $x^3 + 8$ with real coefficients (see the last section of this chapter).

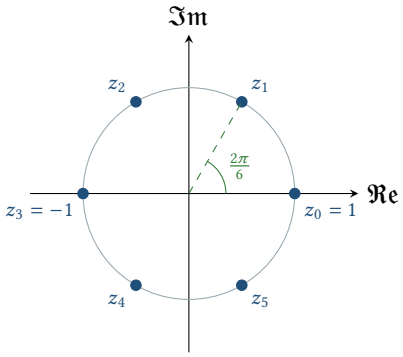
Example 1.33 (Roots of unity). Taking $z = 1$, the n-th roots of 1 are

$$z_k = \cos\tfrac{2k\pi}{n} + i\sin\tfrac{2k\pi}{n}, \qquad k = 0, \ldots, n-1.$$

For instance, the cube roots of unity are

$$1, \qquad -\tfrac{1}{2} + \tfrac{\sqrt{3}}{2}\,i, \qquad -\tfrac{1}{2} - \tfrac{\sqrt{3}}{2}\,i,$$

and the fourth roots of unity are $1,\ i,\ -1,\ -i$. The picture below shows the sixth roots of unity $z_0, \ldots, z_5$: they form a regular hexagon inscribed in the unit circle.



1.6 Polynomials

Definition 1.34. A *polynomial* f of degree $n =: \deg(f)$ with *real coefficients* is an expression

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_1 x + a_0,$$

with $a_0, a_1, \ldots, a_n \in \mathbb{R}$ and $a_n \neq 0$. If we allow $a_0, a_1, \ldots, a_n \in \mathbb{C}$, we speak of a polynomial with *complex coefficients*. The sets of all such polynomials are denoted by $\mathbb{R}[x]$ and $\mathbb{C}[x]$, respectively. The degree of the zero polynomial 0 is taken to be $-\infty$.

Since every real number is a complex number, $\mathbb{R}[x] \subseteq \mathbb{C}[x]$: a polynomial with real coefficients is a special polynomial with complex coefficients. Polynomials are added and multiplied in the usual way, and f can be evaluated at any complex number z, giving $f(z) \in \mathbb{C}$.

Example 1.35. $x + 1$ and $2x^3 + 1$ belong to $\mathbb{R}[x]$ (and hence also to $\mathbb{C}[x]$), while $x^2 + (2 + i)x + i$ belongs to $\mathbb{C}[x]$ but not to $\mathbb{R}[x]$.

Fact 1.36 (Division with remainder). Let $f, g \in \mathbb{C}[x]$ with $g \neq 0$. Then there exist polynomials $q, r \in \mathbb{C}[x]$ such that

$$f = q \cdot g + r, \qquad \deg(r) < \deg(g).$$

Moreover, if $f, g \in \mathbb{R}[x]$, then also $q, r \in \mathbb{R}[x]$.

Example 1.37. For $f(x) = x^2 + 1$ and $g(x) = x + 1$,

$$x^2 + 1 = (x - 1)(x + 1) + 2.$$

Definition 1.38. A polynomial $f \in \mathbb{R}[x]$ is *reducible over* $\mathbb{R}$ if there exist $g, h \in \mathbb{R}[x]$ with $\deg(g) > 0$, $\deg(h) > 0$ and $f = g \cdot h$. It is *irreducible over* $\mathbb{R}$ if

$$f = g \cdot h \implies \deg(g) = 0 \text{ or } \deg(h) = 0 \qquad (g, h \in \mathbb{R}[x]).$$

Reducibility and irreducibility *over* $\mathbb{C}$ of $f \in \mathbb{C}[x]$ are defined in the same way, with $\mathbb{R}[x]$ replaced by $\mathbb{C}[x]$.

Example 1.39. $x + 1$ is irreducible over both $\mathbb{R}$ and $\mathbb{C}$. The polynomial $x^2 + 1$ is irreducible over $\mathbb{R}$, but reducible over $\mathbb{C}$ since $x^2 + 1 = (x - i)(x + i)$.

A complex number a is a *root* of $f \in \mathbb{C}[x]$ if $f(a) = 0$. For $f \in \mathbb{R}[x]$ we distinguish *real roots* ($a \in \mathbb{R}$) from non-real ones.

Theorem 1.40 (Bézout's root–factor theorem). *Let $f \in \mathbb{C}[x]$ and $a \in \mathbb{C}$. Then a is a root of f if and only if $x - a$ divides f, i.e.*

$$f(a) = 0 \iff f(x) = q(x)\,(x - a) \text{ for some } q \in \mathbb{C}[x].$$

If $f \in \mathbb{R}[x]$ and $a \in \mathbb{R}$, then $q \in \mathbb{R}[x]$.

Proof. Division with remainder by $g(x) = x - a$ gives $f(x) = q(x)(x - a) + r$ with $\deg(r) < 1$, so r is a constant.

Substituting $x = a$ yields $r = f(a)$. Hence $f(a) = 0$ if and only if $r = 0$, i.e. if and only if $f(x) = q(x)(x - a)$. □

Corollary 1.41. *A non-zero polynomial $f \in \mathbb{C}[x]$ of degree n has at most n roots in $\mathbb{C}$ (and hence, if $f \in \mathbb{R}[x]$, at most n real roots).*

Theorem 1.42 (Fundamental Theorem of Algebra). *Any polynomial $f \in \mathbb{C}[x]$ of degree $\deg(f) \geq 1$ has a root in $\mathbb{C}$.*

Corollary 1.43. *Any polynomial $f \in \mathbb{C}[x]$ of degree $\deg(f) \geq 1$ factors into linear factors:*

$$f(x) = a(x - z_0)^{k_0}(x - z_1)^{k_1} \cdots (x - z_m)^{k_m},$$

where $k_i > 0$ and $k_0 + k_1 + \ldots + k_m = \deg(f)$. In particular, the only irreducible polynomials of positive degree over $\mathbb{C}$ are those of degree 1.

Theorem 1.44 (Quadratic formula over $\mathbb{C}$). *For a polynomial $f(x) = ax^2 + bx + c$ with $a, b, c \in \mathbb{C}$ ($a \neq 0$) the roots are*

$$z_0 = \frac{-b - \delta}{2a}, \qquad z_1 = \frac{-b + \delta}{2a},$$

where δ is any square root of $\Delta = b^2 - 4ac$.

Example 1.45.

- For $x^2 + 2x + 5$ (real coefficients) we get $\Delta = 4 - 20 = -16$, so we may take $\delta = 4i$, and

  $$z_0 = \frac{-2 - 4i}{2} = -1 - 2i, \qquad z_1 = \frac{-2 + 4i}{2} = -1 + 2i.$$

  The two roots are conjugate — as predicted by the theorem below.
- For $z^2 - 3z + (3 - i)$ we get $\Delta = 9 - 4(3 - i) = -3 + 4i$. Since $(1 + 2i)^2 = -3 + 4i$, we may take $\delta = 1 + 2i$, and

  $$z_0 = \frac{3 - (1 + 2i)}{2} = 1 - i, \qquad z_1 = \frac{3 + (1 + 2i)}{2} = 2 + i.$$

  Here the coefficients are not all real, and indeed the roots are not conjugate to each other.

Theorem 1.46. *Let $f \in \mathbb{R}[x]$. If a complex number z is a root of f, then so is its conjugate $\bar{z}$:*

$$f(z) = 0 \implies f(\bar{z}) = 0.$$

Corollary 1.47. *The irreducible polynomials of positive degree over $\mathbb{R}$ are exactly the polynomials of degree 1 and the polynomials of degree 2 with $\Delta = b^2 - 4ac < 0$. Consequently, every polynomial $f \in \mathbb{R}[x]$ of positive degree factors into a product of real linear factors and real quadratic factors with negative discriminant (the latter pairing up the conjugate non-real roots of f).*

Example 1.48. Consider $f(x) = x^4 - 16$. Over $\mathbb{R}$ it factors as

$$x^4 - 16 = (x^2 - 4)(x^2 + 4) = (x - 2)(x + 2)(x^2 + 4),$$

and the quadratic factor $x^2 + 4$ (with $\Delta = -16 < 0$) is irreducible over $\mathbb{R}$. Over $\mathbb{C}$ the factorisation continues all the way down to linear factors:

$$x^4 - 16 = (x - 2)(x + 2)(x - 2i)(x + 2i).$$

The two non-real roots $\pm 2i$ form a conjugate pair.

Remark 1.49. Nothing in division with remainder or in Bézout's theorem depends on the coefficients being real or complex numbers specifically: all we need is to be able to add, subtract, multiply and divide them. Once *fields* are introduced in the next chapter, every definition and fact in this section up to and including the corollary to Bézout's theorem carries over word for word to polynomials $\mathbb{K}[x]$ with coefficients in any field $\mathbb{K}$. The Fundamental Theorem of Algebra, by contrast, is a special property of $\mathbb{C}$.

2 Groups and fields

2.1 Motivation

Computer science students meet abstraction every day: an *interface* specifies which operations a data type must support and which laws they must obey, and any code written against the interface automatically works for every implementation. This chapter does exactly that to arithmetic. The arithmetic of the complex numbers we have just met — and of $\mathbb{Q}$ and $\mathbb{R}$ before them — obeys the same short list of rules: operations that are associative, have a neutral element, and admit inverses. Abstracting this list leads to two of the most basic algebraic "interfaces": *groups* and *fields*. Every theorem we prove from the axioms alone is proved once, for all implementations simultaneously: for real numbers, for complex numbers and for residues modulo a prime.

This is not abstraction for its own sake — the "exotic" implementations are precisely the ones running your digital life:

- **Cryptography.** Whenever your browser shows a padlock, a key exchange has just been performed in a large finite group (Diffie–Hellman in $\mathbb{Z}_p \setminus \{0\}$, or on an elliptic curve). RSA encryption is arithmetic modulo n, and its security rests on how hard certain computations in these groups are. The finite fields $\mathbb{Z}_p$ defined in this chapter are the entry ticket to all of modern cryptography.
- **Error-correcting codes.** QR codes, scratched CDs, hard-disk RAID arrays and photographs sent back by space probes all survive damage thanks to Reed–Solomon codes — linear algebra over a finite field with 256 elements. The AES cipher that encrypts your Wi-Fi traffic computes in the same field.
- **Bits form a field.** The smallest field, $\mathbb{Z}_2 = \{0, 1\}$, is the arithmetic of a single bit: its addition is the XOR gate and its multiplication is the AND gate. Linear algebra over $\mathbb{Z}_2$ is a standard tool in hashing, checksums and cryptanalysis.

For this course, though, the most important reason is different: fields are exactly the sets of *scalars* over which vector spaces — the central objects of linear algebra — will be built in the following chapters. Whenever a definition below feels dry, remember: we are writing the interface, and the rest of this book is the code that runs against it.

2.2 Groups

Definition 2.1. A *group* is a set G together with a binary operation $*\colon G \times G \to G$ satisfying:

- **(Associativity)** $(a * b) * c = a * (b * c)$ for all $a, b, c \in G$;
- **(Identity)** there is an element $e \in G$ such that $a * e = e * a = a$ for all $a \in G$;
- **(Inverses)** for every $a \in G$ there is an element $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$.

A group is called *abelian* (or *commutative*) if, in addition, $a * b = b * a$ for all $a, b \in G$.

Remark 2.2. When the operation is written additively (denoted $+$) the identity is written 0 and the inverse of a is written $-a$. When it is written multiplicatively (denoted $\cdot$) the identity is written 1 and the inverse of a is written a^{-1} or $\frac{1}{a}$.

Before listing examples we introduce one construction that will recur throughout the course: *modular arithmetic*.

Definition 2.3 (Modular arithmetic). Fix a positive integer n. For an integer a we write $a \bmod n$ for the remainder of the division of a by n: the unique number $r \in \{0, 1, \dots, n-1\}$ such that $a = qn + r$ for some

$q \in \mathbb{Z}$. Two integers a and b are *congruent modulo n* if they leave the same remainder, i.e. if n divides $a - b$. On the set $\mathbb{Z}_n = \{0, 1, \dots, n-1\}$ we define *addition and multiplication modulo n*:

$$a +_n b = (a + b) \bmod n, \qquad a \cdot_n b = (a \cdot b) \bmod n.$$

Example 2.4. $17 \bmod 4 = 1$ (since $17 = 4 \cdot 4 + 1$), $4 \bmod 17 = 4$, and $(-2) \bmod 5 = 3$ (since $-2 = (-1) \cdot 5 + 3$; the remainder is always taken in $\{0, \dots, n-1\}$). In $\mathbb{Z}_6$ we have $4 +_6 5 = 3$ and $4 \cdot_6 5 = 2$.

Example 2.5.

- $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$ and $(\mathbb{C}, +)$ are abelian groups; the identity is 0 and the inverse of a is $-a$.
- $(\mathbb{Q} \setminus \{0\}, \cdot)$, $(\mathbb{R} \setminus \{0\}, \cdot)$ and $(\mathbb{C} \setminus \{0\}, \cdot)$ are abelian groups; the identity is 1 and the inverse of a is $\frac{1}{a}$.
- $(\mathbb{Z}, \cdot)$ is *not* a group: apart from ± 1, no integer has a multiplicative inverse in $\mathbb{Z}$.
- For a fixed n, the set $\mathbb{Z}_n = \{0, 1, \dots, n-1\}$ with addition modulo n is an abelian group.

Example 2.6 (A finite multiplicative group). For a prime p, the non-zero residues $\mathbb{Z}_p \setminus \{0\} = \{1, 2, \dots, p-1\}$ form an abelian group under multiplication modulo p (this is proved in the next section, as part of the theorem that $\mathbb{Z}_p$ is a field). For $p = 5$, for example, $2 \cdot 3 = 6 \equiv 1 \pmod 5$, so 3 is the inverse of 2. Inverses allow us to solve linear equations exactly as in $\mathbb{Q}$ or $\mathbb{R}$: in $\mathbb{Z}_7$ the equation $2 \cdot_7 x = 1$ has the unique solution $x = 4$ (indeed $2 \cdot 4 = 8 \equiv 1 \pmod 7$), and $2 \cdot_7 x = 5$ is solved by multiplying both sides by 4: $x = 4 \cdot_7 5 = 6$.

Example 2.7 (The power-set group). Let X be a set and let 2^X denote the set of all its subsets. The *symmetric difference*

$$A \triangle B = (A \cup B) \setminus (A \cap B)$$

(the set of elements belonging to exactly one of A and B) makes $(2^X, \triangle)$ an abelian group: the identity element is $\emptyset$, and every element is its own inverse, since $A \triangle A = \emptyset$. Identifying a subset $A \subseteq \{1, \dots, n\}$ with its characteristic bit vector, $\triangle$ becomes the bitwise XOR of bit strings.

Example 2.8 (Roots of unity). For a fixed n, the n-th roots of unity from Chapter 1,

$$U_n = \left\{ \cos \tfrac{2k\pi}{n} + i \sin \tfrac{2k\pi}{n} \;\middle|\; k = 0, 1, \dots, n-1 \right\} \subseteq \mathbb{C},$$

form an abelian group under multiplication of complex numbers: multiplying two roots of unity adds their arguments, which produces again a multiple of $\frac{2\pi}{n}$. The identity is 1, and the inverse of a root is its conjugate. Since multiplying the roots amounts to adding the exponents k modulo n, the group U_n is the group $(\mathbb{Z}_n, +_n)$ in disguise.

Fact 2.9. In any group G the identity element is unique, and each element has a unique inverse.

Proof. If e and e' are both identities, then $e = e * e' = e'$. If b and b' are both inverses of a, then

$$b = b * e = b * (a * b') = (b * a) * b' = e * b' = b'. \qquad \square$$

Fact 2.10 (Cancellation laws). In any group G: if $a * b = a * c$, then $b = c$; similarly, if $b * a = c * a$, then $b = c$.

Proof. Multiplying $a * b = a * c$ by a^{-1} on the left gives $a^{-1} * (a * b) = a^{-1} * (a * c)$; by associativity $(a^{-1} * a) * b = (a^{-1} * a) * c$, that is $e * b = e * c$, so $b = c$. The second law is proved symmetrically. $\square$

2.3 Fields

A field is a set carrying *two* compatible operations, addition and multiplication, each of which behaves like an abelian group (with multiplication restricted to the non-zero elements), tied together by distributivity. The fields $\mathbb{Q}$ and $\mathbb{R}$ are the familiar prototypes.

Definition 2.11. A set $\mathbb{K}$ together with two binary operations $+$ and $\cdot$ is called a *field* if it satisfies the following conditions.

- for any $a, b, c \in \mathbb{K}$, $a + b = b + a$ and $a + (b + c) = (a + b) + c$;

- there is an element $0 \in \mathbb{K}$ such that $a + 0 = 0 + a = a$ for any $a \in \mathbb{K}$;
- for any $a \in \mathbb{K}$ there exists $-a \in \mathbb{K}$ such that $a + (-a) = (-a) + a = 0$;
- for any $a, b, c \in \mathbb{K}$, $a \cdot b = b \cdot a$ and $a \cdot (b \cdot c) = (a \cdot b) \cdot c$;
- there is an element $1 \in \mathbb{K}$ such that $a \cdot 1 = 1 \cdot a = a$ for any $a \in \mathbb{K}$;
- for any $a \in \mathbb{K}$ with $a \neq 0$ there is $\frac{1}{a} \in \mathbb{K}$ such that $a \cdot \frac{1}{a} = \frac{1}{a} \cdot a = 1$;
- for any $a, b, c \in \mathbb{K}$, $(a + b) \cdot c = a \cdot c + b \cdot c$.

Remark 2.12. Equivalently: $(\mathbb{K}, +)$ is an abelian group with identity 0, the non-zero elements $(\mathbb{K} \setminus \{0\}, \cdot)$ form an abelian group with identity 1, and multiplication distributes over addition.

Remark 2.13 (Subtraction and division). In any field we write $a - b$ for $a + (-b)$, and $\frac{a}{b}$ (or a/b) for $a \cdot \frac{1}{b}$ whenever $b \neq 0$. Thus all four arithmetic operations make sense in any field, and — as the next fact begins to show — they obey the computational rules familiar from $\mathbb{Q}$ and $\mathbb{R}$.

Example 2.14. $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$, each with their usual addition and multiplication, are fields.

Fact 2.15 (Elementary consequences of the field axioms). In any field $\mathbb{K}$, for all $a, b \in \mathbb{K}$:

- $a \cdot 0 = 0$;
- $(-1) \cdot a = -a$;
- if $a \cdot b = 0$, then $a = 0$ or $b = 0$ (a field has *no zero divisors*).

Proof. First, $a \cdot 0 = a \cdot (0 + 0) = a \cdot 0 + a \cdot 0$, and adding $-(a \cdot 0)$ to both sides gives $0 = a \cdot 0$. Second, $a + (-1) \cdot a = 1 \cdot a + (-1) \cdot a = \big(1 + (-1)\big) \cdot a = 0 \cdot a = 0$, so $(-1) \cdot a$ is the additive inverse of a. Third, if $a \cdot b = 0$ and $a \neq 0$, then multiplying by $\frac{1}{a}$ gives $b = \frac{1}{a} \cdot (a \cdot b) = \frac{1}{a} \cdot 0 = 0$. □

Example 2.16. Consider the set $\mathbb{Z}_2 = \{0, 1\}$ with addition and multiplication given by

$+$	0	1
0	0	1
1	1	0

$\cdot$	0	1
0	0	0
1	0	1

Then $\mathbb{Z}_2$ with these operations is a field (the smallest one).

Example 2.17. More generally, consider $\mathbb{Z}_n$ with addition and multiplication modulo n.

- $\mathbb{Z}_4$ is *not* a field: the products $2 \cdot 0 = 0$, $2 \cdot 1 = 2$, $2 \cdot 2 = 0$ and $2 \cdot 3 = 2$ (all modulo 4) never equal 1, so the element 2 has no multiplicative inverse. Note that $\mathbb{Z}_4$ also violates the no-zero-divisors property: $2 \cdot_4 2 = 0$ although $2 \neq 0$.
- $\mathbb{Z}_5$ *is* a field: modulo 5 we have $1 \cdot 1 = 1$, $2 \cdot 3 = 1$ and $4 \cdot 4 = 1$, so every non-zero element has an inverse.

Theorem 2.18. *$\mathbb{Z}_n$, with addition and multiplication modulo n, is a field if and only if n is a prime number.*

The proof is left to the reader as an exercise.

Remark 2.19. The theorem only says that inverses exist; it does not say how to find them efficiently. In practice inverses modulo p are computed with the *extended Euclidean algorithm* — a staple of every cryptographic library.

Example 2.20 (A field between $\mathbb{Q}$ and $\mathbb{R}$). The set

$$\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\} \subseteq \mathbb{R}$$

with the usual addition and multiplication is a field. The only non-obvious axiom is the existence of multiplicative inverses, and it follows from the same trick we used for complex numbers: if $a + b\sqrt{2} \neq 0$, then

$$\frac{1}{a + b\sqrt{2}} = \frac{a - b\sqrt{2}}{(a + b\sqrt{2})(a - b\sqrt{2})} = \frac{a}{a^2 - 2b^2} - \frac{b}{a^2 - 2b^2}\sqrt{2},$$

where $a^2 - 2b^2 \neq 0$ because $\sqrt{2}$ is irrational.

> *Warning.* Not every set with an addition and a multiplication is a field. For example, $\mathbb{Z}$ with the standard operations is *not* a field, because only 1 and -1 have multiplicative inverses in $\mathbb{Z}$.

In the chapters that follow, $\mathbb{K}$ always denotes a field; the reader may safely keep $\mathbb{R}$ or $\mathbb{C}$ in mind throughout.

3 Vector spaces

3.1 Motivation

If there is one mathematical habit that defines modern computing, it is this: *turn your data into a vector.* A greyscale image of size 1000×1000 is a list of a million pixel brightnesses — a vector in $\mathbb{R}^{1\,000\,000}$. An audio clip is a vector of samples. A user of a streaming service is a vector of ratings. A word, inside a modern language model, is a vector of a few hundred coordinates learned from text. Once data is a vector, geometry becomes a computational tool: similar users are *nearby* vectors, recommendation is finding neighbours, and classification is separating regions of a space that we cannot draw but can compute in.

The operations we axiomatise in this chapter are not abstractions invented for their own sake — each is something computers do to data billions of times a second.

- *Scaling.* Turning up the volume multiplies every audio sample by the same number; dimming a photograph does the same to every pixel. Both are the product $\lambda \cdot v$ of a scalar and a vector.
- *Addition and the opposite vector.* Mixing two audio tracks adds their sample vectors coordinate by coordinate. Noise-cancelling headphones, in their idealised form, record the ambient noise n and play $-n$, so that the ear receives $s + n + (-n) = s$: the axiom $v + (-v) = \mathbf{0}$, at work inside your ears.
- *Linear combinations.* A cross-fade from image a to image b in a video is the family $(1 - t)\,a + t\,b$ for t running from 0 to 1 — a straight line segment in $\mathbb{R}^{1\,000\,000}$. And every colour on your screen is a combination $r \cdot (1, 0, 0) + g \cdot (0, 1, 0) + b \cdot (0, 0, 1)$ of pure red, green and blue: the first *basis* you ever used, probably without knowing it.

The deeper notions of the chapter answer a question every computer scientist should ask about data: *how many numbers does it really take to describe it?*

- A *spanning set* is a collection of building blocks capable of expressing everything in sight.
- *Linear independence* means no building block is redundant: none can be reconstructed from the others, so none can be dropped for free.
- A *basis* is a minimal, lossless set of building blocks, and the *dimension* — the size of any basis — is the honest number of degrees of freedom of the data.

Real data rarely uses all of its apparent dimensions: photographs of faces occupy a tiny sliver of $\mathbb{R}^{1\,000\,000}$, and that sliver is well approximated by low-dimensional *subspaces*. This single observation is why image compression, denoising and recommender systems work at all, and we will be able to *compute* such dimensions once matrix rank is available in Chapter 4.

One more remark on abstraction. We will not define vectors as "arrows" or as "tuples of numbers", but as elements of *any* set equipped with a well-behaved addition and scaling — programming against an interface again, as in the previous chapter. The payoff is immediate: tuples, polynomials, matrices and functions all satisfy the axioms, so every theorem proved below holds for all of them at once.

3.2 Vector spaces

You are already familiar with the two-dimensional vector space

$$\mathbb{R}^2 = \{(x, y) \mid x, y \in \mathbb{R}\}.$$

In $\mathbb{R}^2$ vectors can be added and multiplied by a number called a *scalar*:

$$(x_1, y_1) \oplus (x_2, y_2) = (x_1 + x_2,\ y_1 + y_2), \qquad \lambda \odot (x, y) = (\lambda x,\ \lambda y) \text{ for } \lambda \in \mathbb{R}.$$

The abstract notion of a vector space axiomatises exactly these two operations.

Definition 3.1. Let $\mathbb{K}$ be a field. A set $\mathbf{V}$ together with two operations:

- *vector addition* $+ \colon \mathbf{V} \times \mathbf{V} \to \mathbf{V}$ and
- *scalar multiplication* $\cdot \colon \mathbb{K} \times \mathbf{V} \to \mathbf{V}$

is called a *vector space* over the field $\mathbb{K}$ if it satisfies the following conditions:

- for any $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbf{V}$ we have $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ and $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$;
- there is $\mathbf{0} \in \mathbf{V}$, called the *zero vector*, such that for any $\mathbf{v} \in \mathbf{V}$ we have $\mathbf{v} + \mathbf{0} = \mathbf{0} + \mathbf{v} = \mathbf{v}$;
- for any $\mathbf{v} \in \mathbf{V}$ there is $-\mathbf{v} \in \mathbf{V}$ such that $\mathbf{v} + (-\mathbf{v}) = (-\mathbf{v}) + \mathbf{v} = \mathbf{0}$;
- for any $\mathbf{u}, \mathbf{v} \in \mathbf{V}$ and any scalar $\lambda \in \mathbb{K}$ we have $\lambda \cdot (\mathbf{u} + \mathbf{v}) = \lambda \cdot \mathbf{u} + \lambda \cdot \mathbf{v}$;
- for any $\mathbf{u} \in \mathbf{V}$ and any scalars $\lambda_1, \lambda_2 \in \mathbb{K}$ we have $(\lambda_1 + \lambda_2) \cdot \mathbf{u} = \lambda_1 \cdot \mathbf{u} + \lambda_2 \cdot \mathbf{u}$;
- for any $\mathbf{u} \in \mathbf{V}$ and any scalars $\lambda_1, \lambda_2 \in \mathbb{K}$ we have $(\lambda_1 \lambda_2) \cdot \mathbf{u} = \lambda_1 \cdot (\lambda_2 \cdot \mathbf{u})$;
- for any $\mathbf{u} \in \mathbf{V}$ we have $1 \cdot \mathbf{u} = \mathbf{u}$.

The elements of $\mathbf{V}$ are called *vectors* and the elements of $\mathbb{K}$ are called *scalars*.

Fact 3.2 (Elementary consequences of the axioms). Let $\mathbf{V}$ be a vector space over $\mathbb{K}$. For any $\mathbf{v} \in \mathbf{V}$ and $\lambda \in \mathbb{K}$:

$$0 \cdot \mathbf{v} = \mathbf{0}, \qquad \lambda \cdot \mathbf{0} = \mathbf{0}, \qquad (-1) \cdot \mathbf{v} = -\mathbf{v}, \qquad \text{and} \qquad \lambda \cdot \mathbf{v} = \mathbf{0} \implies \lambda = 0 \ \text{ or } \ \mathbf{v} = \mathbf{0}.$$

Proof. We prove the first equality; the others are proved similarly. By distributivity,

$$0 \cdot \mathbf{v} = (0 + 0) \cdot \mathbf{v} = 0 \cdot \mathbf{v} + 0 \cdot \mathbf{v},$$

and adding $-(0 \cdot \mathbf{v})$ to both sides gives $\mathbf{0} = 0 \cdot \mathbf{v}$. □

Example 3.3 (The space $\mathbb{K}^n$). Let $\mathbb{K}$ be any field and let $n \in \mathbb{N}$. Then

$$\mathbb{K}^n = \{(x_1, \ldots, x_n) \mid x_i \in \mathbb{K}\}$$

together with vector addition

$$(x_1, \ldots, x_n) + (y_1, \ldots, y_n) = (x_1 + y_1, \ldots, x_n + y_n)$$

and scalar multiplication

$$\lambda \cdot (x_1, \ldots, x_n) = (\lambda x_1, \ldots, \lambda x_n) \qquad (\lambda \in \mathbb{K})$$

forms a vector space over $\mathbb{K}$. The zero vector is $\mathbf{0} = (0, \ldots, 0)$.
In the case $\mathbb{K} = \mathbb{R}$ and $n = 2$ we recover the well-known space $\mathbb{R}^2$. Taking $\mathbb{K} = \mathbb{C}$ we obtain $\mathbb{C}^n = \{(z_1, \ldots, z_n) \mid z_i \in \mathbb{C}\}$.

Example 3.4 (Bit vectors). The field $\mathbb{K}$ in the previous example may also be finite. Taking $\mathbb{K} = \mathbb{Z}_2$ gives the space $\mathbb{Z}_2^n$ of *bit vectors*: vector addition is bitwise XOR, e.g. in $\mathbb{Z}_2^4$,

$$(1, 0, 1, 1) + (1, 1, 0, 1) = (0, 1, 1, 0),$$

and the only scalars are 0 and 1. Note that every vector is its own additive inverse. Identifying a subset of $\{1, \ldots, n\}$ with its characteristic bit vector, we may equally view $\mathbb{Z}_2^n$ as the set $2^{\{1,\ldots,n\}}$ of all subsets of $\{1, \ldots, n\}$, with the symmetric difference as vector addition. Vector spaces over $\mathbb{Z}_2$ are the natural home of error-correcting codes and appear throughout cryptography.

Example 3.5 (Complex numbers as a real vector space). Recall that the set of complex numbers is $\mathbb{C} = \mathbb{R} \times \mathbb{R} = \{(a, b) \mid a, b \in \mathbb{R}\}$. Together with the standard addition of complex numbers and multiplication by real scalars, $\mathbb{C}$ can be viewed as a vector space over the field $\mathbb{R}$.

Example 3.6 (Polynomials). Let $\mathbb{K}$ be a field. The set $\mathbb{K}[x]$ of all polynomials over $\mathbb{K}$, with the standard polynomial addition and multiplication by a constant from $\mathbb{K}$, forms a vector space over $\mathbb{K}$. The zero vector is the zero polynomial 0. Similarly, the set $\mathbb{K}_n[x]$ of polynomials of degree less than or equal to n over $\mathbb{K}$ is a vector space.

Example 3.7 (Function spaces). Let X be a non-empty set and $\mathbb{K}$ a field. The set $\mathbb{K}^X$ of all functions $f\colon X \to \mathbb{K}$, with the pointwise operations

$$(f+g)(x) = f(x) + g(x), \qquad (\lambda \cdot f)(x) = \lambda \cdot f(x),$$

is a vector space over $\mathbb{K}$. The zero vector is the constant function 0. For $X = \mathbb{R}$ and $\mathbb{K} = \mathbb{R}$ we obtain the space of all real functions of one real variable.

3.3 Subspaces

Definition 3.8. Let $\mathbf{V}$ be a vector space over $\mathbb{K}$. A subset $\mathbf{W} \subseteq \mathbf{V}$ is called a *subspace* of $\mathbf{V}$ if $\mathbf{W}$ is itself a vector space over $\mathbb{K}$ under the same operations of vector addition and scalar multiplication.

To put it differently, a subset $\mathbf{W} \subseteq \mathbf{V}$ is a subspace of $\mathbf{V}$ if

- for any $\mathbf{w}_1, \mathbf{w}_2 \in \mathbf{W}$ we have $\mathbf{w}_1 + \mathbf{w}_2 \in \mathbf{W}$;
- $\mathbf{0} \in \mathbf{W}$;
- for any $\mathbf{w} \in \mathbf{W}$ we have $-\mathbf{w} \in \mathbf{W}$;
- for any scalar $\lambda \in \mathbb{K}$ and any $\mathbf{w} \in \mathbf{W}$ we have $\lambda \cdot \mathbf{w} \in \mathbf{W}$.

Remark 3.9. The list can be shortened: if $\mathbf{W} \neq \emptyset$ is closed under addition and under scalar multiplication, the remaining conditions come for free, since $\mathbf{0} = 0 \cdot \mathbf{w}$ and $-\mathbf{w} = (-1) \cdot \mathbf{w}$ for any $\mathbf{w} \in \mathbf{W}$. In practice one therefore checks three things: $\mathbf{0} \in \mathbf{W}$ (in particular, $\mathbf{W}$ is non-empty), and $\mathbf{W}$ is closed under $+$ and under scalar multiplication. Note that every vector space $\mathbf{V}$ has two *trivial* subspaces: $\{\mathbf{0}\}$ and $\mathbf{V}$ itself.

Example 3.10. Consider the vector space $\mathbb{R}^3 = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}$. The set

$$W = \{(x, y, z) \mid 2x + y + 3z = 0\}$$

is a subspace of $\mathbb{R}^3$. Indeed, $\mathbf{0} = (0, 0, 0) \in W$; if $\mathbf{w}_1 = (x_1, y_1, z_1)$ and $\mathbf{w}_2 = (x_2, y_2, z_2)$ both satisfy the equation, then so does their sum,

$$2(x_1 + x_2) + (y_1 + y_2) + 3(z_1 + z_2) = \underbrace{(2x_1 + y_1 + 3z_1)}_{=0} + \underbrace{(2x_2 + y_2 + 3z_2)}_{=0} = 0,$$

and so does every multiple: $2(\lambda x_1) + \lambda y_1 + 3(\lambda z_1) = \lambda \cdot 0 = 0$, which covers both $\lambda \mathbf{w}_1 \in W$ and $-\mathbf{w}_1 \in W$. On the other hand, the set $W' = \{(x, y, z) \mid x^2 = 1\}$ is *not* a subspace of $\mathbb{R}^3$: it does not contain the zero vector, and it is not closed under addition either, since $(1, 0, 0) \in W'$ but $(1, 0, 0) + (1, 0, 0) = (2, 0, 0) \notin W'$.

Example 3.11. Consider the vector space $\mathbb{C}$ over $\mathbb{R}$. The set $\{z \in \mathbb{C} \mid \mathfrak{Re}(z) = 0\}$ is a subspace of $\mathbb{C}$.

Warning. The intersection of two subspaces of $\mathbf{V}$ is always a subspace, but their *union* usually is not. In $\mathbb{R}^2$ the two axes $W_1 = \{(x, 0) \mid x \in \mathbb{R}\}$ and $W_2 = \{(0, y) \mid y \in \mathbb{R}\}$ are subspaces, yet $(1, 0) + (0, 1) = (1, 1) \notin W_1 \cup W_2$, so $W_1 \cup W_2$ is not closed under addition.

3.4 Linear combinations and span

Definition 3.12. Let $\mathbf{V}$ be a vector space over a field $\mathbb{K}$, let $a_1, \ldots, a_n \in \mathbb{K}$ and $\mathbf{v}_1, \ldots, \mathbf{v}_n \in \mathbf{V}$. The vector

$$a_1\mathbf{v}_1 + \ldots + a_n\mathbf{v}_n \in \mathbf{V}$$

is called the *linear combination* of the vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ with coefficients $a_1, \ldots, a_n$.

Example 3.13. In $\mathbb{R}^2$ the vector $(2, 3)$ is a linear combination of $(1, 1)$ and $(0, 1)$ with coefficients 2 and 1:

$$(2, 3) = 2 \cdot (1, 1) + 1 \cdot (0, 1).$$

Definition 3.14. Let $\mathbf{V}$ be a vector space over a field $\mathbb{K}$ and let $S \subseteq \mathbf{V}$ be a set of vectors. By the *span* of S we mean the set

$$\operatorname{span}(S) = \{a_1\mathbf{v}_1 + \ldots + a_n\mathbf{v}_n \mid a_i \in \mathbb{K} \text{ and } \mathbf{v}_i \in S\} \subseteq \mathbf{V}.$$

In other words, $\operatorname{span}(S)$ is the set of all possible linear combinations of vectors from S. By convention, $\operatorname{span}(\emptyset) = \{\mathbf{0}\}$.

Example 3.15. Consider $\mathbb{R}^3$ and the two vectors $(1, 2, 0), (0, 0, 3) \in \mathbb{R}^3$. Then

$$\operatorname{span}(\{(1, 2, 0), (0, 0, 3)\}) = \{a(1, 2, 0) + b(0, 0, 3) \mid a, b \in \mathbb{R}\} = \{(a, 2a, 3b) \mid a, b \in \mathbb{R}\}.$$

Remark 3.16 (Geometric picture). In $\mathbb{R}^3$, the span of a single non-zero vector $\mathbf{v}$ is the line through the origin in the direction of $\mathbf{v}$, and the span of two vectors neither of which is a multiple of the other is the plane through the origin containing both. In fact the subspaces of $\mathbb{R}^3$ are exactly: $\{\mathbf{0}\}$, the lines through the origin, the planes through the origin, and $\mathbb{R}^3$ itself — a fact we will be able to justify once the notion of dimension is available at the end of this chapter.

Theorem 3.17. *Let $\mathbf{V}$ be a vector space over a field $\mathbb{K}$ and let $\mathbf{v}_1, \ldots, \mathbf{v}_n \in \mathbf{V}$. Then the set*

$$\operatorname{span}(\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}) = \{a_1\mathbf{v}_1 + \ldots + a_n\mathbf{v}_n \mid a_i \in \mathbb{K}\}$$

is a subspace of $\mathbf{V}$ containing the vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$.

Proof. Take $\mathbf{u}, \mathbf{w} \in \operatorname{span}(\{\mathbf{v}_1, \ldots, \mathbf{v}_n\})$, that is

$$\mathbf{u} = a_1\mathbf{v}_1 + \ldots + a_n\mathbf{v}_n, \qquad \mathbf{w} = b_1\mathbf{v}_1 + \ldots + b_n\mathbf{v}_n,$$

for some $a_i, b_i \in \mathbb{K}$. We check the four conditions for a subspace.

- $\mathbf{u} + \mathbf{w} \in \operatorname{span}(\{\mathbf{v}_1, \ldots, \mathbf{v}_n\})$, because
$$\mathbf{u} + \mathbf{w} = (a_1 + b_1)\mathbf{v}_1 + \ldots + (a_n + b_n)\mathbf{v}_n.$$
- $-\mathbf{u} \in \operatorname{span}(\{\mathbf{v}_1, \ldots, \mathbf{v}_n\})$, because
$$-\mathbf{u} = (-a_1)\mathbf{v}_1 + \ldots + (-a_n)\mathbf{v}_n.$$
- $\mathbf{0} \in \operatorname{span}(\{\mathbf{v}_1, \ldots, \mathbf{v}_n\})$, because $\mathbf{0} = 0 \cdot \mathbf{v}_1 + \ldots + 0 \cdot \mathbf{v}_n$.
- $\lambda \cdot \mathbf{u} \in \operatorname{span}(\{\mathbf{v}_1, \ldots, \mathbf{v}_n\})$ for any $\lambda \in \mathbb{K}$, because
$$\lambda \cdot \mathbf{u} = (\lambda a_1)\mathbf{v}_1 + \ldots + (\lambda a_n)\mathbf{v}_n.$$

□

Fact 3.18 (Properties of the span). Let $\mathbf{V}$ be a vector space and let $S, T \subseteq \mathbf{V}$. Then:

- $S \subseteq \operatorname{span}(S)$;
- if $S \subseteq T$, then $\operatorname{span}(S) \subseteq \operatorname{span}(T)$;
- $\operatorname{span}(\operatorname{span}(S)) = \operatorname{span}(S)$;
- $\operatorname{span}(S)$ is the *smallest* subspace of $\mathbf{V}$ containing S: if W is a subspace of $\mathbf{V}$ and $S \subseteq W$, then $\operatorname{span}(S) \subseteq W$.

Example 3.19.

- $\mathbb{C}$ over $\mathbb{R}$:
$$\operatorname{span}(\{1, i\}) = \{a + bi \mid a, b \in \mathbb{R}\} = \mathbb{C}, \qquad \operatorname{span}(\{i, 2 + i\}) = \mathbb{C},$$
$$\operatorname{span}(\{3 + i\}) = \{3a + ai \mid a \in \mathbb{R}\} \neq \mathbb{C}.$$
- $\mathbb{R}[x]$ over $\mathbb{R}$:
$$\operatorname{span}\big(\{x^2, x, 1\}\big) = \operatorname{span}\big(\{x^2, x, 1, x^2 + 5\}\big) = \{ax^2 + bx + c \mid a, b, c \in \mathbb{R}\},$$
the space of quadratic functions.

Example 3.20 (Testing membership in a span). Is $(1, 2, 3) \in \operatorname{span}(\{(1, 1, 1), (1, 0, 1)\})$ in $\mathbb{R}^3$? We look for $a, b \in \mathbb{R}$ with
$$a(1, 1, 1) + b(1, 0, 1) = (1, 2, 3), \qquad \text{i.e.} \qquad a + b = 1, \quad a = 2, \quad a + b = 3.$$
The first and third equations contradict each other, so no such a, b exist: $(1, 2, 3) \notin \operatorname{span}(\{(1, 1, 1), (1, 0, 1)\})$. On the other hand, for the vector $(3, 1, 3)$ the same equations give $a = 1$, $b = 2$, consistently, hence
$$(3, 1, 3) = 1 \cdot (1, 1, 1) + 2 \cdot (1, 0, 1) \in \operatorname{span}(\{(1, 1, 1), (1, 0, 1)\}).$$
Deciding membership in a span always amounts to solving a system of linear equations; the systematic theory of such systems is developed in Chapter 5.

3.5 Linear independence and basis

Definition 3.21. Let $\mathbf{V}$ be a vector space over a field $\mathbb{K}$. A set $S \subseteq \mathbf{V}$ of vectors is *linearly independent* if for any $\mathbf{v}_1, \ldots, \mathbf{v}_n \in S$ and any $a_1, \ldots, a_n \in \mathbb{K}$
$$a_1\mathbf{v}_1 + \ldots + a_n\mathbf{v}_n = \mathbf{0} \implies a_1 = \ldots = a_n = 0.$$
Otherwise, the set S is *linearly dependent*.

Example 3.22. In $\mathbb{R}^2$ over $\mathbb{R}$ the set $\{(0, 1), (1, 1)\}$ is linearly independent, because the linear combination $a(0, 1) + b(1, 1) = (b, a + b)$ equals the zero vector $(0, 0)$ only if $b = 0$ and $a + b = 0$, that is $a = b = 0$.

Example 3.23.
- $\mathbb{C}$ over $\mathbb{R}$: $\{1, i\}$ is linearly independent, since $a \cdot 1 + b \cdot i = 0 \implies a = b = 0$.
- $\mathbb{R}[x]$ over $\mathbb{R}$: $\{x + 1, x - 1, 2\}$ is linearly dependent, because
$$(x + 1) + (-1)(x - 1) + (-1) \cdot 2 = 0.$$
- $\mathbb{R}[x]$ over $\mathbb{R}$: the set $\{1, x, x^2, x^3, \ldots\}$ is linearly independent.

Example 3.24 (Testing linear independence). Are the vectors $(1, 2, 3)$, $(2, 1, 0)$, $(1, -4, -9)$ linearly independent in $\mathbb{R}^3$? We solve
$$a(1, 2, 3) + b(2, 1, 0) + c(1, -4, -9) = (0, 0, 0), \qquad \text{i.e.} \qquad \begin{aligned} a + 2b + c &= 0, \\ 2a + b - 4c &= 0, \\ 3a - 9c &= 0. \end{aligned}$$
The third equation gives $a = 3c$; substituting into the first gives $b = -2c$, and the second equation is then satisfied automatically. Taking $c = 1$ produces a non-trivial combination
$$3 \cdot (1, 2, 3) - 2 \cdot (2, 1, 0) + 1 \cdot (1, -4, -9) = (0, 0, 0),$$
so the vectors are linearly *dependent*.

Fact 3.25. Let $\mathbf{V}$ be a vector space over $\mathbb{K}$. The set $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\} \subseteq \mathbf{V}$ is linearly dependent if and only if at least one of the vectors $\mathbf{v}_i$ can be expressed as a linear combination of the others.

Fact 3.26. Let $\mathbf{V}$ be a vector space over $\mathbb{K}$. If the set $S \subseteq \mathbf{V}$ is linearly independent, then any subset $S' \subseteq S$ is also linearly independent. Equivalently, any set containing a linearly dependent subset is itself linearly dependent. In particular, any set containing the zero vector is linearly dependent, since $1 \cdot \mathbf{0} = \mathbf{0}$ is a non-trivial linear combination equal to $\mathbf{0}$.

Definition 3.27. Let $\mathbf{V}$ be a vector space over a field $\mathbb{K}$. A set $B \subseteq \mathbf{V}$ of vectors is called a *basis* of $\mathbf{V}$ if

- B is linearly independent, and
- $\operatorname{span}(B) = \mathbf{V}$.

Example 3.28.

- For the vector space $\mathbb{R}^2$ over $\mathbb{R}$ the set $\{(1,0),(1,1)\}$ is a basis.
- $\mathbb{C}$ over $\mathbb{R}$: the set $\{1, i\}$ is a basis of $\mathbb{C}$; so is, for instance, $\{1, i+2\}$.
- For the vector space $\mathbb{K}^n$ over $\mathbb{K}$ the set

$$\{(1,0,\ldots,0),(0,1,\ldots,0),\ldots,(0,0,\ldots,1)\}$$

  is a basis, called the *standard basis* of $\mathbb{K}^n$; its elements are denoted by $e_1, e_2, \ldots, e_n$.
- $\mathbb{K}_n[x]$ over $\mathbb{K}$: the set $\{1, x, x^2, x^3, \ldots, x^n\}$ is a basis of $\mathbb{K}_n[x]$.

Theorem 3.29. *Let $\mathbf{V}$ be a vector space over $\mathbb{K}$. A set $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\} \subseteq \mathbf{V}$ is a basis of $\mathbf{V}$ if and only if every vector $\mathbf{w} \in \mathbf{V}$ can be expressed* uniquely *as a linear combination of $\mathbf{v}_1, \ldots, \mathbf{v}_n$.*

Proof. ($\Rightarrow$) Suppose $\mathbf{w} = a_1\mathbf{v}_1 + \ldots + a_n\mathbf{v}_n$ and $\mathbf{w} = b_1\mathbf{v}_1 + \ldots + b_n\mathbf{v}_n$. Then

$$(a_1 - b_1)\mathbf{v}_1 + \ldots + (a_n - b_n)\mathbf{v}_n = \mathbf{0}.$$

Since $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is linearly independent, $a_1 - b_1 = 0, \ldots, a_n - b_n = 0$, that is $a_1 = b_1, \ldots, a_n = b_n$.

($\Leftarrow$) Since every $\mathbf{w} \in \mathbf{V}$ is a linear combination of $\mathbf{v}_1, \ldots, \mathbf{v}_n$, we have $\operatorname{span}(\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}) = \mathbf{V}$. It remains to show linear independence. The zero vector has the representation $\mathbf{0} = 0 \cdot \mathbf{v}_1 + \ldots + 0 \cdot \mathbf{v}_n$. Since the representation of every vector is unique,

$$a_1\mathbf{v}_1 + \ldots + a_n\mathbf{v}_n = \mathbf{0} \implies a_1 = a_2 = \ldots = a_n = 0. \qquad \square$$

Definition 3.30. Let $B = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ be a basis of $\mathbf{V}$ (with a fixed order of its elements) and let $\mathbf{w} \in \mathbf{V}$. The unique scalars $a_1, \ldots, a_n \in \mathbb{K}$ with $\mathbf{w} = a_1\mathbf{v}_1 + \ldots + a_n\mathbf{v}_n$ are called the *coordinates* of $\mathbf{w}$ in the basis B.

Example 3.31. In $\mathbb{R}^2$, take the basis $B = \{(1,0),(1,1)\}$ and $\mathbf{w} = (2,3)$. Solving $(2,3) = a(1,0) + b(1,1)$ gives $b = 3$ and $a = -1$, so the coordinates of $(2,3)$ in B are $(-1,3)$. In the standard basis $\{(1,0),(0,1)\}$ the coordinates of $(2,3)$ are simply $(2,3)$.

Example 3.32. In $\mathbb{R}_2[x]$ with the basis $B = \{1, x, x^2\}$, the coordinates of the polynomial $f(x) = 3 - x + 2x^2$ are $(3,-1,2)$. This is the real power of coordinates: once a basis is fixed, *any* vector — a polynomial, a matrix, a function — is faithfully encoded by a tuple of scalars, and computing with vectors reduces to computing with tuples in $\mathbb{K}^n$. This bridge between the abstract spaces of this chapter and concrete columns of numbers is what the matrix machinery of the following chapters runs on.

3.6 Dimension

Theorem 3.33. *Let a vector space $\mathbf{V}$ have a finite basis. Then any two bases of $\mathbf{V}$ have the same size.*

Definition 3.34. If a vector space $\mathbf{V}$ has a finite basis, then the *dimension* of $\mathbf{V}$ is the size of any basis of $\mathbf{V}$. Otherwise the dimension is defined to be ∞. We denote this number by $\dim(\mathbf{V})$.

Example 3.35.

- $\dim(\{\mathbf{0}\}) = 0$;
- $\dim(\mathbb{K}^n) = n$;
- $\dim(\mathbb{K}[x]) = \infty$;
- $\dim(\mathbb{K}_n[x]) = n + 1$;
- $\dim(\mathbb{C} \text{ over } \mathbb{R}) = 2$;
- $\dim(\mathbb{C} \text{ over } \mathbb{C}) = 1$.

Theorem 3.36. *If* $\mathbf{W}$ *is a subspace of a vector space* $\mathbf{V}$*, then*

- $\dim(\mathbf{W}) \leq \dim(\mathbf{V})$*;*
- *if* $\mathbf{V}$ *is finite dimensional, then* $\dim(\mathbf{W}) = \dim(\mathbf{V})$ *implies* $\mathbf{W} = \mathbf{V}$.

Theorem 3.37. *Let* $\mathbf{V}$ *be a vector space with* $\dim(\mathbf{V}) = n$ *and let* $S \subseteq \mathbf{V}$*. Then:*

- *if* $\operatorname{span}(S) = \mathbf{V}$*, then* $|S| \geq n$*;*
- *if* S *is linearly independent and* $|S| = n$*, then* S *is a basis;*
- *if* $\operatorname{span}(S) = \mathbf{V}$ *and* $|S| = n$*, then* S *is a basis;*
- *if* $|S| > n$*, then* S *is linearly dependent.*

Theorem 3.38. *Let* $\mathbf{V}$ *be a vector space with* $\dim(\mathbf{V}) = n < \infty$*. Then:*

- *every linearly independent set* $S \subseteq \mathbf{V}$ *can be extended to a basis of* $\mathbf{V}$*;*
- *every set* $S \subseteq \mathbf{V}$ *with* $\operatorname{span}(S) = \mathbf{V}$ *contains a basis of* $\mathbf{V}$.

Example 3.39. The set $\{(1, 1, 0)\}$ is linearly independent in $\mathbb{R}^3$; it can be extended to a basis, e.g. $\{(1, 1, 0), (0, 1, 0), (0, 0, 1)\}$: a quick check shows that these three vectors are linearly independent, and by the previous theorem three linearly independent vectors in a 3-dimensional space form a basis.

4 Matrices

4.1 Motivation

If vectors are the standard container for data, *matrices* are the standard container for everything that is two-dimensional — or that *acts* on data. This makes them, with little exaggeration, the central data structure of computing:

- a greyscale **image** is a matrix of pixel intensities, and image filters (blur, sharpen, edge detection) are matrix operations;
- a **graph** — a road network, a social network, the web — is stored as its *adjacency matrix*, whose ij-entry records the connection between node i and node j; counting walks, routing flows and ranking nodes all become matrix computations (Chapter 7 makes this concrete with Google's PageRank);
- a layer of a **neural network** is a matrix of weights: an input vector is multiplied by the matrix, a simple non-linear function is applied, and the result is passed to the next layer. "Training" a network means nothing else than adjusting the entries of its matrices;
- a **spreadsheet**, a database table, the term–document table inside a search engine — matrices again.

The strangest-looking definition of this chapter — the row-by-column rule for *multiplying* matrices — is also its most consequential one. It is designed precisely so that multiplication of matrices corresponds to *composition* of transformations (we prove this in Chapter 6), and its central role in graphics and machine learning explains a striking fact about modern hardware: GPUs, the chips that power both video games and the training of large AI models, are at heart machines optimised for one task — multiplying enormous matrices fast. Even the theory is still alive: the true asymptotic complexity of matrix multiplication (naively $O(n^3)$, improved to about $O(n^{2.81})$ by Strassen's recursive algorithm) remains a famous open problem of theoretical computer science.

Towards the end of the chapter we develop *elementary operations*, *echelon forms* and the notion of *rank*. Rank measures the genuine information content of a matrix: a 1000×1000 matrix of rank 20 can be reconstructed from roughly 40 000 numbers instead of a million. Low rank is the compressible structure hiding in most real-world data — and the algorithm that computes it, row reduction, is the same engine that will solve systems of equations in Chapter 5 and invert matrices later in this chapter.

The chapter closes with the *determinant*, which compresses an entire $n \times n$ matrix into a single scalar that answers a remarkable number of yes/no questions at once. Is the matrix invertible? Does the system $AX = B$ have exactly one solution? Do these n vectors form a basis of $\mathbb{K}^n$? Does the transformation collapse space onto something flatter? All of these questions are equivalent, and all reduce to checking $\det(A) \neq 0$ — which is why a determinant test appears in virtually every algorithm that must know whether a matrix is "degenerate".

For a computer scientist the determinant has a second life as a *geometric* quantity:

- In 2D, $|\det(A)|$ is the **area** of the parallelogram spanned by the rows of A; in 3D it is a volume. A transformation with determinant 6 magnifies every area six-fold; determinant 0 means space gets flattened.
- The **sign** of the determinant detects *orientation*. The test "does point C lie to the left or to the right of the line through A and B?" is the sign of a 2×2 determinant; it is the basic primitive of computational geometry, executed millions of times inside convex-hull algorithms, triangulations, collision detection and map matching. In 3D rendering the same sign test (*backface culling*) lets the GPU skip the triangles that face away from the camera.
- In graphics and machine learning, determinants of *Jacobian* matrices track how transformations distort volume — the bookkeeping device behind texture mapping and behind generative "normalizing flow" models.

Part of the motivation is also a computational warning. The Laplace expansion defined below is mathematically fundamental but computationally explosive: applied naively it performs on the order of $n!$ operations. The behaviour of determinants under row operations, established in this chapter, brings the cost down to that of Gaussian elimination, $O(n^3)$ — one of the nicest examples in this course of theory rescuing an algorithm. The same warning applies to the elegant closed formulas (the adjugate formula for A^{-1} in this chapter, and Cramer's rule in the next): they are invaluable for insight and perfectly practical for small matrices, but for large ones the method of choice is, once again, row reduction.

4.2 Matrices

Definition 4.1. A *matrix of size $m \times n$ over a field* $\mathbb{K}$ (or simply a *matrix*) is a rectangular table of elements $a_{ij} \in \mathbb{K}$ for $i = 1, \dots, m$, $j = 1, \dots, n$:

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}.$$

We denote this matrix by $(a_{ij})_{m\times n}$ or simply (a_{ij}). The element a_{ij} is called the ij*-entry*. Matrices are usually denoted by capital letters $A, B, \dots$. Two matrices A, B are *equal* (written $A = B$) if they have the same size and the same ij-entries. The set of matrices of size $m \times n$ over $\mathbb{K}$ is denoted by $\mathbb{K}^n_m$.

Definition 4.2. Given a matrix $(a_{ij})_{m\times n}$, the n-tuples

$$(a_{11}, a_{12}, \dots, a_{1n}),\ \dots,\ (a_{m1}, a_{m2}, \dots, a_{mn})$$

are its *rows*, and the m-tuples

$$\begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix}, \dots, \begin{pmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{pmatrix}$$

are its *columns*.

Example 4.3. The matrix

$$\begin{pmatrix} 1 & 0 & 2 \\ -3 & \frac{1}{2} & 4 \end{pmatrix}$$

is a 2×3 matrix over $\mathbb{R}$. Its rows are $(1, 0, 2)$ and $(-3, \frac{1}{2}, 4)$, and its columns are

$$\begin{pmatrix} 1 \\ -3 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ \frac{1}{2} \end{pmatrix}, \quad \begin{pmatrix} 2 \\ 4 \end{pmatrix}.$$

Fact 4.4. Let A be an $m \times n$ matrix over a field $\mathbb{K}$. The rows of A are members of the vector space $\mathbb{K}^n$ and the columns are members of $\mathbb{K}^m$.

Definition 4.5. An $m \times n$ matrix is called a *square matrix* if $m = n$. A square matrix is *symmetric* if $a_{ij} = a_{ji}$ for all $i, j = 1, \dots, n$. The n-tuple $(a_{11}, a_{22}, \dots, a_{nn})$ is called the *main diagonal* of a square matrix $(a_{ij})_{n\times n}$.

Example 4.6. The following are square matrices, the second one symmetric:

$$\begin{pmatrix} 1 & 0 \\ -3 & \frac{1}{2} \end{pmatrix}, \qquad \begin{pmatrix} 1 & -3 \\ -3 & 1 \end{pmatrix}.$$

Definition 4.7. A square matrix $A = (a_{ij})_{n\times n}$ is called

- *diagonal* if $a_{ij} = 0$ for all $i \neq j$;
- *upper triangular* if $a_{ij} = 0$ for all $i > j$ (all entries below the main diagonal are zero);

- *lower triangular* if $a_{ij} = 0$ for all $i < j$.

Example 4.8. The matrices

$$\begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 5 & 0 \\ 0 & 2 & -1 \\ 0 & 0 & 3 \end{pmatrix}$$

are diagonal and upper triangular, respectively. Every diagonal matrix is both upper and lower triangular.

4.3 Matrix operations

Definition 4.9 (Addition). Let $A = (a_{ij})$ and $B = (b_{ij})$ be two matrices of the same size $m \times n$. The *sum* of A and B, denoted $A + B$, is the matrix

$$A + B = (a_{ij} + b_{ij})_{m \times n}.$$

Definition 4.10 (Scalar multiplication). The *product* of a matrix $A = (a_{ij})$ by a scalar $k \in \mathbb{K}$, denoted $k \cdot A$, is the matrix $k \cdot A = (k\, a_{ij})$. We write $-A := (-1) \cdot A$ and $A - B := A + (-B)$.

Example 4.11.

$$\begin{pmatrix} 1 & 0 & 2 \\ -3 & \frac{1}{2} & 4 \end{pmatrix} + \begin{pmatrix} -3 & 1 & 4 \\ 2 & \frac{3}{2} & 0 \end{pmatrix} = \begin{pmatrix} -2 & 1 & 6 \\ -1 & 2 & 4 \end{pmatrix}, \qquad 3 \cdot \begin{pmatrix} 1 & 0 & 2 \\ -3 & \frac{1}{2} & 4 \end{pmatrix} = \begin{pmatrix} 3 & 0 & 6 \\ -9 & \frac{3}{2} & 12 \end{pmatrix}.$$

Warning. Addition of matrices of different sizes is *not* defined.

Definition 4.12. The $m \times n$ matrix with all entries equal to zero is called the *zero matrix* and is denoted by $0_{m \times n}$, or simply 0 when the size is clear from the context.

Theorem 4.13. *Let A, B, C be $m \times n$ matrices over a field $\mathbb{K}$ and let $k_1, k_2 \in \mathbb{K}$. Then*

$$\begin{aligned} &A + 0 = 0 + A = A, && k_1(A + B) = k_1 \cdot A + k_1 \cdot B, \\ &(A + B) + C = A + (B + C), && (k_1 + k_2)A = k_1 \cdot A + k_2 \cdot A, \\ &A + B = B + A, && (k_1 k_2)A = k_1(k_2 \cdot A), \\ &A - A = 0, && 1 \cdot A = A. \end{aligned}$$

Definition 4.14 (Matrix multiplication). Let A be a $k \times m$ matrix and B an $m \times n$ matrix. The *product* $A \cdot B$ is the $k \times n$ matrix $C = (c_{ij})$ whose entries are

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \ldots + a_{im}b_{mj}, \qquad i = 1, \ldots, k, \quad j = 1, \ldots, n.$$

Warning. The product $A \cdot B$ is defined *only* when the number of columns of A equals the number of rows of B.

Example 4.15. Let

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 2 & 1 & 3 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Here A is 2×3 and B is 3×2, so both products are defined: $A \cdot B$ is 2×2 and $B \cdot A$ is 3×3. We compute

$$A \cdot B = \begin{pmatrix} 1 \cdot 1 + 0 \cdot 0 + 2 \cdot 1 & 1 \cdot 2 + 0 \cdot 1 + 2 \cdot 0 \\ 2 \cdot 1 + 1 \cdot 0 + 3 \cdot 1 & 2 \cdot 2 + 1 \cdot 1 + 3 \cdot 0 \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 5 & 5 \end{pmatrix}, \qquad B \cdot A = \begin{pmatrix} 5 & 2 & 8 \\ 2 & 1 & 3 \\ 1 & 0 & 2 \end{pmatrix}.$$

In particular, $A \cdot B$ and $B \cdot A$ need not even have the same size.

Remark 4.16 (Matrix–vector products). An important special case is when the second factor is a single column: the product of an $m \times n$ matrix A with a column vector $\mathbf{v} \in \mathbb{K}^n$ is a column vector $A\mathbf{v} \in \mathbb{K}^m$. Writing the

definition out reveals a useful reading:

$$A\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = x_1 \cdot \big(\text{1st column of } A\big) + \ldots + x_n \cdot \big(n\text{-th column of } A\big),$$

i.e. $A\mathbf{v}$ is the *linear combination of the columns of* A with coefficients the entries of $\mathbf{v}$. This observation will be used repeatedly in Chapters 5 and 6.

Remark 4.17. All of matrix algebra works verbatim over *any* field $\mathbb{K}$ — including the finite fields $\mathbb{Z}_p$ — and the same is true of everything that follows in this book: row reduction, rank, systems of linear equations, determinants and inverses. Over $\mathbb{Z}_p$ one simply reduces modulo p at every step.

Example 4.18 (Matrices over a finite field). Over the field $\mathbb{Z}_5$,

$$\begin{pmatrix} 2 & 3 \\ 4 & 1 \end{pmatrix}\begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 2+6 & 8+9 \\ 4+2 & 16+3 \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 1 & 4 \end{pmatrix},$$

since $8 \equiv 3$, $17 \equiv 2$, $6 \equiv 1$ and $19 \equiv 4$ modulo 5.

Definition 4.19. The square matrix of size $n \times n$ of the form

$$\begin{pmatrix} 1 & 0 & \ldots & 0 \\ 0 & 1 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & 1 \end{pmatrix}$$

is called the *identity matrix* and is denoted by $I_{n\times n}$ or I.

Theorem 4.20 (Properties of multiplication). *Let A, A_1 be $k \times m$ matrices, B, B_1 be $m \times n$ matrices and C be an $n \times o$ matrix. Then*

$$\begin{aligned} &(A \cdot B) \cdot C = A \cdot (B \cdot C), \\ &A \cdot I_{m\times m} = I_{k\times k} \cdot A = A, \\ &(A + A_1) \cdot B = A \cdot B + A_1 \cdot B, \\ &A \cdot (B + B_1) = A \cdot B + A \cdot B_1. \end{aligned}$$

Warning. For $A, B \in \mathbb{K}_n^n$ multiplication is in general *not commutative*: sometimes $A \cdot B \neq B \cdot A$.

Example 4.21. Even for square matrices the order of multiplication matters:

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Warning. A product of two non-zero matrices can be the zero matrix: for $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ we have $A \neq 0$ but $A \cdot A = 0$. Consequently, one cannot cancel in matrix equations: $A \cdot B = A \cdot C$ does *not* imply $B = C$.

Definition 4.22 (Matrix powers). For a square matrix A we set $A^0 = I$ and $A^k = A^{k-1} \cdot A$ for $k \geq 1$. Since matrix multiplication is associative, the usual rules $A^k \cdot A^l = A^{k+l}$ and $(A^k)^l = A^{kl}$ hold.

Example 4.23 (Counting walks in a graph). Let A be the *adjacency matrix* of a graph: $a_{ij} = 1$ if there is an edge from node i to node j, and $a_{ij} = 0$ otherwise. Then the ij-entry of A^k counts the walks of length k from i to j. For $k = 2$ this is immediate from the definition of the product: the entry $\sum_m a_{im}\, a_{mj}$ gains 1 exactly for those intermediate nodes m with a step $i \to m$ and a step $m \to j$; the general case follows by induction. This is the walk-counting claim of the motivation section made precise.

Definition 4.24 (Transpose). Let $A = (a_{ij})$ be an $m \times n$ matrix. The *transpose* of A, denoted A^T, is the $n \times m$

matrix obtained by interchanging rows and columns:

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}^T = \begin{pmatrix} a_{11} & a_{21} & \dots & a_{m1} \\ a_{12} & a_{22} & \dots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \dots & a_{mn} \end{pmatrix}.$$

Example 4.25.

$$\begin{pmatrix} 1 & 0 & 2 \\ -3 & \frac{1}{2} & 4 \end{pmatrix}^T = \begin{pmatrix} 1 & -3 \\ 0 & \frac{1}{2} \\ 2 & 4 \end{pmatrix}.$$

Fact 4.26. A square matrix A is symmetric if and only if $A^T = A$.

Theorem 4.27 (Properties of the transpose). *Let A, B be matrices of appropriate sizes and k a scalar. Then*

$$(A+B)^T = A^T + B^T, \qquad (A^T)^T = A, \qquad (k \cdot A)^T = k \cdot A^T, \qquad (A \cdot B)^T = B^T \cdot A^T.$$

4.4 Elementary operations

Definition 4.28. By *elementary row operations* on a matrix A we mean the following:

1. **(Row switching)** the i-th and j-th rows are interchanged ($R_i \leftrightarrow R_j$);
2. **(Row scaling)** each element of the i-th row is multiplied by a *nonzero* scalar $k \in \mathbb{K}$ ($kR_i \to R_i$);
3. **(Row addition)** the i-th row is replaced by the sum of the i-th row and a multiple of the j-th row ($R_i + k \cdot R_j \to R_i$).

Elementary column operations are defined analogously.

Example 4.29.

$$\begin{pmatrix} -3 & 1 & 4 & 2 \\ 2 & \frac{3}{2} & 0 & 3 \\ -3 & 1 & 2 & 1 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 2 & \frac{3}{2} & 0 & 3 \\ -3 & 1 & 4 & 2 \\ -3 & 1 & 2 & 1 \end{pmatrix} \xrightarrow{3R_2 \to R_2} \begin{pmatrix} 2 & \frac{3}{2} & 0 & 3 \\ -9 & 3 & 12 & 6 \\ -3 & 1 & 2 & 1 \end{pmatrix} \xrightarrow{R_2 + 2R_1 \to R_2} \begin{pmatrix} 2 & \frac{3}{2} & 0 & 3 \\ -5 & 6 & 12 & 12 \\ -3 & 1 & 2 & 1 \end{pmatrix}.$$

Theorem 4.30. *Each elementary row (column) operation is invertible, and its inverse is again an elementary row (resp. column) operation.*

Proof.
1. The inverse of $R_i \leftrightarrow R_j$ is $R_j \leftrightarrow R_i$.
2. The inverse of $k \cdot R_i \to R_i$ (with $k \neq 0$) is $\frac{1}{k} \cdot R_i \to R_i$.
3. The inverse of $R_i + k \cdot R_j \to R_i$ is $R_i - k \cdot R_j \to R_i$.

Replacing the letter R by C gives the proof for elementary column operations. □

Definition 4.31. Two matrices A and B of the same size are *row equivalent* if one can be obtained from the other using elementary row operations.

4.5 Row echelon form and rank

Definition 4.32. A matrix A is in *row echelon form* if

1. all zero rows are at the bottom, and
2. the first nonzero entry in the i-th row — called the *leading entry* (or *pivot*) of that row — lies to the right of the leading entry of the row above it.

Example 4.33. The matrix

$$\begin{pmatrix} 2 & 1 & 0 & 3 \\ 0 & 6 & 12 & 12 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

is in row echelon form.

Fact 4.34. Any matrix A is row equivalent to a matrix in row echelon form.

Example 4.35.

$$\begin{pmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{pmatrix} \xrightarrow{R_2-2R_1\to R_2} \begin{pmatrix} 1 & 0 & 2 \\ 0 & -1 & -1 \\ 4 & 1 & 8 \end{pmatrix} \xrightarrow{R_3-4R_1\to R_3} \begin{pmatrix} 1 & 0 & 2 \\ 0 & -1 & -1 \\ 0 & 1 & 0 \end{pmatrix} \xrightarrow{R_3+R_2\to R_3} \begin{pmatrix} 1 & 0 & 2 \\ 0 & -1 & -1 \\ 0 & 0 & -1 \end{pmatrix}.$$

Definition 4.36. Let A be an $m \times n$ matrix over $\mathbb{K}$. The *row rank* of A is the dimension of the subspace of $\mathbb{K}^n$ spanned by the rows of A. The *column rank* of A is the dimension of the subspace of $\mathbb{K}^m$ spanned by the columns of A.

Theorem 4.37. *For every matrix A the row rank of A equals the column rank of A.*

Definition 4.38. The *rank* of a matrix A, denoted $\mathrm{r}(A)$, is its row rank (equivalently, its column rank).

Fact 4.39. For any $m \times n$ matrix A we have $\mathrm{r}(A) = \mathrm{r}(A^T)$ and $\mathrm{r}(A) \leq \min(m, n)$: the rank is bounded both by the number of rows and by the number of columns.

Example 4.40. Consider $A = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 4 \end{smallmatrix}\right)$. Then

$$\dim\big(\operatorname{span}(\{(1,2),(2,4)\})\big) = \dim\{a(1,2)+b(2,4) \mid a,b \in \mathbb{R}\} = \dim\{(a+2b)(1,2) \mid a,b \in \mathbb{R}\} = \dim\big(\operatorname{span}(\{(1,2)\})\big) = 1.$$

Hence $\mathrm{r}(A) = 1$.

Theorem 4.41. *If a matrix A is row equivalent to a matrix B, then $\mathrm{r}(A) = \mathrm{r}(B)$.*

Theorem 4.42. *Let A be a matrix. The number of nonzero rows in a row echelon form which is row equivalent to A equals $\mathrm{r}(A)$.*

Example 4.43. We compute the rank of a 3×4 matrix by row reduction:

$$A = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 3 & 1 \\ 3 & 6 & 4 & 1 \end{pmatrix} \xrightarrow[R_3-3R_1\to R_3]{R_2-2R_1\to R_2} \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_3-R_2\to R_3} \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The row echelon form has two nonzero rows, so $\mathrm{r}(A) = 2$.

Theorem 4.44. *The dimension of the subspace W of $\mathbb{K}^n$ spanned by vectors $\mathbf{v}_1, \ldots, \mathbf{v}_m$ equals the rank of the matrix*

$$\begin{pmatrix} \mathbf{v}_1 \\ \vdots \\ \mathbf{v}_m \end{pmatrix}.$$

Example 4.45. What is the dimension of $W = \operatorname{span}(\{(1,2,1),(2,4,3),(3,6,4)\}) \subseteq \mathbb{R}^3$? Writing the vectors as the rows of a matrix and row reducing,

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 6 & 4 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix},$$

we get $\dim(W) = 2$. Moreover, the nonzero rows $(1,2,1)$ and $(0,0,1)$ of the echelon form constitute a basis of W.

4.6 Determinants

To every square matrix $A \in \mathbb{K}_n^n$ over a field $\mathbb{K}$ one assigns a scalar from $\mathbb{K}$, denoted $|A|$ or $\det(A)$ and called *the determinant of A*. For matrices of sizes 1×1, 2×2 and 3×3 it is given by the explicit formulas

$$|a_{11}| = a_{11},$$

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21},$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{12}a_{21}a_{33} - a_{11}a_{23}a_{32}.$$

Remark 4.46. The 3×3 formula is known as the *rule of Sarrus*: copy the first two columns to the right of the matrix; the three "descending" diagonal products are added and the three "ascending" ones subtracted. Beware: the rule has *no* analogue for $n \geq 4$ — larger determinants are computed by the Laplace expansion defined below or, more efficiently, by row reduction.

Example 4.47.

$$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = 1 \cdot 4 - 2 \cdot 3 = -2,$$

and, applying the 3×3 formula term by term,

$$\begin{vmatrix} 2 & 1 & 3 \\ 0 & 1 & -1 \\ 1 & 2 & 1 \end{vmatrix} = 2 \cdot 1 \cdot 1 + 1 \cdot (-1) \cdot 1 + 3 \cdot 0 \cdot 2 - 3 \cdot 1 \cdot 1 - 1 \cdot 0 \cdot 1 - 2 \cdot (-1) \cdot 2 = 2 - 1 + 0 - 3 - 0 + 4 = 2.$$

Definition 4.48 (Laplace expansion). Let A be an $n \times n$ matrix and let A_{ij} denote the matrix obtained from A by deleting the i-th row and the j-th column. For any fixed column index j between 1 and n,

$$\det(A) = \sum_{i=1}^{n} (-1)^{i+j} a_{ij} \cdot \det(A_{ij}).$$

This formula is the *Laplace expansion along the j-th column*. In particular, for $j = 1$,

$$\det(A) = \sum_{i=1}^{n} (-1)^{i+1} a_{i1} \cdot \det(A_{i1}).$$

Remark 4.49. It is a non-trivial fact, which we accept without proof, that the result does not depend on the chosen column j — so the recursive definition is consistent, and for sizes 2 and 3 it reproduces the explicit formulas above. Moreover, since $\det(A) = \det(A^T)$ (see the properties below), the determinant may equally well be expanded along any fixed *row*: $\det(A) = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} \cdot \det(A_{ij})$ for any fixed row index i. In practice one expands along the row or column containing the most zeros.

Example 4.50. We compute the determinant by expanding along the first column:

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = 1 \cdot \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} - 4 \cdot \begin{vmatrix} 2 & 3 \\ 8 & 9 \end{vmatrix} + 7 \cdot \begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix} = 0.$$

Fact 4.51 (Properties of determinants). Let A be a square matrix of size $n \times n$. Then

1. $\det(A) = \det(A^T)$;
2. $\det(A) = 0$ if A has a zero row (or column) or two identical rows (or columns);
3. $\det(A) = 0$ if and only if $\mathrm{r}(A) < n$;
4. $\det(A) = -\det(B)$ if B is obtained from A by a single row switch ($R_i \leftrightarrow R_j$);
5. $\det(A) = \det(B)$ if B is obtained from A by a single row addition ($R_i + k \cdot R_j \to R_i$);

6. $\det(B) = k \cdot \det(A)$ if B is obtained from A by the row scaling $kR_i \to R_i$.

The last three properties also hold for the corresponding elementary column operations.

Example 4.52. Using row operations to reach a triangular form:

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} \overset{R_2-4R_1}{=} \begin{vmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 7 & 8 & 9 \end{vmatrix} \overset{R_3-7R_1}{=} \begin{vmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{vmatrix} = 2 \cdot \begin{vmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -3 & -6 \end{vmatrix} = 2 \cdot 0 = 0.$$

Fact 4.53. The determinant of an upper (or lower) triangular matrix equals the product of the entries on its main diagonal. In particular this applies to diagonal matrices, and $\det(I_{n\times n}) = 1$.

Example 4.54.

$$\begin{vmatrix} 2 & 7 & -1 \\ 0 & 3 & 5 \\ 0 & 0 & -4 \end{vmatrix} = 2 \cdot 3 \cdot (-4) = -24.$$

Combined with row reduction, this fact gives an efficient general method of computing determinants: reduce the matrix to a triangular form, keeping track of how each operation changes the determinant, exactly as in the example above.

Example 4.55 (A 4×4 determinant). Row additions do not change the determinant, so

$$\begin{vmatrix} 1 & 2 & 0 & 1 \\ 2 & 5 & 2 & 3 \\ 1 & 2 & 1 & 2 \\ 3 & 6 & 0 & 4 \end{vmatrix} \overset{\substack{R_2-2R_1 \\ R_3-R_1 \\ R_4-3R_1}}{=} \begin{vmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{vmatrix} = 1 \cdot 1 \cdot 1 \cdot 1 = 1.$$

For larger matrices this approach is dramatically faster than a direct Laplace expansion.

Fact 4.56 (Multiplicativity). Let A and B be square matrices of size $n \times n$. Then $\det(A \cdot B) = \det(A) \cdot \det(B)$.

Warning. The determinant is *not* additive: in general $\det(A+B) \neq \det(A) + \det(B)$. Note also that $\det(k \cdot A) = k^n \det(A)$ for an $n \times n$ matrix A: scaling the whole matrix scales *each* of its n rows.

Remark 4.57 (Geometric meaning). For a real 2×2 matrix A, the absolute value $|\det(A)|$ is the area of the parallelogram spanned by the two rows (equivalently, the two columns) of A; the sign of the determinant records the orientation of this pair of vectors. Similarly, for a real 3×3 matrix, $|\det(A)|$ is the volume of the parallelepiped spanned by the rows.

4.7 Matrix inversion

Definition 4.58. Let A be a square matrix of size $n \times n$. A matrix B of the same size is called an *inverse* of A if

$$A \cdot B = I_{n\times n}, \qquad B \cdot A = I_{n\times n}.$$

If A has an inverse, it is unique; it is denoted by A^{-1}.

Example 4.59. For $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 2 \end{smallmatrix}\right)$ one checks that $A^{-1} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{smallmatrix}\right)$.

Warning. Not every matrix is invertible.

Theorem 4.60. *A square matrix A is invertible if and only if $\det(A) \neq 0$. If A is invertible, then*

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

Proof. We prove the second statement. Since $\det(I) = 1$ and $\det(A \cdot B) = \det(A) \cdot \det(B)$, and since $A \cdot A^{-1} = I$

for invertible A, we get

$$1 = \det(I) = \det(A \cdot A^{-1}) = \det(A) \cdot \det(A^{-1}).$$ □

Theorem 4.61 (Properties of the inverse). *Let A, B be invertible matrices of size $n \times n$ and let $k \neq 0$ be a scalar. Then A^{-1}, $A \cdot B$, A^T and $k \cdot A$ are all invertible, and*

$$(A^{-1})^{-1} = A, \qquad (A \cdot B)^{-1} = B^{-1} \cdot A^{-1}, \qquad (A^T)^{-1} = (A^{-1})^T, \qquad (k \cdot A)^{-1} = \tfrac{1}{k} \cdot A^{-1}.$$

Warning. Note the reversed order in $(A \cdot B)^{-1} = B^{-1} \cdot A^{-1}$ — the same reversal as in the transpose formula $(A \cdot B)^T = B^T \cdot A^T$.

Example 4.62 (The general linear group). Recall the notion of a group from Chapter 2. By the theorem above, a product of two invertible $n \times n$ matrices is invertible, and so is the inverse of an invertible matrix; moreover, matrix multiplication is associative and I is invertible. Hence the set of all invertible $n \times n$ matrices over $\mathbb{K}$, with matrix multiplication, is a group, called the *general linear group* $GL_n(\mathbb{K})$. For $n \geq 2$ it is *not* abelian; for instance, in $GL_2(\mathbb{R})$,

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \neq \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

This is our first example of a non-abelian group.

Fact 4.63 (Inversion by row reduction). Let A be a square matrix of size $n \times n$. Form the $n \times 2n$ matrix $C = (A \mid I_{n\times n})$ and row reduce it to the form $(I_{n\times n} \mid B)$. Such a reduction is possible if and only if A is invertible, and the resulting matrix B is the inverse of A.

Example 4.64.

$$\left(\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ 2 & 2 & 0 & 1 \end{array}\right) \xrightarrow{R_2 - 2R_1 \to R_2} \left(\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ 0 & 2 & -2 & 1 \end{array}\right) \xrightarrow{\frac{1}{2}R_2 \to R_2} \left(\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & \frac{1}{2} \end{array}\right),$$

so $\left(\begin{smallmatrix} 1 & 0 \\ 2 & 2 \end{smallmatrix}\right)^{-1} = \left(\begin{smallmatrix} 1 & 0 \\ -1 & \frac{1}{2} \end{smallmatrix}\right)$.

Example 4.65 (A 3×3 inverse). We invert $A = \left(\begin{smallmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{smallmatrix}\right)$:

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_2 - R_1 \to R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_3 - R_2 \to R_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & 2 & 1 & -1 & 1 \end{array}\right)$$

$$\xrightarrow{\frac{1}{2}R_3 \to R_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & 1 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{array}\right) \xrightarrow[R_1 - R_3 \to R_1]{R_2 + R_3 \to R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & 0 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{array}\right),$$

so

$$A^{-1} = \frac{1}{2}\begin{pmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix}.$$

One can (and always should) verify the result by checking that $A \cdot A^{-1} = I$.

Fact 4.66 (Inversion by determinants). Let A be an invertible square matrix and let $A_{k,m}$ denote the matrix obtained from A by deleting its k-th row and m-th column. Then

$$A^{-1} = \frac{1}{|A|} \cdot \begin{pmatrix} (-1)^{1+1}|A_{11}| & (-1)^{1+2}|A_{12}| & \dots & (-1)^{1+n}|A_{1n}| \\ (-1)^{2+1}|A_{21}| & (-1)^{2+2}|A_{22}| & \dots & (-1)^{2+n}|A_{2n}| \\ \vdots & \vdots & \ddots & \vdots \\ (-1)^{n+1}|A_{n1}| & (-1)^{n+2}|A_{n2}| & \dots & (-1)^{n+n}|A_{nn}| \end{pmatrix}^T.$$

Example 4.67.

$$\begin{pmatrix} 1 & 0 \\ 2 & 2 \end{pmatrix}^{-1} = \frac{1}{2}\begin{pmatrix} |(2)| & -|(2)| \\ -|(0)| & |(1)| \end{pmatrix}^{T} = \frac{1}{2}\begin{pmatrix} 2 & 0 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -1 & \frac{1}{2} \end{pmatrix}.$$

Fact 4.68 (Inverse of a 2×2 matrix). Let $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ with $\det(A) = ad - bc \neq 0$. Then

$$A^{-1} = \frac{1}{ad - bc} \cdot \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

5 Systems of linear equations

5.1 Motivation

Solving systems of linear equations is the oldest problem of linear algebra — Chinese mathematicians were row reducing two thousand years ago — and, measured in CPU-hours, it may well be the computation the world performs most. When engineers simulate a bridge, a chip or tomorrow's weather, the physics is chopped into millions of small interacting pieces, and the interactions are linear equations; the supercomputer's job is to solve the resulting system. Concrete examples surround you:

- **GPS.** Your position is computed from your distances to several satellites; after a standard algebraic trick the unknown coordinates satisfy a small linear system, which your phone solves several times per second.
- **Computer graphics and game physics.** Finding where a ray of light meets a surface, or resolving the forces when a stack of crates topples, means setting up and solving linear systems — many thousands of them per frame.
- **Electronics.** Kirchhoff's laws turn any circuit diagram into a linear system relating its currents and voltages; circuit simulators such as SPICE are, at their core, linear-system solvers.
- **Medical imaging.** A CT scanner measures how much X-ray energy survives along thousands of lines through the body; reconstructing the image of the inside is solving a huge linear system for the unknown tissue densities.
- **Machine learning.** Fitting a linear model to data by least squares — the workhorse of statistics — leads to the so-called normal equations: a linear system in the model's parameters.

This chapter presents *Gaussian elimination*, the algorithm behind all of the above. It deserves the same respect as any classic algorithm you know: it is provably correct (elementary row operations preserve the solution set), it is fast (about $\frac{2}{3}n^3$ arithmetic operations for n equations with n unknowns), and it is universal. Just as valuable is the *structure theory* that comes with it: a linear system has either no solution, exactly one, or infinitely many — never, say, exactly two — and the Kronecker–Capelli theorem decides which case occurs by comparing two ranks. In the infinite case the solutions are beautifully organised: one particular solution plus a *subspace* of $\mathbb{K}^n$ — the first place in the course where the abstract theory of Chapter 3 pays off directly.

5.2 Systems of linear equations

Definition 5.1. A *system of m linear equations with n unknowns* $x_1, \dots, x_n$ is a set of equations

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1, \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2, \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m, \end{aligned}$$

with $a_{ij}, b_i \in \mathbb{K}$. A *solution* of this system is a tuple of values $x_1 = k_1, \dots, x_n = k_n$ which satisfies all m equalities.

Example 5.2. The system

$$\begin{aligned} 3x + y + z &= 1, \\ x + y &= 0 \end{aligned}$$

has 2 equations and 3 unknowns. One checks that $x = 0, y = 0, z = 1$ and $x = -1, y = 1, z = 3$ are both solutions (there are others). On the other hand the system

$$\begin{aligned} x + y &= 1, \\ x + y &= 0 \end{aligned}$$

has no solutions, since a solution would imply $0 = 1$.

Observation 5.3 (Matrix representation). A system of m linear equations with unknowns $x_1, \ldots, x_n$ can be written as a single matrix equation

$$A \cdot X = B,$$

where

$$A = \begin{pmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{pmatrix}, \qquad X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \qquad B = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}.$$

Definition 5.4. Two systems $A_1X = B_1$ and $A_2X = B_2$ of m linear equations with n unknowns are *equivalent* if they have the same solution set.

Example 5.5. The system $\{\, x - y = 1,\ x + y = 0 \,\}$ is equivalent to $\{\, 2x - 2y = 2,\ x + y = 0 \,\}$.

Definition 5.6. The matrix

$$(A \mid B) = \left(\begin{array}{cccc|c} a_{11} & a_{12} & \ldots & a_{1n} & b_1 \\ a_{21} & a_{22} & \ldots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} & b_m \end{array}\right)$$

is called the *augmented matrix* of the system $A \cdot X = B$.

Example 5.7. The matrix $\left(\begin{array}{cc|c} 2 & -2 & 2 \\ 1 & 1 & 0 \end{array}\right)$ is the augmented matrix of the system $\{\, 2x - 2y = 2,\ x + y = 0 \,\}$.

Theorem 5.8. *Two systems $A_1X = B_1$ and $A_2X = B_2$ of m linear equations with n unknowns are equivalent if and only if their augmented matrices $(A_1 \mid B_1)$ and $(A_2 \mid B_2)$ are row equivalent.*

Recall that two matrices are row equivalent if one can be obtained from the other by elementary row operations: row switching ($R_i \leftrightarrow R_j$), row scaling by a nonzero scalar ($kR_i \to R_i$) and row addition ($R_i + k \cdot R_j \to R_i$).

5.3 Solving systems

Definition 5.9. A system $AX = B$ of linear equations is in *echelon form* if its augmented matrix $(A \mid B)$ is in row echelon form.

Since every matrix is row equivalent to a matrix in row echelon form, for any system $AX = B$ there exists an equivalent system $A'X = B'$ in echelon form, that is, a system in echelon form with the same solution set. This gives a general method.

Theorem 5.10 (Solving a system). *To solve a system $AX = B$:*

- *write its augmented matrix $(A \mid B)$;*
- *find a matrix $(A' \mid B')$ in row echelon form which is row equivalent to $(A \mid B)$;*
- *write the new system $A'X = B'$ and read off its solutions.*

Remark 5.11. This procedure is known as *Gaussian elimination*. Reading the solutions off an echelon system, from the last equation upwards, is called *back substitution*.

Definition 5.12. Let $A'X = B'$ be a system in echelon form. The unknowns multiplying the leading entries of the rows of A' are called *leading variables*; the remaining unknowns are the *free variables*. To read off the solutions one assigns arbitrary parameter values to the free variables and computes the leading variables by back substitution.

Example 5.13 (A unique solution). Consider the system

$$\begin{aligned} 2x + 4y - z &= 11,\\ -4x - 3y + 3z &= 20,\\ 2x + 4y + 2z &= 2, \end{aligned} \qquad \text{with augmented matrix} \qquad \left(\begin{array}{ccc|c} 2 & 4 & -1 & 11\\ -4 & -3 & 3 & 20\\ 2 & 4 & 2 & 2 \end{array}\right).$$

Using elementary row operations we obtain the row echelon form

$$\left(\begin{array}{ccc|c} 2 & 4 & -1 & 11\\ 0 & 5 & 1 & 2\\ 0 & 0 & 3 & -9 \end{array}\right), \qquad \text{i.e.} \qquad \begin{aligned} 2x + 4y - z &= 11,\\ 5y + z &= 2,\\ 3z &= -9. \end{aligned}$$

The third equation gives $z = -3$; the second then gives $5y - 3 = 2$, so $y = 1$; finally $x = 2$. The system has the unique solution $x = 2,\ y = 1,\ z = -3$.

Example 5.14 (Infinitely many solutions). Consider the system

$$\begin{aligned} x + 4y - 3z + 2t &= 5,\\ 2x + 8y - 5z \qquad &= 12, \end{aligned} \qquad \text{with augmented matrix} \qquad \left(\begin{array}{cccc|c} 1 & 4 & -3 & 2 & 5\\ 2 & 8 & -5 & 0 & 12 \end{array}\right).$$

By row reduction we obtain

$$\left(\begin{array}{cccc|c} 1 & 4 & -3 & 2 & 5\\ 0 & 0 & 1 & -4 & 2 \end{array}\right), \qquad \text{i.e.} \qquad \begin{aligned} x + 4y - 3z + 2t &= 5,\\ z - 4t &= 2. \end{aligned}$$

This echelon system has 2 equations and 4 unknowns, hence more than one solution. We assign arbitrary values to the free variables $y = a$ and $t = b$. Substitution gives $z = 2 + 4b$ and $x = 11 - 4a + 10b$. The solution depends on two parameters:

$$x = 11 - 4a + 10b, \quad y = a, \quad z = 2 + 4b, \quad t = b.$$

Example 5.15 (No solution). Consider the system

$$\begin{aligned} 2x + 4y &= 0,\\ 2x + 4y &= 1, \end{aligned} \qquad \text{with augmented matrix} \qquad \left(\begin{array}{cc|c} 2 & 4 & 0\\ 2 & 4 & 1 \end{array}\right) \longrightarrow \left(\begin{array}{cc|c} 2 & 4 & 0\\ 0 & 0 & 1 \end{array}\right).$$

The last row reads $0 = 1$, so this system has no solutions.

Theorem 5.16 (Kronecker–Capelli). *A system $AX = B$ of linear equations has a solution if and only if* $\mathrm{r}(A) = \mathrm{r}(A \mid B)$.

Corollary 5.17 (Number of solutions). *Let $AX = B$ be a solvable system with n unknowns and let $r = \mathrm{r}(A) = \mathrm{r}(A \mid B)$. If $r = n$, the system has exactly one solution. If $r < n$, the solutions are described by $n - r$ free parameters: over an infinite field such as $\mathbb{R}$ or $\mathbb{C}$ this means infinitely many solutions, and over a finite field $\mathbb{Z}_p$ it means exactly p^{n-r} of them. In particular, a system over $\mathbb{R}$ has no solution, exactly one solution, or infinitely many — never, say, exactly two.*

Example 5.18. For the system from the previous example,

$$(A \mid B) = \left(\begin{array}{cc|c} 2 & 4 & 0\\ 2 & 4 & 1 \end{array}\right) \longrightarrow \left(\begin{array}{cc|c} 2 & 4 & 0\\ 0 & 0 & 1 \end{array}\right),$$

we read off $\mathrm{r}(A) = 1$ and $\mathrm{r}(A \mid B) = 2$. The two ranks differ, which confirms that the system has no solutions.

Example 5.19 (A system with a parameter). For which values of the parameter $p \in \mathbb{R}$ does the system

$$\begin{aligned} x + y + z &= 1, \\ x + 2y + 3z &= 2, \\ 2x + 3y + 4z &= p \end{aligned}$$

have solutions? Row reducing the augmented matrix,

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 2 \\ 2 & 3 & 4 & p \end{array}\right) \xrightarrow[R_3-2R_1\to R_3]{R_2-R_1\to R_2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & 2 & p-2 \end{array}\right) \xrightarrow{R_3-R_2\to R_3} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & p-3 \end{array}\right).$$

If $p \neq 3$, the last row reads $0 = p - 3 \neq 0$ and there are no solutions (equivalently, $\mathrm{r}(A) = 2 < 3 = \mathrm{r}(A \mid B)$). If $p = 3$, then $\mathrm{r}(A) = \mathrm{r}(A \mid B) = 2$ and the system has infinitely many solutions: taking $z = a$ we get $y = 1 - 2a$ and $x = a$.

5.4 Homogeneous systems

Definition 5.20. A system of linear equations is called *homogeneous* if it has the form $A \cdot X = \mathbf{0}$.

A homogeneous system *always* has the solution $x_1 = 0, x_2 = 0, \ldots, x_n = 0$. Any other solution (if one exists) is called a *non-trivial* solution.

Fact 5.21. A homogeneous system with fewer equations than unknowns ($m < n$) always has a non-trivial solution: indeed $\mathrm{r}(A) \leq m < n$, so after row reduction at least one variable is free.

Example 5.22. The homogeneous system

$$\begin{aligned} x + 2y - 3z + w &= 0, \\ x - 3y + z - 2w &= 0, \\ 2x + y - 3z + 5w &= 0 \end{aligned}$$

has 3 equations and 4 unknowns; therefore it has non-trivial solutions.

Example 5.23. The homogeneous system

$$\begin{aligned} x + y - z &= 0, \\ 2x - 3y + z &= 0, \\ x - 4y + 2z &= 0 \end{aligned} \qquad \text{reduces to} \qquad \begin{aligned} x + y - z &= 0, \\ -5y + 3z &= 0. \end{aligned}$$

It has a non-trivial solution: assigning $z = a$ gives $y = \frac{3}{5}a$ and $x = \frac{2}{5}a$.

Example 5.24. The homogeneous system

$$\begin{aligned} x + y - z &= 0, \\ 2x + 4y - z &= 0, \\ 3x + 2y + 2z &= 0 \end{aligned} \qquad \text{reduces to} \qquad \begin{aligned} x + y - z &= 0, \\ 2y + z &= 0, \\ 11z &= 0. \end{aligned}$$

Here the only solution is the trivial one $x = y = z = 0$.

Theorem 5.25 (Solution space). *Let $AX = \mathbf{0}$ be a system of m homogeneous linear equations with n unknowns. Then the set $W = \{\mathbf{v} \in \mathbb{K}^n \mid A\mathbf{v} = \mathbf{0}\}$ of solutions is a subspace of the vector space $\mathbb{K}^n$. Moreover,*

$$\dim(W) = n - \mathrm{r}(A).$$

Proof. We prove that W is a subspace. Take $\mathbf{v}_1, \mathbf{v}_2 \in W$, so $A\mathbf{v}_1 = \mathbf{0}$ and $A\mathbf{v}_2 = \mathbf{0}$. Then

$$A(\mathbf{v}_1 + \mathbf{v}_2) = A\mathbf{v}_1 + A\mathbf{v}_2 = \mathbf{0} + \mathbf{0} = \mathbf{0},$$

hence $\mathbf{v}_1 + \mathbf{v}_2 \in W$. Similarly, for any $k \in \mathbb{K}$ we have $A(k\mathbf{v}_1) = kA\mathbf{v}_1 = \mathbf{0}$, so $k \cdot \mathbf{v}_1 \in W$. □

Remark 5.26. The dimension formula is visible in the algorithm itself: solving $AX = \mathbf{0}$ by Gaussian elimination leaves $\mathrm{r}(A)$ leading variables and $n - \mathrm{r}(A)$ free variables, and setting one free variable to 1 and the others to 0 — as in the next example — produces a basis of W with $n - \mathrm{r}(A)$ elements.

Example 5.27. Consider the homogeneous system

$$\begin{aligned} x + 2y + 3z &= 0, \\ -2x - 4y - 6z &= 0, \end{aligned} \qquad \text{which reduces to} \qquad x + 2y + 3z = 0.$$

The non-leading variables are y and z. Putting $y = a$ and $z = b$, the solutions are

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -2a - 3b \\ a \\ b \end{pmatrix},$$

so the solution space is

$$W = \left\{ \begin{pmatrix} -2a - 3b \\ a \\ b \end{pmatrix} \;\middle|\; a, b \in \mathbb{R} \right\}.$$

Taking $(a, b) = (1, 0)$ and $(a, b) = (0, 1)$ yields two vectors

$$\begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}, \qquad \begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix}$$

which form a basis of W.

5.5 Homogeneous and non-homogeneous systems

Theorem 5.28. *Let $AX = B$ be an arbitrary system of linear equations, let U be its solution set and let $\mathbf{v}_0 \in U$ be a particular solution. Then*

$$U = \mathbf{v}_0 + W = \{\mathbf{v}_0 + \mathbf{w} \mid \mathbf{w} \in W\},$$

where W is the solution space of the associated homogeneous system $AX = \mathbf{0}$.

Proof. Every element of $\mathbf{v}_0 + W$ is a solution of $AX = B$, since

$$A(\mathbf{v}_0 + \mathbf{w}) = A\mathbf{v}_0 + A\mathbf{w} = B + \mathbf{0} = B.$$

Conversely, if $A\mathbf{v} = B$, put $\mathbf{w} = \mathbf{v} - \mathbf{v}_0$. Then $\mathbf{v} = \mathbf{v}_0 + \mathbf{w}$ and

$$A\mathbf{w} = A(\mathbf{v} - \mathbf{v}_0) = A\mathbf{v} - A\mathbf{v}_0 = B - B = \mathbf{0},$$

so $\mathbf{w} \in W$ and $\mathbf{v} \in \mathbf{v}_0 + W$. □

Example 5.29. Consider the equation $x + 2y + 3z = 6$ (a system of one equation with three unknowns). A particular solution is $\mathbf{v}_0 = (1, 1, 1)$, and the associated homogeneous equation $x + 2y + 3z = 0$ has the solution space

$$W = \mathrm{span}(\{(-2, 1, 0),\ (-3, 0, 1)\}),$$

as computed in an earlier example. Hence every solution of $x + 2y + 3z = 6$ has the form

$$(1, 1, 1) + a\,(-2, 1, 0) + b\,(-3, 0, 1), \qquad a, b \in \mathbb{R}.$$

5.6 Determinants and systems of linear equations

Theorem 5.30. *Let $AX = B$ be a system of n linear equations with n unknowns. Then $AX = B$ has a unique solution if and only if* $\det(A) \neq 0$.

Theorem 5.31 (Cramer's rule). *Consider a system $AX = B$ with n equations and n unknowns, where A is a square matrix of size $n \times n$. Let $A_{|i}$ denote the matrix obtained from A by replacing its i-th column with the column B. If* $\det(A) \neq 0$, *then*

$$x_1 = \frac{\det(A_{|1})}{\det(A)}, \qquad x_2 = \frac{\det(A_{|2})}{\det(A)}, \qquad \dots, \qquad x_n = \frac{\det(A_{|n})}{\det(A)}.$$

Example 5.32. Consider the system $\{\, x + y = 1,\ x + 2y = 0 \,\}$. Here

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad \det(A) = 1,$$

$$A_{|1} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix},\ \det(A_{|1}) = 2, \qquad A_{|2} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix},\ \det(A_{|2}) = -1.$$

Hence $x = \frac{2}{1} = 2$ and $y = \frac{-1}{1} = -1$.

Example 5.33 (Cramer's rule for a 3×3 system). Consider the system

$$\begin{aligned} x + y + z &= 2, \\ x - y + z &= 0, \\ x + y - z &= 0. \end{aligned}$$

We compute

$$\det(A) = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix} = 4, \qquad \det(A_{|1}) = \begin{vmatrix} 2 & 1 & 1 \\ 0 & -1 & 1 \\ 0 & 1 & -1 \end{vmatrix} = 0,$$

$$\det(A_{|2}) = \begin{vmatrix} 1 & 2 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & -1 \end{vmatrix} = 4, \qquad \det(A_{|3}) = \begin{vmatrix} 1 & 1 & 2 \\ 1 & -1 & 0 \\ 1 & 1 & 0 \end{vmatrix} = 4.$$

Hence $x = \frac{0}{4} = 0$, $y = \frac{4}{4} = 1$ and $z = \frac{4}{4} = 1$; a direct check confirms that $(0, 1, 1)$ solves the system.

5.7 Summary: invertibility criteria

The chapters so far have produced a remarkable list of ways of saying the same thing. It deserves to be collected in one place.

Theorem 5.34 (The invertible matrix theorem). *For a square matrix A of size $n \times n$ over a field $\mathbb{K}$ the following conditions are equivalent:*

1. *A is invertible;*
2. *$\det(A) \neq 0$;*
3. *$\mathrm{r}(A) = n$;*
4. *the rows of A (equivalently, the columns of A) are linearly independent, i.e. they form a basis of $\mathbb{K}^n$;*
5. *the homogeneous system $AX = \mathbf{0}$ has only the trivial solution;*
6. *for every $B \in \mathbb{K}^n$ the system $AX = B$ has exactly one solution, namely $X = A^{-1}B$.*

Remark 5.35. Each chapter has added a line to this list, and Chapter 7 will add one more: 0 *is not an eigenvalue of A*. Checking any one of these conditions checks them all — this versatility is what makes the determinant test so ubiquitous in algorithms.

6 Linear mappings

6.1 Motivation

Having studied vector spaces — the objects of linear algebra — we now study the *functions* between them. Not all functions, though: only the *linear* ones, those that respect addition and scaling. This restriction looks severe, and that is exactly its charm. A linear mapping $\mathbb{R}^n \to \mathbb{R}^m$ is completely determined by its values on the n basis vectors, so it can be stored as a matrix and evaluated by matrix multiplication. Compare: an arbitrary function on a million-dimensional space is unknowable, while a linear one is a finite table of numbers. Linearity is the price we pay to make functions computable — and an astonishing amount of computing fits within that budget:

- **3D graphics.** Every frame, a game engine pushes millions of vertices through the same pipeline: rotate the model, place it in the world, position the camera, project onto the screen. Each stage is a linear mapping, and the whole pipeline is their composition — which, by a central theorem of this chapter, is again a single matrix, precomputed once per frame and then applied cheaply to every vertex. This is why the graphics industry runs on matrix multiplication.
- **Neural networks.** Each layer of a neural network is a linear mapping followed by a fixed, very simple non-linear function. All of the network's learned knowledge is stored in the matrices of its linear parts.
- **Signals, robots, cameras.** Audio filters, the coordinate changes between a robot's joints and the room it stands in, and the projection performed by a camera are all linear mappings.

The second half of the chapter introduces the *kernel* of a mapping (everything that is sent to zero) and its *image* (everything it can produce). These measure the two ways a transformation loses information. The kernel is what becomes indistinguishable from nothing: projecting a 3D scene onto a 2D screen has the viewing direction in its kernel — which is precisely why a single photograph cannot tell you how far away things are. The image tells us which outputs are achievable at all. The chapter culminates in the *rank–nullity theorem*, a conservation law for dimensions: what survives (the image) plus what is lost (the kernel) always adds up to what you started with. It is the theoretical heart of the question "can this transformation be undone?" — tying back to the invertible matrices of Chapter 4 — and of data compression, where losing the right information is the whole point.

6.2 Linear mappings

Definition 6.1. Let $\mathbf{V}$ and $\mathbf{W}$ be two vector spaces over a field $\mathbb{K}$. A mapping $F\colon \mathbf{V} \to \mathbf{W}$ is called a *linear mapping* if it satisfies the following conditions:

- $F(\mathbf{v}_1 + \mathbf{v}_2) = F(\mathbf{v}_1) + F(\mathbf{v}_2)$ for any $\mathbf{v}_1, \mathbf{v}_2 \in \mathbf{V}$;
- $F(\lambda \cdot \mathbf{v}) = \lambda \cdot F(\mathbf{v})$ for any $\mathbf{v} \in \mathbf{V}$ and $\lambda \in \mathbb{K}$.

For $\mathbf{V} = \mathbf{W} = \mathbb{R}$, every linear mapping from $\mathbb{R}$ to $\mathbb{R}$ is of the form $y = a \cdot x$ for some $a \in \mathbb{R}$. Standard examples of linear mappings include rotation by a fixed angle and scaling (length multiplication).

Theorem 6.2. *For any linear mapping $F\colon \mathbf{V} \to \mathbf{W}$ we have $F(\mathbf{0}) = \mathbf{0}$.*

Proof. $F(\mathbf{0}) = F(\mathbf{0} + \mathbf{0}) = F(\mathbf{0}) + F(\mathbf{0})$, and adding $-F(\mathbf{0})$ to both sides gives $\mathbf{0} = F(\mathbf{0})$. □

Theorem 6.3. *A mapping $F\colon \mathbf{V} \to \mathbf{W}$ is linear if and only if*

$$F(\lambda \cdot \mathbf{v}_1 + \mathbf{v}_2) = \lambda \cdot F(\mathbf{v}_1) + F(\mathbf{v}_2) \quad \text{for any } \mathbf{v}_1, \mathbf{v}_2 \in \mathbf{V} \text{ and any } \lambda \in \mathbb{K}.$$

Example 6.4. The mapping $f\colon \mathbb{R}^3 \to \mathbb{R}^2$ given by $f(x,y,z) = (x+y,\ z-x)$ is linear. Indeed, for $\mathbf{v}_1 = (x_1,y_1,z_1)$ and $\mathbf{v}_2 = (x_2,y_2,z_2)$,

$$f(\mathbf{v}_1+\mathbf{v}_2) = \big((x_1+x_2)+(y_1+y_2),\ (z_1+z_2)-(x_1+x_2)\big) = (x_1+y_1,\ z_1-x_1)+(x_2+y_2,\ z_2-x_2) = f(\mathbf{v}_1)+f(\mathbf{v}_2),$$

and similarly $f(\lambda x,\lambda y,\lambda z) = (\lambda x+\lambda y,\ \lambda z-\lambda x) = \lambda\cdot f(x,y,z)$.

Warning. The mapping $f\colon \mathbb{R}^3 \to \mathbb{R}^2$ given by $f(x,y,z) = (x+y,\ 2z+1)$ is *not* linear: indeed $f(0,0,0) = (0,1) \neq (0,0)$. The mapping $g(x,y,z) = (x^2,\ y)$ is not linear either, even though $g(0,0,0) = (0,0)$: we have $g(2,0,0) = (4,0)$, whereas $2\cdot g(1,0,0) = (2,0)$.

Example 6.5 (Differentiation is linear). Linear mappings need not involve tuples of numbers. On the vector space $\mathbb{R}_n[x]$ of polynomials of degree at most n, the differentiation map

$$D\colon \mathbb{R}_n[x] \to \mathbb{R}_n[x], \qquad D(f) = f',$$

is linear, since $(f+g)' = f'+g'$ and $(\lambda f)' = \lambda\cdot f'$. The same is true of many other operations of calculus — integration over a fixed interval, evaluation at a fixed point — which is why linear algebra is the natural language of so much of analysis.

6.3 Linear mappings and matrices

We focus on linear mappings $F\colon \mathbb{K}^n \to \mathbb{K}^m$; the general case is similar but symbolically more involved. We write all vectors as columns.

Theorem 6.6. *Let A be an $m\times n$ matrix over $\mathbb{K}$. The mapping $F\colon \mathbb{K}^n \to \mathbb{K}^m$ defined by*

$$F(\mathbf{v}) = A\cdot\mathbf{v} \qquad (\mathbf{v}\in\mathbb{K}^n)$$

is a linear mapping.

Example 6.7. For $A = \left(\begin{smallmatrix} 1 & 2 & 0 \\ -1 & 1 & 1 \end{smallmatrix}\right)$ the mapping $F(\mathbf{v}) = A\cdot\mathbf{v}$ is, written out explicitly,

$$F\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x+2y \\ -x+y+z \end{pmatrix}, \qquad \text{i.e.} \qquad F(x,y,z) = (x+2y,\ -x+y+z).$$

Conversely, every linear mapping $\mathbb{K}^n \to \mathbb{K}^m$ arises from a matrix in this way.

Theorem 6.8. *Let $F\colon \mathbb{K}^n \to \mathbb{K}^m$ be a linear mapping and let $M(F)$ be the matrix whose columns are the images of the standard basis vectors:*

$$M(F) = \big(F(e_1)\ \ F(e_2)\ \ \ldots\ \ F(e_n)\big), \qquad e_1 = (1,0,\ldots,0)^T,\ \ldots,\ e_n = (0,\ldots,0,1)^T.$$

Then $F(\mathbf{v}) = M(F)\cdot\mathbf{v}$ for every $\mathbf{v}\in\mathbb{K}^n$.

Example 6.9. Consider the linear mapping $F\colon \mathbb{R}^3 \to \mathbb{R}^2$ given by $F(x,y,z) = (x+y,\ z-x)$. Then

$$F(1,0,0) = (1,-1), \quad F(0,1,0) = (1,0), \quad F(0,0,1) = (0,1),$$

so

$$M(F) = \begin{pmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}.$$

The standard basis is not special here: a linear mapping may be prescribed freely on *any* basis.

Theorem 6.10. *Let $\mathbf{V}$ and $\mathbf{W}$ be vector spaces over $\mathbb{K}$, let $\{\mathbf{v}_1,\ldots,\mathbf{v}_n\}$ be a basis of $\mathbf{V}$ and let $\mathbf{w}_1,\ldots,\mathbf{w}_n \in \mathbf{W}$ be arbitrary vectors. Then there exists exactly one linear mapping $F\colon \mathbf{V}\to\mathbf{W}$ with $F(\mathbf{v}_i) = \mathbf{w}_i$ for $i = 1,\ldots,n$, namely*

$$F(a_1\mathbf{v}_1+\ldots+a_n\mathbf{v}_n) = a_1\mathbf{w}_1+\ldots+a_n\mathbf{w}_n.$$

Example 6.11. Find the formula for the linear mapping $F\colon \mathbb{R}^2 \to \mathbb{R}^2$ such that $F(1,1) = (2,0)$ and $F(1,-1) = (0,4)$. Writing an arbitrary vector in the basis $\{(1,1),(1,-1)\}$,

$$(x,y) = \tfrac{x+y}{2}\,(1,1) + \tfrac{x-y}{2}\,(1,-1),$$

we get, by linearity,

$$F(x,y) = \tfrac{x+y}{2}\,(2,0) + \tfrac{x-y}{2}\,(0,4) = (x+y,\ 2x-2y).$$

A quick check: $F(1,1) = (2,0)$ and $F(1,-1) = (0,4)$, as required.

6.4 Geometric interpretation

Given a vector $\mathbf{v} \in \mathbb{K}^n$ and an $m \times n$ matrix A, the product $A\mathbf{v}$ may be thought of as the image of $\mathbf{v}$ under the mapping A.

Example 6.12 (Rotation). Fix $\alpha \in [0, 2\pi)$ and consider the matrix

$$R_\alpha = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}.$$

For a vector $\mathbf{v} = \left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right)$ the image

$$R_\alpha \mathbf{v} = \begin{pmatrix} x\cos\alpha - y\sin\alpha \\ x\sin\alpha + y\cos\alpha \end{pmatrix}$$

is the vector $\mathbf{v}$ rotated about the origin by the angle α.

Example 6.13 (Scaling). Consider the matrix $A = \left(\begin{smallmatrix} 3 & 0 \\ 0 & 2 \end{smallmatrix}\right)$. For a vector $\mathbf{v} = \left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right)$ the image is $A\mathbf{v} = \left(\begin{smallmatrix} 3x \\ 2y \end{smallmatrix}\right)$, i.e. $\mathbf{v}$ stretched by the factor 3 in the x-direction and 2 in the y-direction.

Example 6.14 (Reflection and projection). The matrix $\left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ maps $\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right)$ to $\left(\begin{smallmatrix} x \\ -y \end{smallmatrix}\right)$: it is the *reflection* across the x-axis. The matrix $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ maps $\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right)$ to $\left(\begin{smallmatrix} x \\ 0 \end{smallmatrix}\right)$: it is the *orthogonal projection* onto the x-axis.

6.5 Composition of linear mappings

Theorem 6.15. *Let $F\colon \mathbb{K}^n \to \mathbb{K}^m$ and $G\colon \mathbb{K}^m \to \mathbb{K}^k$ be linear mappings. Then the composition $G \circ F\colon \mathbb{K}^n \to \mathbb{K}^k$ is linear, and*

$$M(G \circ F) = M(G) \cdot M(F).$$

Remark 6.16. Matrix multiplication is precisely composition of linear mappings — this is the real reason behind its seemingly complicated definition, and also an explanation of why it fails to be commutative: composing mappings in different orders usually produces different mappings.

Example 6.17. Composing the rotation by α with the rotation by β gives the rotation by $\alpha+\beta$, so $R_\beta \cdot R_\alpha = R_{\alpha+\beta}$, that is,

$$\begin{pmatrix} \cos\beta & -\sin\beta \\ \sin\beta & \cos\beta \end{pmatrix}\begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix} = \begin{pmatrix} \cos(\alpha+\beta) & -\sin(\alpha+\beta) \\ \sin(\alpha+\beta) & \cos(\alpha+\beta) \end{pmatrix}.$$

Comparing the entries of both sides recovers the trigonometric angle-addition formulas:

$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta, \qquad \sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta.$$

6.6 Kernel and image

Definition 6.18. Let $F\colon \mathbf{V} \to \mathbf{W}$ be a linear mapping. The *kernel* and the *image* of F are the sets

$$\ker(F) = \{\mathbf{v} \in \mathbf{V} \mid F(\mathbf{v}) = \mathbf{0}\} \subseteq \mathbf{V}, \qquad \operatorname{im}(F) = \{F(\mathbf{v}) \mid \mathbf{v} \in \mathbf{V}\} \subseteq \mathbf{W}.$$

Theorem 6.19. $\ker(F)$ *is a subspace of* $\mathbf{V}$ *and* $\operatorname{im}(F)$ *is a subspace of* $\mathbf{W}$.

Proof. If $\mathbf{v}_1, \mathbf{v}_2 \in \ker(F)$, then $F(\mathbf{v}_1 + \mathbf{v}_2) = F(\mathbf{v}_1) + F(\mathbf{v}_2) = \mathbf{0} + \mathbf{0} = \mathbf{0}$ and $F(\lambda \mathbf{v}_1) = \lambda \cdot F(\mathbf{v}_1) = \mathbf{0}$, so $\ker(F)$ is closed under addition and scalar multiplication; moreover $F(\mathbf{0}) = \mathbf{0}$ gives $\mathbf{0} \in \ker(F)$. For the image: $F(\mathbf{v}_1) + F(\mathbf{v}_2) = F(\mathbf{v}_1 + \mathbf{v}_2) \in \operatorname{im}(F)$ and $\lambda \cdot F(\mathbf{v}_1) = F(\lambda \mathbf{v}_1) \in \operatorname{im}(F)$. □

Remark 6.20. Let $F\colon \mathbb{K}^n \to \mathbb{K}^m$ be linear with matrix $A = M(F)$. Then the kernel of F is the solution space of the homogeneous system $AX = \mathbf{0}$, and the image of F is the span of the columns of A. In particular, $\dim \operatorname{im}(F) = \mathrm{r}(A)$.

Definition 6.21. Let $F\colon \mathbf{V} \to \mathbf{W}$ be a linear mapping. The *rank* of F is $\mathrm{r}(F) = \dim \operatorname{im}(F)$, and the *nullity* of F is $\dim \ker(F)$.

Theorem 6.22 (Rank–nullity theorem). *Let* $F\colon \mathbf{V} \to \mathbf{W}$ *be a linear mapping with* $\dim(\mathbf{V}) < \infty$. *Then*

$$\dim \ker(F) + \dim \operatorname{im}(F) = \dim(\mathbf{V}).$$

Example 6.23. Let $F\colon \mathbb{R}^3 \to \mathbb{R}^2$, $F(x, y, z) = (x + y,\ z - x)$, with the matrix $M(F) = \left(\begin{smallmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \end{smallmatrix}\right)$ computed earlier. The kernel consists of the solutions of $x + y = 0$, $z - x = 0$, so

$$\ker(F) = \{(t, -t, t) \mid t \in \mathbb{R}\} = \operatorname{span}(\{(1, -1, 1)\}), \qquad \dim \ker(F) = 1.$$

Since $\mathrm{r}\, M(F) = 2$, the image is a 2-dimensional subspace of $\mathbb{R}^2$, i.e. $\operatorname{im}(F) = \mathbb{R}^2$ and F is surjective. The rank–nullity theorem confirms the count: $1 + 2 = 3 = \dim(\mathbb{R}^3)$.

Example 6.24 (Kernel and image of differentiation). For the differentiation map $D\colon \mathbb{R}_n[x] \to \mathbb{R}_n[x]$ with $n \geq 1$: the kernel consists of the polynomials with zero derivative, i.e. the constants, so $\dim \ker(D) = 1$. The image consists of exactly the polynomials of degree at most $n - 1$ (every such polynomial is the derivative of one of its antiderivatives, which again lies in $\mathbb{R}_n[x]$), so $\dim \operatorname{im}(D) = n$. The rank–nullity theorem checks out: $1 + n = n + 1 = \dim(\mathbb{R}_n[x])$.

Fact 6.25. A linear mapping $F\colon \mathbf{V} \to \mathbf{W}$ is injective (one-to-one) if and only if $\ker(F) = \{\mathbf{0}\}$.

Proof. If F is injective, then $F(\mathbf{v}) = \mathbf{0} = F(\mathbf{0})$ forces $\mathbf{v} = \mathbf{0}$. Conversely, assume $\ker(F) = \{\mathbf{0}\}$ and $F(\mathbf{v}_1) = F(\mathbf{v}_2)$. Then $F(\mathbf{v}_1 - \mathbf{v}_2) = \mathbf{0}$, so $\mathbf{v}_1 - \mathbf{v}_2 \in \ker(F) = \{\mathbf{0}\}$, that is $\mathbf{v}_1 = \mathbf{v}_2$. □

6.7 Matrices in arbitrary bases

The matrix $M(F)$ was built from the standard basis of $\mathbb{K}^n$. Often another basis is better suited to the problem — finding a basis in which the matrix of a mapping becomes as simple as possible is the theme of Chapter 7 — so we now allow arbitrary bases. Recall from Chapter 3 that every vector has unique coordinates in a fixed basis.

Definition 6.26. Let $F\colon \mathbf{V} \to \mathbf{W}$ be a linear mapping, let $S = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a basis of $\mathbf{V}$ and let $R = (\mathbf{w}_1, \ldots, \mathbf{w}_m)$ be a basis of $\mathbf{W}$. The *matrix of* F *with respect to the bases* S *and* R, denoted $M_S^R(F)$, is the $m \times n$ matrix whose j-th column consists of the coordinates of $F(\mathbf{v}_j)$ in the basis R. For an operator $F\colon \mathbf{V} \to \mathbf{V}$ considered with a single basis S we write $M_S(F) = M_S^S(F)$.

Theorem 6.27. *With the notation above: if* $x \in \mathbb{K}^n$ *is the column of coordinates of* $\mathbf{v}$ *in the basis* S, *then* $M_S^R(F) \cdot x$ *is the column of coordinates of* $F(\mathbf{v})$ *in the basis* R.

Example 6.28. Let $F\colon \mathbb{R}^2 \to \mathbb{R}^2$, $F(x, y) = (x + y,\ x - y)$, and let $S = \big((1, 1), (1, 0)\big)$.

- Taking R to be the standard basis: $F(1, 1) = (2, 0)$ and $F(1, 0) = (1, 1)$, so $M_S^R(F) = \left(\begin{smallmatrix} 2 & 1 \\ 0 & 1 \end{smallmatrix}\right)$.
- Taking the basis S on both sides, we express the images in S: $F(1, 1) = (2, 0) = 0 \cdot (1, 1) + 2 \cdot (1, 0)$ and $F(1, 0) = (1, 1) = 1 \cdot (1, 1) + 0 \cdot (1, 0)$, so $M_S(F) = \left(\begin{smallmatrix} 0 & 1 \\ 2 & 0 \end{smallmatrix}\right)$.

Definition 6.29. Let $S = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ and R be two bases of $\mathbf{V}$. The *change-of-basis matrix* from S to R is the matrix P whose j-th column consists of the coordinates of $\mathbf{v}_j$ in the basis R — in other words, $P = M_S^R(\mathrm{id})$, the matrix of the identity mapping.

Theorem 6.30 (Change of basis). *Let $F\colon \mathbf{V} \to \mathbf{V}$ be a linear operator, let S and R be bases of $\mathbf{V}$, and let P be the change-of-basis matrix from S to R. Then P is invertible and*

$$M_S(F) = P^{-1} \cdot M_R(F) \cdot P.$$

Definition 6.31. Square matrices A and B are called *similar* if $A = P^{-1} \cdot B \cdot P$ for some invertible matrix P. By the theorem above, similar matrices are exactly the matrices of one and the same linear operator written in different bases.

Example 6.32. Let $F\colon \mathbb{R}^2 \to \mathbb{R}^2$ have the matrix $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ in the standard basis R, and take $S = \big((1,1),(1,-1)\big)$. The change-of-basis matrix and its inverse are

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad P^{-1} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

and we compute

$$M_S(F) = P^{-1} A P = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

In the basis S the mapping is simply "stretch the first coordinate by 3, keep the second": the matrix became *diagonal*. Finding such bases systematically is precisely the subject of the next chapter.

7 Eigenvalues and eigenvectors

7.1 Motivation

A matrix acting on a space moves every vector somewhere else, and at first sight the motion is a mess. Eigenvectors are the directions in which the mess disappears: along an eigenvector the matrix acts as mere scaling by a number — the eigenvalue. Finding these special directions means finding the natural coordinate system of a transformation, the axes along which its behaviour is simplest. It is hard to overstate how often computing profits from this one idea:

- **Ranking the web**. Google's founding insight was that the importance of a web page is an eigenvector computation: importance flows along links, and a stable assignment of importances is precisely an eigenvector of the web's matrix for the eigenvalue 1. Section 7.6 works this out in detail, and Section 7.7 continues into present-day AI.
- **Dimensionality reduction**. Principal Component Analysis (PCA), a staple of data science, computes the eigenvectors of a data set's covariance matrix: they are the directions along which the data genuinely varies, and the eigenvalues say by how much. Keeping only the few directions with the largest eigenvalues compresses the data with minimal loss — the classic "eigenfaces" method recognised human faces this way.
- **Repeated processes**. Whatever evolves in discrete steps by a fixed linear rule — a Markov chain, a population model, the pair of previous terms in the Fibonacci recurrence — evolves by matrix powers A^k. Eigenvalues make such powers computable in closed form (see the section on diagonalisation), and the largest eigenvalue governs the long-run fate of the process: growth, decay or convergence to a steady state. In essence this is also why eigenvalues decide the stability of physical systems, and why they appear when engineers ask whether a bridge will resonate in the wind.
- **Graphs and clustering**. The eigenvalues of matrices associated with a graph reveal its global shape: how well connected it is and whether it splits into communities. *Spectral clustering* and much of modern network analysis are built on this.

Within this book, eigenvalues are also a grand finale in a literal sense: computing them exercises everything that came before. The eigenvalue condition is a determinant equation (Chapter 4); its solutions are the roots of a polynomial, which may force us into the complex numbers of Chapter 1; and finding the eigenvectors for each eigenvalue means solving a homogeneous linear system (Chapter 5) whose solution set is a subspace (Chapter 3).

7.2 Eigenvalues and eigenvectors

Definition 7.1. Let A be an $n \times n$ matrix over the field $\mathbb{K}$. A scalar $\lambda \in \mathbb{K}$ is called an *eigenvalue* of A if there is a non-zero vector $\mathbf{v} \in \mathbb{K}^n$ such that

$$A\mathbf{v} = \lambda \cdot \mathbf{v}.$$

In computations one uses the following equivalent characterisation.

Theorem 7.2. *Let A be an $n \times n$ matrix over the field $\mathbb{K}$. A scalar $\lambda \in \mathbb{K}$ is an eigenvalue of A if and only if*

$$\det(A - \lambda \cdot I) = 0.$$

Proof. ($\Rightarrow$) Let $\mathbf{v}$ be a non-zero vector with $A\mathbf{v} = \lambda \cdot \mathbf{v}$. Then

$$A\mathbf{v} - \lambda \cdot \mathbf{v} = \mathbf{0}, \qquad (A - \lambda \cdot I) \cdot \mathbf{v} = \mathbf{0},$$

so the system $(A - \lambda \cdot I)X = \mathbf{0}$ has a non-zero solution. This happens precisely when $\det(A - \lambda \cdot I) = 0$.

($\Leftarrow$) Assume $\det(A - \lambda \cdot I) = 0$. Then the system $(A - \lambda \cdot I)X = \mathbf{0}$ has a non-zero solution $\mathbf{v}$, and

$$(A - \lambda \cdot I) \cdot \mathbf{v} = \mathbf{0}, \qquad A\mathbf{v} - \lambda \cdot \mathbf{v} = \mathbf{0}, \qquad A\mathbf{v} = \lambda \cdot \mathbf{v}.$$ □

Definition 7.3. The polynomial

$$p_A(\lambda) = \det(A - \lambda \cdot I)$$

(a polynomial of degree n in the variable λ) is called the *characteristic polynomial* of A. The eigenvalues of A are exactly the roots of p_A lying in $\mathbb{K}$.

Fact 7.4. Similar matrices have the same characteristic polynomial: if $B = P^{-1}AP$, then

$$p_B(\lambda) = \det(P^{-1}AP - \lambda I) = \det\left(P^{-1}(A - \lambda I)P\right) = \det(P^{-1}) \cdot \det(A - \lambda I) \cdot \det(P) = p_A(\lambda).$$

Remark 7.5 (Eigenvalues of a linear operator). For a linear operator $F\colon \mathbf{V} \to \mathbf{V}$, eigenvalues and eigenvectors are defined by the same condition $F(\mathbf{v}) = \lambda \cdot \mathbf{v}$ (with $\mathbf{v} \neq \mathbf{0}$). They are computed from the matrix of F in *any* basis S: the matrices $M_S(F)$ in different bases are similar (Chapter 6), so by the fact above they all have the same characteristic polynomial and hence the same eigenvalues.

Definition 7.6. A non-zero vector $\mathbf{v} \in \mathbb{K}^n$ is called an *eigenvector* for the eigenvalue λ if $A \cdot \mathbf{v} = \lambda \cdot \mathbf{v}$.

Theorem 7.7. *Let λ be an eigenvalue of A. The set W_λ of all eigenvectors for λ (together with $\mathbf{0}$) is a subspace of $\mathbb{K}^n$.*

Proof. Note that $\mathbf{0} \in W_\lambda$. For $\mathbf{v}_1, \mathbf{v}_2 \in W_\lambda$,

$$A(\mathbf{v}_1 + \mathbf{v}_2) = A\mathbf{v}_1 + A\mathbf{v}_2 = \lambda \cdot \mathbf{v}_1 + \lambda \cdot \mathbf{v}_2 = \lambda \cdot (\mathbf{v}_1 + \mathbf{v}_2),$$

so $\mathbf{v}_1 + \mathbf{v}_2 \in W_\lambda$. Similarly, for $\mathbf{v} \in W_\lambda$ and $k \in \mathbb{K}$ we have $A(k\mathbf{v}) = kA\mathbf{v} = k\lambda\mathbf{v} = \lambda(k\mathbf{v})$, so $k \cdot \mathbf{v} \in W_\lambda$. □

Remark 7.8. The subspace W_λ is called the *eigenspace* of λ. It is exactly the solution space of the homogeneous system $(A - \lambda I)X = \mathbf{0}$, so by the dimension formula of Chapter 5

$$\dim(W_\lambda) = n - \mathrm{r}(A - \lambda I),$$

and this dimension is at least 1 for every eigenvalue λ.

7.3 Computing eigenvalues and eigenvectors

The recipe is always the same: find the roots of the characteristic polynomial $p_A(\lambda)$, and for each root λ solve the homogeneous system $(A - \lambda I)X = \mathbf{0}$ to find the eigenvectors.

Example 7.9 (A 2×2 matrix). Let $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ over $\mathbb{R}$. Then

$$p_A(\lambda) = \begin{vmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{vmatrix} = (2 - \lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3),$$

so the eigenvalues are $\lambda = 1$ and $\lambda = 3$.

Eigenvectors for $\lambda = 1$. The system $(A - I)X = \mathbf{0}$ reads $x + y = 0$ (twice), so

$$W_1 = \left\{ \begin{pmatrix} t \\ -t \end{pmatrix} \;\middle|\; t \in \mathbb{R} \right\}.$$

Eigenvectors for $\lambda = 3$. The system $(A - 3I)X = \mathbf{0}$ reads $-x + y = 0$, so

$$W_3 = \left\{ \begin{pmatrix} t \\ t \end{pmatrix} \;\middle|\; t \in \mathbb{R} \right\}.$$

Indeed, $A \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Example 7.10 (A 3×3 matrix). Let

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 0 & -1 & 3 \\ 0 & 1 & 1 \end{pmatrix}, \qquad A - \lambda \cdot I = \begin{pmatrix} 1-\lambda & 0 & 2 \\ 0 & -1-\lambda & 3 \\ 0 & 1 & 1-\lambda \end{pmatrix}.$$

Expanding along the first column,

$$\det(A - \lambda \cdot I) = (1-\lambda) \cdot \begin{vmatrix} -1-\lambda & 3 \\ 1 & 1-\lambda \end{vmatrix} = (1-\lambda)\big((-1-\lambda)(1-\lambda) - 3\big) = (1-\lambda)(\lambda^2 - 4) = (1-\lambda)(\lambda-2)(\lambda+2).$$

The eigenvalues of A are therefore $\lambda = 1,\ 2,\ -2$.

Eigenvectors for $\lambda = 1$. They are the solutions of

$$\begin{pmatrix} 0 & 0 & 2 \\ 0 & -2 & 3 \\ 0 & 1 & 0 \end{pmatrix} X = \mathbf{0} \xrightarrow{\text{row reduction}} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} X = \mathbf{0}, \qquad W_1 = \left\{ \begin{pmatrix} x \\ 0 \\ 0 \end{pmatrix} \;\middle|\; x \in \mathbb{R} \right\}.$$

Eigenvectors for $\lambda = 2$. They are the solutions of

$$\begin{pmatrix} -1 & 0 & 2 \\ 0 & -3 & 3 \\ 0 & 1 & -1 \end{pmatrix} X = \mathbf{0} \xrightarrow{\text{row reduction}} \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} X = \mathbf{0}, \qquad W_2 = \left\{ \begin{pmatrix} 2z \\ z \\ z \end{pmatrix} \;\middle|\; z \in \mathbb{R} \right\}.$$

Eigenvectors for $\lambda = -2$. They are the solutions of

$$\begin{pmatrix} 3 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 1 & 3 \end{pmatrix} X = \mathbf{0} \xrightarrow{\text{row reduction}} \begin{pmatrix} 1 & 0 & \frac{2}{3} \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix} X = \mathbf{0}, \qquad W_{-2} = \left\{ \begin{pmatrix} -\frac{2}{3}z \\ -3z \\ z \end{pmatrix} \;\middle|\; z \in \mathbb{R} \right\}.$$

Fact 7.11. For an $n \times n$ matrix A there are at most n distinct eigenvalues of A (the characteristic polynomial has degree n, so it has at most n roots).

Fact 7.12. The eigenvalues of an upper (or lower) triangular matrix are exactly its diagonal entries: for a triangular A the matrix $A - \lambda I$ is again triangular, so

$$p_A(\lambda) = (a_{11} - \lambda)(a_{22} - \lambda) \cdots (a_{nn} - \lambda).$$

Remark 7.13. It may happen that a matrix A has no eigenvalues over a field $\mathbb{K}$, or that some eigenvalues are multiple.

Example 7.14 (No real eigenvalues). Consider $A = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ over $\mathbb{R}$. Then

$$\det(A - \lambda \cdot I) = \begin{vmatrix} -\lambda & -1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 + 1.$$

The equation $\lambda^2 + 1 = 0$ has no solution over $\mathbb{R}$, so A has no real eigenvalues.

Example 7.15 (Complex eigenvalues). The same matrix $A = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ — the rotation by $\frac{\pi}{2}$ — viewed over $\mathbb{K} = \mathbb{C}$ does have eigenvalues: $\lambda^2 + 1 = 0$ gives $\lambda = \pm i$. For $\lambda = i$ the system $(A - iI)X = \mathbf{0}$ reads $-ix - y = 0$, so $(1, -i)^T$ is an eigenvector. By the Fundamental Theorem of Algebra, over $\mathbb{C}$ every square matrix has at least one eigenvalue.

Example 7.16 (A multiple eigenvalue). For the 3×3 identity matrix I,

$$\det(I - \lambda \cdot I) = \begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & 1-\lambda & 0 \\ 0 & 0 & 1-\lambda \end{vmatrix} = (1-\lambda)^3,$$

so the only eigenvalue is $\lambda = 1$ (with multiplicity 3).

Fact 7.17. If λ is an eigenvalue of A, then it is also an eigenvalue of A^T.

Fact 7.18 (Trace and determinant). Let A be an $n \times n$ matrix over $\mathbb{C}$ with eigenvalues $\lambda_1, \ldots, \lambda_n$ listed with multiplicities. Then

$$\lambda_1 \lambda_2 \cdots \lambda_n = \det(A), \qquad \lambda_1 + \lambda_2 + \ldots + \lambda_n = a_{11} + a_{22} + \ldots + a_{nn}$$

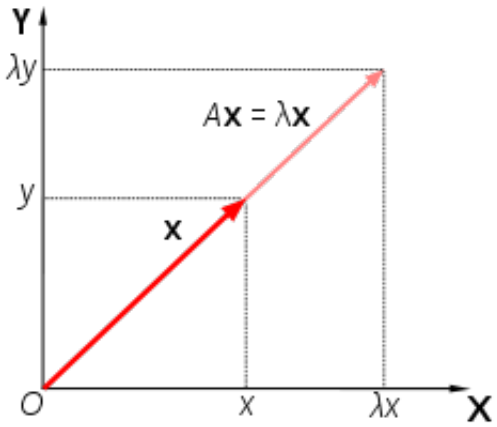
(the sum of the diagonal entries of A is called the *trace* of A). For a 2×2 matrix these two quantities already determine the characteristic polynomial:

$$p_A(\lambda) = \lambda^2 - (a_{11} + a_{22})\,\lambda + \det(A).$$

Theorem 7.19. *Eigenvectors corresponding to pairwise distinct eigenvalues are linearly independent: if $\lambda_1, \ldots, \lambda_k$ are distinct eigenvalues of A and $\mathbf{v}_1, \ldots, \mathbf{v}_k$ are corresponding eigenvectors, then the set $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is linearly independent.*

7.4 Geometric interpretation

Eigenvectors of a matrix A are the directions left unchanged (up to scaling) by the linear mapping $\mathbf{v} \mapsto A\mathbf{v}$. The figure below illustrates the action of a shear mapping: the horizontal vector is an eigenvector (its direction is preserved), whereas the other vector changes direction.



7.5 Diagonalisation

Definition 7.20. An $n \times n$ matrix A over $\mathbb{K}$ is *diagonalisable* if there exists an invertible matrix P such that $P^{-1} \cdot A \cdot P$ is a diagonal matrix.

Theorem 7.21. *An $n \times n$ matrix A over $\mathbb{K}$ is diagonalisable if and only if $\mathbb{K}^n$ has a basis $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ consisting of eigenvectors of A. In that case one may take $P = (\mathbf{v}_1\ \mathbf{v}_2\ \ldots\ \mathbf{v}_n)$, the matrix with the eigenvectors as columns,*

and then

$$P^{-1}AP = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix} =: D,$$

where λ_i is the eigenvalue corresponding to the eigenvector $\mathbf{v}_i$.

Remark 7.22. In the language of Chapter 6: P is the change-of-basis matrix from the eigenvector basis to the standard basis, and $D = P^{-1}AP$ is the matrix of the operator $\mathbf{v} \mapsto A\mathbf{v}$ written in the eigenvector basis. "Diagonalisable" therefore means "similar to a diagonal matrix", and the closing example of the previous chapter, with $M_S(F) = \left(\begin{smallmatrix} 3 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, was secretly a diagonalisation.

Corollary 7.23. *If an $n \times n$ matrix has n distinct eigenvalues in $\mathbb{K}$, then it is diagonalisable (eigenvectors for distinct eigenvalues are linearly independent, and n linearly independent vectors form a basis of $\mathbb{K}^n$).*

Theorem 7.24 (Diagonalisability criterion). *Let A be an $n \times n$ matrix over $\mathbb{K}$ with distinct eigenvalues $\lambda_1, \dots, \lambda_k \in \mathbb{K}$. Then A is diagonalisable over $\mathbb{K}$ if and only if the eigenspace dimensions add up to n:*

$$\dim(W_{\lambda_1}) + \dots + \dim(W_{\lambda_k}) = n.$$

In that case a basis of $\mathbb{K}^n$ consisting of eigenvectors is obtained by putting together bases of the individual eigenspaces.

Example 7.25 (Powers of a matrix). Diagonalisation makes it easy to compute powers of a matrix: if $A = PDP^{-1}$, then the inner factors cancel in pairs,

$$A^k = (PDP^{-1})(PDP^{-1}) \cdots (PDP^{-1}) = P\,D^k\,P^{-1},$$

and the power D^k of a diagonal matrix is computed entrywise on the diagonal. For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ we found the eigenvalues 1 and 3 with eigenvectors $(1,-1)^T$ and $(1,1)^T$, so

$$P = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \qquad D = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}, \qquad P^{-1} = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix},$$

and therefore, for every $k \in \mathbb{N}$,

$$A^k = P\begin{pmatrix} 1 & 0 \\ 0 & 3^k \end{pmatrix}P^{-1} = \frac{1}{2}\begin{pmatrix} 3^k+1 & 3^k-1 \\ 3^k-1 & 3^k+1 \end{pmatrix}.$$

For $k = 1$ this indeed recovers A.

Warning. Not every matrix is diagonalisable, even over $\mathbb{C}$: for $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ the only eigenvalue is $\lambda = 1$, and its eigenspace $W_1 = \operatorname{span}(\{(1,0)^T\})$ is one-dimensional — there is no basis of $\mathbb{C}^2$ consisting of eigenvectors of A.

7.6 Application: PageRank

This section is a case study: a complete tour, with computations, of the algorithm that made web search work. It exercises nearly everything developed in this book — graphs stored as matrices (Chapter 4), homogeneous systems (Chapter 5), and eigenvectors — and the effort is well invested: the eigenvector computed below ranked the web for billions of users.

7.6.1 The ranking problem

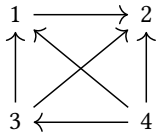
By the late 1990s the web had grown to hundreds of millions of pages, and for any query a search engine could easily find tens of thousands of pages containing the right keywords. The hard problem was not *finding* pages but *ordering* them. Early engines ranked pages by their own content — for instance, by how often the

keyword appeared — a criterion that a page author fully controls, and which was therefore trivially manipulated (a spammer simply repeats the keyword a thousand times in invisible text).

The insight of Sergey Brin and Larry Page (1998) was to rank a page by what the *rest of the web* thinks of it. A link from page i to page j is a vote of i for j. Not all votes are equal: a link from an important page should weigh more than a link from an obscure one, and a page voting for a thousand targets should confer less on each than a page voting for two. This makes the definition of importance *circular* — the importance of a page depends on the importances of the pages linking to it, which depend on… — and the whole point of this section is that linear algebra resolves the circularity: a self-consistent assignment of importances is precisely an eigenvector for the eigenvalue $\lambda = 1$.

7.6.2 The web graph and the random surfer

A *web graph* models the web: web pages are nodes and links are directed edges (arrows). For example:



Definition 7.26. In a web graph G, for a vertex v let l_v denote the number of outgoing edges starting at v.

In the example above $l_1 = 1,\ l_2 = 0,\ l_3 = 2,\ l_4 = 3$.

Instead of reasoning about "importance" directly, we follow Brin and Page and imagine a *random surfer*: a user who starts on some page and endlessly clicks links, choosing at every step one of the outgoing links of the current page uniformly at random. Pages where the surfer spends much of their time are important; pages the surfer rarely reaches are not. The surfer's behaviour is encoded in a matrix.

Definition 7.27. For a web graph G define its *stochastic matrix* S whose ij-entry is

$$s_{ij} = \frac{1}{l_i} \quad \text{whenever there is a link from } i \text{ to } j, \qquad s_{ij} = 0 \text{ otherwise.}$$

If $l_i = 0$, we set $s_{ii} = 1$.

The entry s_{ij} is the probability that a surfer currently on page i moves to page j in one click. For the graph above,

$$S = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \end{pmatrix}.$$

Fact 7.28. For the stochastic matrix S of a web graph G:

- $0 \le s_{ij} \le 1$ for all i, j;
- every row of S sums to 1, that is, $S \cdot (1, \dots, 1)^T = (1, \dots, 1)^T$;
- $\lambda = 1$ is an eigenvalue of S (and of S^T).

Definition 7.29. A vector $w \in \mathbb{R}^n$ is a *probability vector* if its entries are non-negative and sum to 1.

If the i-th entry of a probability vector w is the probability that the surfer is currently on page i, then one click later the probability of being on page j is $\sum_i w_i\, s_{ij}$ — that is, the new distribution is the vector $S^T w$.

Fact 7.30. If w is a probability vector, then so is $S^T w$.

Proof. The entries of $S^T w$ are sums of products of non-negative numbers, hence non-negative. Their total is

$$\sum_j (S^T w)_j = \sum_j \sum_i s_{ij} w_i = \sum_i w_i \sum_j s_{ij} = \sum_i w_i \cdot 1 = 1,$$

using the fact that every row of S sums to 1. □

Remark 7.31. A process that moves between finitely many states with fixed transition probabilities, forgetting everything about its past except the current state, is called a *Markov chain*. The random surfer is the most famous Markov chain in computer science, but the machinery below applies to any of them — queues, board games, speech models, randomised algorithms.

7.6.3 PageRank as an eigenvector

A distribution w deserves to be called *the* importance ranking if it is *stationary*: one more click leaves it unchanged.

Definition 7.32. A *PageRank vector* of the web graph G is a probability vector w with

$$S^T w = w,$$

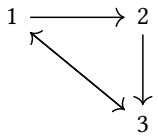
that is, an eigenvector of S^T for the eigenvalue $\lambda = 1$, normalised so that its entries sum to 1.

Written entry by entry, the condition $S^T w = w$ reads

$$w_j = \sum_{i \to j} \frac{w_i}{l_i},$$

where the sum runs over all pages i linking to j. This is exactly the voting scheme we asked for: page j receives the importance of each page i that links to it, diluted by the number l_i of votes that i casts. The circular definition of importance has become an eigenvector equation — and finding eigenvectors for a *known* eigenvalue $\lambda = 1$ is a homogeneous linear system, which we know how to solve.

Example 7.33 (A complete PageRank computation). Consider the web graph



with $S = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$.

The system $S^T w = w$ reads

$$w_1 = w_3, \qquad w_2 = \tfrac{1}{2} w_1, \qquad w_3 = \tfrac{1}{2} w_1 + w_2,$$

and its solution space is spanned by $(1, \frac{1}{2}, 1)$. Normalising the sum to 1 gives the PageRank vector

$$w = \left(\tfrac{2}{5},\ \tfrac{1}{5},\ \tfrac{2}{5}\right).$$

Pages 1 and 3 tie for first place and page 2 comes last — note that this is *not* just counting in-links (pages 1 and 2 have one in-link each, yet their ranks differ): page 1's single vote comes undiluted from the important page 3, while page 2 receives only half a vote from page 1.

Example 7.34 (A rank sink). Now run the same computation on the four-page graph from the beginning of this section. The system $S^T w = w$ gives, from the last coordinate upwards,

$$w_4 = 0, \qquad w_3 = \tfrac{1}{3} w_4 = 0, \qquad w_1 = \tfrac{1}{2} w_3 + \tfrac{1}{3} w_4 = 0,$$

and w_2 arbitrary — so the unique PageRank vector is $w = (0, 1, 0, 0)$. All importance has drained into page 2: it has no outgoing links, so the random surfer, once arrived, is stuck there forever (recall $s_{22} = 1$). Such a page is called a *rank sink*, and the verdict "every other page has importance exactly zero" is clearly useless for ranking. A second failure mode: if the web graph consists of several disconnected pieces, the eigenspace of $\lambda = 1$ has dimension greater than 1, and there are *many* PageRank vectors — the ranking is not even well defined.

7.6.4 The Google matrix

Both failure modes are cured by one modification of the model: with a small probability the surfer gets bored and, instead of following a link, jumps to a completely random page.

Definition 7.35 (Google matrix). Let α be the *damping factor* (e.g. $\alpha = 0.85$) and let v be a *personalisation vector*: a probability vector describing where the surfer jumps when bored (for the uniform choice, $v = (\frac{1}{n}, \dots, \frac{1}{n})$). Put

$$G = \alpha \cdot S + (1-\alpha) \cdot (1, \dots, 1)^T \cdot v^T.$$

In words: with probability α the surfer follows a random link of the current page (the αS term), and with probability $1-\alpha$ they *teleport* to a random page drawn according to v. Note that G is again a stochastic matrix: its entries are non-negative and

$$G\cdot(1,\dots,1)^T = \alpha\, S(1,\dots,1)^T + (1-\alpha)(1,\dots,1)^T\big(v^T(1,\dots,1)^T\big) = \alpha(1,\dots,1)^T + (1-\alpha)(1,\dots,1)^T = (1,\dots,1)^T,$$

since the entries of v sum to 1. Unlike S, however, all entries of G are strictly positive — every page can reach every other page in one step — and this is exactly what guarantees a unique answer.

Fact 7.36. The matrix G^T has a *unique* eigenvector π for $\lambda = 1$ (that is, $G^T\pi = \pi$) whose entries are non-negative and sum to 1. The i-th entry of π is the *PageRank* of web page i.

Remark 7.37. The deep reason is the *Perron–Frobenius theorem*: a square matrix with strictly positive entries has a unique largest eigenvalue, which is positive and simple, and its eigenvector can be chosen with strictly positive entries. For a stochastic matrix that largest eigenvalue is $\lambda = 1$. The theorem is beyond the scope of this course, but everything else in this section is not.

Example 7.38 (Damping defeats the sink). Take the smallest possible rank sink: two pages, one link $1 \to 2$, and page 2 dangling ($s_{22} = 1$), so

$$S = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}, \qquad \text{and, with } v = \left(\tfrac{1}{2}, \tfrac{1}{2}\right), \qquad G = \begin{pmatrix} \frac{1-\alpha}{2} & \alpha + \frac{1-\alpha}{2} \\ \frac{1-\alpha}{2} & \alpha + \frac{1-\alpha}{2} \end{pmatrix}.$$

Solving $G^T\pi = \pi$ with $\pi_1 + \pi_2 = 1$ gives (both rows of G are equal, which makes the computation immediate)

$$\pi = \left(\frac{1-\alpha}{2},\ \frac{1+\alpha}{2}\right) \overset{\alpha=0.85}{=} (0.075,\ 0.925).$$

Without damping the ranking was $(0, 1)$; the teleportation guarantees every page a baseline importance of at least $(1-\alpha)\, v_i$, while the sink, though still first, no longer erases the rest of the web.

7.6.5 Computing PageRank at scale: the power method

For a web graph with n in the billions, nobody solves $(G^T - I)X = \mathbf{0}$ by row reduction — Gaussian elimination costs on the order of n^3 operations, and G, unlike S, has no zero entries to exploit. Instead one uses the oldest eigenvector algorithm there is: simply let the surfer surf.

Fact 7.39 (Power method). Let G be a Google matrix with damping factor α and let $w^{(0)}$ be *any* probability vector. Define

$$w^{(k+1)} = G^T w^{(k)}.$$

Then the sequence $w^{(k)}$ converges to the PageRank vector π, and the error shrinks at least as fast as α^k.

Remark 7.40 (Why this is fast). Each step is one matrix–vector product. By the definition of G,

$$G^T w = \alpha\, S^T w + (1-\alpha)\, v\, \big((1,\dots,1)\cdot w\big) = \alpha\, S^T w + (1-\alpha)\, v,$$

so the dense matrix G never needs to be formed: one multiplies by the *sparse* matrix S^T — whose number of non-zero entries is the number of links, a small multiple of n rather than n^2 — and adds a fixed vector. With $\alpha = 0.85$, the error factor α^k drops below 10^{-4} after about 60 iterations, independently of n. Sixty sparse products: that is the whole algorithm that launched Google.

Example 7.41 (Power iteration in action). For the three-page graph computed earlier (PageRank $(0.4, 0.2, 0.4)$), start the undamped iteration $w^{(k+1)} = S^T w^{(k)}$ from $w^{(0)} = (1, 0, 0)$ — the surfer starts on page 1 with certainty:

k	0	1	2	3	4	5	6	7	8	$\to$
w_1	1	0	0.5	0.5	0.25	0.5	0.375	0.375	0.438	0.4
w_2	0	0.5	0	0.25	0.25	0.125	0.25	0.188	0.188	0.2
w_3	0	0.5	0.5	0.25	0.5	0.375	0.375	0.438	0.375	0.4

The iterates oscillate around the answer and settle towards it. (With damping the convergence is faster and guaranteed; the undamped iteration can even fail to converge — for the two-page cycle $1 \to 2 \to 1$ starting at $(1, 0)$ it alternates between $(1, 0)$ and $(0, 1)$ forever, which is one more reason for the Google matrix.)

Remark 7.42 (Beyond the web). The idea "importance = eigenvector of a suitably chosen matrix" — *eigenvector centrality* — has long outgrown web search. The same computation ranks scientific papers by citations, finds influential users in social networks, identifies key proteins in biological interaction networks, and seeds the ranking of players and teams in sports analytics. Whenever influence flows along the edges of a graph, a stationary eigenvector is the natural measure of it.

7.7 Outlook: linear algebra inside modern AI

PageRank is a twentieth-century success story of linear algebra in computing. This closing section sketches the twenty-first: modern artificial intelligence is, at the level of actual computation, largely the linear algebra of this course executed at enormous scale. None of what follows is examinable; all of it is true, and it is a good answer to the question "why did I have to learn this?"

7.7.1 Embeddings: coordinates for meaning

Machine learning begins by turning objects into vectors — here this is called an *embedding*. Words, images, songs, users and products are all represented as elements of $\mathbb{R}^d$ for some dimension d (typically a few hundred to a few thousand), chosen so that geometric relations mirror semantic ones: similar objects lie close together, and meaningful differences correspond to directions. The structure is regular enough that *linear maps* (Chapter 6) act on meaning: word embeddings trained separately on English and Spanish text turn out to be related, to a good approximation, by a single matrix W with

$$W \cdot x_{cat} \approx x_{gato}, \qquad W \cdot x_{five} \approx x_{cinco}, \ \ldots$$

learned from a few thousand dictionary pairs and then used to translate words never seen in training. An embedding of a vocabulary of 50 000 words into $\mathbb{R}^{768}$ is nothing but a $50\,000 \times 768$ matrix, learned from data, whose rows are the word vectors.

Example 7.43 (Recommendation as low rank). A streaming service stores its ratings as a matrix R with one row per user and one column per film — say $10^6 \times 10^4$, ten billion entries, almost all unknown. The classical recommender-system insight is that R is approximately of *low rank* (Chapter 4): tastes are governed by a few dozen hidden factors. One therefore looks for matrices U (of size $10^6 \times 50$) and V (of size 50×10^4) with

$$R \approx U \cdot V.$$

The i-th row of U is a 50-dimensional embedding of user i, the j-th column of V an embedding of film j, and the predicted rating is the ij-entry of the product. Ten billion entries have been compressed into about fifty million — and the compression is the point: the model fills in the missing ratings, i.e. it *recommends*.

7.7.2 Deep networks: matrices with a pinch of nonlinearity

A neural network transforms an input vector through a sequence of *layers*. A single layer computes

$$x \longmapsto \varphi(A\,x + b),$$

where A is a matrix of learned weights, b is a learned vector, and φ is a fixed, very simple non-linear function applied to each coordinate — the most popular choice is $\varphi(t) = \max(t, 0)$. Everything the network has learned is stored in the matrices A and vectors b of its layers; a layer from $\mathbb{R}^{768}$ to $\mathbb{R}^{768}$ alone carries $768^2 = 589\,824$ learnable matrix entries, and *training* the network means iteratively nudging these entries to reduce the error on data.

Why is the non-linear φ indispensable? Chapter 6 already contains the answer.

Observation 7.44. Suppose the φ's are omitted, so each layer is $x \mapsto A_i x + b_i$. Then the composition of any number of layers collapses:

$$A_2(A_1 x + b_1) + b_2 = (A_2 A_1)\, x + (A_2 b_1 + b_2),$$

which is a mapping of exactly the same form, with matrix $A_2 A_1$ (this is the composition theorem $M(G \circ F) = M(G) \cdot M(F)$ at work). By induction, a hundred-layer purely linear network computes the same function as a single layer — depth buys nothing. All the expressive power of deep learning beyond linear maps is created by the cheap non-linearity squeezed between the matrices; all the *knowledge* sits in the matrices themselves.

7.7.3 Attention: the transformer as matrix products

The architecture behind modern language models, the *transformer*, is an exercise in matrix multiplication. A text of n tokens (word fragments) enters as an $n \times d$ matrix X whose rows are the token embeddings. The signature operation, *attention*, computes from learned $d \times d$ matrices A, B, C the products

$$\underbrace{(XA)\,(XB)^T}_{n \times n} \qquad \text{and then} \qquad \underbrace{\big(\text{normalised scores}\big) \cdot (XC)}_{n \times d}.$$

The first product is an $n \times n$ matrix of interaction scores — entry ij measures how relevant token j is to token i — and the second mixes the tokens accordingly. A large model stacks dozens of such blocks, and generating every single word of a chatbot's answer is a march through these products. This is why AI progress is tied to hardware that multiplies matrices fast (GPUs), and why the sizes of models are quoted in *parameters*: the total count of learned matrix entries.

7.7.4 Principal directions: eigenvalues meet data

Where do the eigenvalues of this chapter enter? Let a data set be a cloud of points in $\mathbb{R}^d$ — one vector per measured object. Statistics summarises the cloud's shape in its *covariance matrix* C: a symmetric $d \times d$ matrix whose ij-entry measures how coordinates i and j vary together. The following fact (a form of the *spectral theorem*, proved in a second course) makes symmetric matrices the best-behaved matrices of all.

Fact 7.45. A symmetric matrix $C \in \mathbb{R}^d_d$ has d real eigenvalues (with multiplicities), and $\mathbb{R}^d$ has a basis consisting of eigenvectors of C. In particular, every symmetric real matrix is diagonalisable.

The eigenvectors of the covariance matrix are the *principal directions* of the data: the eigenvector with the largest eigenvalue points in the direction along which the data varies most, the next one in the most-varying direction after that, and so on, while each eigenvalue measures the amount of variation in its direction. *Principal Component Analysis* (PCA) keeps only the few directions with the largest eigenvalues and discards the rest — compressing, say, 10 000-dimensional face images to 50 coordinates with little loss. The classical *eigenfaces* method recognised faces this way, and PCA remains a first tool of every data scientist. Fittingly, the standard algorithm for finding the top principal direction of a huge data set is an old acquaintance: the power method of the PageRank section, applied to C instead of G^T.

Remark 7.46 (Where to go from here). Two pillars of applied linear algebra are missing from this course and complete the picture: *inner products* (which formalise angles, lengths and orthogonality, and turn "similar embeddings" into a computation) and the *singular value decomposition* (the definitive version of low-rank approximation, behind PCA and the recommender factorisation above). They are the natural next chapter — but the objects they build on are exactly the ones now in your toolkit: vector spaces, bases, matrices, rank, linear mappings and eigenvectors.

E Exercises

The exercises below come from the ten tutorial series accompanying the course (by A. Pilitowska and T. Brengos), collected here in the notation of this book; the three complex-numbers series are merged into a single section.

E.1 Complex numbers

Exercise E.1.1. Calculate: a) $(2+3i)(4-2i)$ b) $(1+2i)(2i-1)$ c) $(3+i)(3-i)$ d) $(a+bi)(a-bi)$

Exercise E.1.2. Calculate: a) $\dfrac{2-3i}{3+4i}$ b) $\dfrac{i}{1+i}$ c) $\dfrac{3-4i}{i}$

Exercise E.1.3. Calculate: a) $\dfrac{1+4i}{4-i}$ b) $\dfrac{(3+4i)^4}{(3-4i)^3}$

Exercise E.1.4. Show that for every two complex numbers z and w:

a) $\overline{z+w} = \bar{z} + \bar{w}$ and $\overline{zw} = \bar{z}\,\bar{w}$

b) $|zw| = |z|\,|w|$

c) $\left|\dfrac{z}{w}\right| = \dfrac{|z|}{|w|}$

Exercise E.1.5. Show that multiplication of complex numbers is associative.

Exercise E.1.6. Show that multiplication of complex numbers is distributive with respect to addition.

Exercise E.1.7. Describe geometrically the sets of points $z \in \mathbb{C}$ satisfying the following conditions:

a) $|z - i + 3| = 5$

b) $|z - i + 3| > 5$

c) $|z - 1 + 3i| \leq 5$

d) $2 \leq |z + i| < 4$

e) $\mathfrak{Re}\big((z-i)^2\big) = 0$

f) $\mathfrak{Re}(z^2) = 2$ and $\big(\mathfrak{Im}(z+i)\big)^2 = 1$

g) $\left|\dfrac{z+3}{z-2i}\right| \geq 1$

h) $\mathfrak{Re}(z+1) < 0$ and $|i - z| \leq 3$

Exercise E.1.8. Find polar forms of the following complex numbers:

a) $1 + i$

b) -3 (negative 3)

c) $-i$

d) $2 + 2i$

e) $3 + 3\sqrt{3}\,i$

f) $z = \sin\alpha + i\cos\alpha$

g) $1 - i\sqrt{3}$

h) $-z$, if $z = \cos\alpha + i\sin\alpha$

i) $\sqrt{3} - i$

j) $\bar{z}$, if $z = \cos\alpha + i\sin\alpha$

k) $\frac{1}{2} + \frac{\sqrt{3}}{2}\,i$

Exercise E.1.9. Calculate:

a) $\sqrt[3]{(1+i)^6}$

b) $\left(\dfrac{1-i}{1+i}\right)^{222}$

c) $\left(\dfrac{1}{2} + i\,\dfrac{\sqrt{3}}{2}\right)^{124}$

d) $\left(\dfrac{2+2i}{\sqrt{3}-i}\right)^8$

e) $\sqrt[6]{1}$

f) $\sqrt[4]{-1}$

g) $\sqrt[4]{(3-5i)^4}$

h) $\sqrt[6]{-1}$

Exercise E.1.10. Describe geometrically the sets of all the following roots: a) $\sqrt[6]{1}$ b) $\sqrt[4]{-1}$ c) $\sqrt[8]{1}$ d) $\sqrt[6]{-1}$

Exercise E.1.11. Use De Moivre's formula to:

(a) express $\sin 5\alpha$ in terms of $\sin\alpha$ and $\cos\alpha$,

(b) express $\cos 4\alpha$ in terms of $\sin\alpha$ and $\cos\alpha$.

Exercise E.1.12. Prove that
$$\frac{\cos\alpha + i\sin\alpha}{\cos\beta + i\sin\beta} = \cos(\alpha-\beta) + i\sin(\alpha-\beta).$$

Exercise E.1.13. Find the polar form of: a) $2+\sqrt{2}+i\sqrt{2}$ b) $2+i+i\sqrt{3}$

Exercise E.1.14. Calculate: a) $\cos\frac{\pi}{12}$ b) $\cos 105^\circ$ c) $\cos 75^\circ$ d) $\cos(-15^\circ)$

Exercise E.1.15. Calculate $(1+\cos\alpha+i\sin\alpha)^{10}$ for $\alpha\in[0,\pi)$.

Exercise E.1.16. Let $f(x)\in\mathbb{R}[x]$. Show that z is a root of f if and only if $\bar{z}$ is a root of f.

Exercise E.1.17. Factor each of the following polynomials into a product of factors of degree at most 2: a) x^6+1 b) x^4+1 c) x^6+x^3+1 d) x^5+1

Exercise E.1.18. Prove that every polynomial of an odd degree, with all real coefficients, has at least one real root.

Exercise E.1.19. Knowing that $1-2i$ is a root of the polynomial $z^4+2z^3+2z^2+10z+25$, find all remaining roots.

Exercise E.1.20. Knowing that $2+i$ is a root of the polynomial $z^4-2z^3-6z^2+22z-15$, find all remaining roots.

Exercise E.1.21. Solve the equations: a) $x^2+x+1=0$ b) $2x^2+ix+3=0$ c) $x^2+2x-i\sqrt{3}=0$

E.2 Operations modulo n. Groups and fields

Exercise E.2.1. Calculate: a) $17 \bmod 4$ b) $4 \bmod 17$ c) $(-2) \bmod 5$

Exercise E.2.2. Let n be a positive integer. For integers x and y we write $x \equiv y \pmod{n}$, and say that x is *congruent to y modulo n*, if n divides $x - y$. Solve the equations:

a) $2x \equiv 1 \pmod{7}$

b) $2x \equiv 1 \pmod{6}$

c) $5x \equiv 1 \pmod{9}$

d) $x^2 \equiv 3 \pmod{11}$

e) $2x + 3 \equiv 0 \pmod{5}$

f) $x + k \equiv 0 \pmod{n}$

Exercise E.2.3. Prove that for all $n, k, p \in \mathbb{Z}$ we have the following:

a) $(n \bmod k) \bmod k = n \bmod k$

b) $(n + p) \bmod k = \big((n \bmod k) + (p \bmod k)\big) \bmod k$

c) $(np) \bmod k = \big((n \bmod k)\,(p \bmod k)\big) \bmod k$

Exercise E.2.4. Recall that for $a, b \in \mathbb{N}$ the operations of addition and multiplication modulo n are defined by $a +_n b = (a + b) \bmod n$ and $a \cdot_n b = (a \cdot b) \bmod n$. Prove that:

a) addition modulo n is associative, i.e. $(a +_n b) +_n c = a +_n (b +_n c)$ for any $a, b, c \in \mathbb{N}$;

b) multiplication modulo n is associative, i.e. $(a \cdot_n b) \cdot_n c = a \cdot_n (b \cdot_n c)$ for any $a, b, c \in \mathbb{N}$;

c) multiplication modulo n is distributive with respect to addition modulo n, i.e. $(a +_n b) \cdot_n c = (a \cdot_n c) +_n (b \cdot_n c)$ for any $a, b, c \in \mathbb{N}$.

Exercise E.2.5. State which of the following are groups: a) $(\mathbb{Z}_n, +_n)$ b) $(\{1, 2, 3, 4, 6, 12\}, \gcd)$

Exercise E.2.6. State which of the following are groups: a) $(\{z \in \mathbb{C} \mid |z| = 1\}, +)$ b) $(\{z \in \mathbb{C} \mid |z| = 1\}, \cdot)$

Exercise E.2.7. Show that for every n all complex roots of 1 of order n form a group under multiplication. Write down the table of the group of all complex roots of 1 of order 4.

Exercise E.2.8. Show that $(2^X, \triangle)$ is a group, where $\triangle$ denotes the symmetric difference. Construct the group table of $(2^{\{a,b,c\}}, \triangle)$.

Exercise E.2.9. Show that $(\mathbb{Z}_n \setminus \{0\}, \cdot_n)$ is a group if and only if n is a prime number.

Exercise E.2.10. Verify whether $(\mathbb{R}_+, \bullet)$ is a group, where $a \bullet b = a^2 b^2$.

Exercise E.2.11. Let $p \in \mathbb{N}$ be a prime number. Show that $(\mathbb{Z}_p, +_p, \cdot_p)$ is a field.

Exercise E.2.12. Determine which of the following structures are fields. Do this in two steps: first verify whether the set is a group with respect to the first operation, and then whether the set without the identity element of the first operation is a group with respect to the second operation. Here 2^X carries the operations $\cup, \cap$ and the symmetric difference $\triangle$, while $\mathbb{R}^{\mathbb{R}}$ denotes the set of all functions $\mathbb{R} \to \mathbb{R}$, with pointwise addition $+$ and composition $\circ$. a) $(2^X, \cup, \cap)$ b) $(\mathbb{R}^{\mathbb{R}}, +, \circ)$ c) $(2^X, \cap, \cup)$ d) $(\mathbb{R}^{\mathbb{R}}, \circ, +)$ e) $(2^X, \cap, \triangle)$ f) $(2^X, \triangle, \cap)$ g) $(2^X, \cup, \triangle)$ h) $(2^X, \triangle, \cup)$

Exercise E.2.13. Verify whether $(\mathbb{R}, \oplus, \odot)$ is a field, where $a \oplus b = a + b + 1$ and $a \odot b = ab + a + b$.

E.3 Vector spaces

Exercise E.3.1. Determine whether or not the following sets are vector spaces over the indicated fields:

(a) $\mathbb{Z}$ over $\mathbb{Z}_2$, with addition of integers, and scaling defined as $0 \cdot z = 0$, $1 \cdot z = z$ for every $z \in \mathbb{Z}$

(b) $\mathbb{Z}_4$ over $\mathbb{Z}_2$, with addition modulo 4, and scaling defined as $0 \cdot z = 0$, $1 \cdot z = z$ for every $z \in \mathbb{Z}_4$

(c) $\mathbb{C}$ over $\mathbb{R}$, with addition of complex numbers, and scaling by real numbers

(d) $\mathbb{R}$ over $\mathbb{C}$, with addition of real numbers, and scaling by complex numbers

(e) all subsets of a finite set X over $\mathbb{Z}_2$, with the symmetric difference as vector addition, and with scaling defined as $0 \cdot A = \emptyset$ and $1 \cdot A = A$ for every $A \subseteq X$

(f) $\mathbb{R}_+$ over $\mathbb{R}$, with addition of real numbers, and scaling defined as $p \odot r = r^p$ for every $r \in \mathbb{R}_+$ and $p \in \mathbb{R}$

Exercise E.3.2. Decide which of the following subsets are subspaces:

(a) $\{(x, y) \in \mathbb{R}^2 \mid x \geq 0,\ y \geq 0\} \subseteq \mathbb{R}^2$

(b) $\{(x, y, z) \in \mathbb{R}^3 \mid yz \leq 0\} \subseteq \mathbb{R}^3$

(c) $\{(x, y, z) \in \mathbb{R}^3 \mid x + y + z = x - y = 0\} \subseteq \mathbb{R}^3$

(d) $\{(x, y) \in \mathbb{R}^2 \mid |x - y| \leq 1\} \subseteq \mathbb{R}^2$

Exercise E.3.3. Prove that in a vector space $\mathbf{V}$ over a field $\mathbb{K}$, for every $\mathbf{v}, \mathbf{u} \in \mathbf{V}$ and $\lambda \in \mathbb{K}$,

$$\lambda \cdot (\mathbf{v} - \mathbf{u}) = \lambda \cdot \mathbf{v} - \lambda \cdot \mathbf{u}.$$

Exercise E.3.4. The intersection of any collection of subspaces of a vector space $\mathbf{V}$ is a subspace of $\mathbf{V}$. Show that this is not necessarily true for the union of subspaces.

Exercise E.3.5. The real plane $\mathbb{R}^2$ is a vector space over $\mathbb{R}$. Describe, in geometrical terms, all subspaces of $\mathbb{R}^2$.

Exercise E.3.6. Let $\mathbf{V}$ be a vector space over a field $\mathbb{K}$ and let $S, T \subseteq \mathbf{V}$. Prove that:

(a) $S \subseteq \operatorname{span}(S)$

(b) $S \subseteq T \implies \operatorname{span}(S) \subseteq \operatorname{span}(T)$

(c) $\operatorname{span}(\operatorname{span}(S)) = \operatorname{span}(S)$

(d) $\mathbf{v} \in \operatorname{span}(S) \iff \operatorname{span}(S) = \operatorname{span}(S \cup \{\mathbf{v}\})$.

E.4 Linear independence. Bases and dimension

Exercise E.4.1. Decide whether the given sequence of vectors is linearly independent in the indicated vector space:

(a) $\big((1,0,1),(1,1,0),(0,1,1)\big)$ in $\mathbb{R}^3$ over $\mathbb{R}$

(b) $\big((1,0,1),(1,1,0),(0,1,1)\big)$ in $\mathbb{Z}_2^3$ over $\mathbb{Z}_2$

Exercise E.4.2. Let $\mathbf{V}$ be any vector space over a field $\mathbb{K}$ and let $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n, \mathbf{u} \in \mathbf{V}$. Decide whether the given sequence of vectors is linearly independent:

(a) $(\mathbf{v}_1,\ \mathbf{v}_1+\mathbf{v}_2,\ \mathbf{v}_1+\mathbf{v}_2+\mathbf{v}_3,\ \ldots,\ \mathbf{v}_1+\ldots+\mathbf{v}_{n-1}+\mathbf{v}_n)$ if the sequence $(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$ is linearly independent

(b) $(\mathbf{v}_1+\mathbf{v}_2,\ \mathbf{v}_2+\mathbf{v}_3,\ \ldots,\ \mathbf{v}_{n-1}+\mathbf{v}_n,\ \mathbf{v}_n+\mathbf{v}_1)$ if the sequence $(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$ is linearly independent

(c) $(\mathbf{v}_1+\mathbf{u},\ \mathbf{v}_2+\mathbf{u},\ \ldots,\ \mathbf{v}_n+\mathbf{u})$ if the sequence $(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$ is linearly independent and $\mathbf{u} \in \mathbf{V}$ is any vector

(d) S, where S is a subsequence of a linearly independent sequence T of vectors from $\mathbf{V}$

(e) S, where S is a sequence of vectors from $\mathbf{V}$ containing a linearly independent subsequence T

(f) $(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$, where $\mathbf{v}_i \in \operatorname{span}(S_i)$ and $\mathbf{v}_i \neq \mathbf{0}$ for $i = 1, 2, \ldots, n$, the sets $S_i \subseteq \mathbf{V}$ are finite and pairwise disjoint, and the set $S_1 \cup S_2 \cup \ldots \cup S_n$ consists of linearly independent vectors.

Exercise E.4.3. Find the dimension of each of the following vector spaces:

(a) $\mathbb{C}$ over $\mathbb{R}$

(b) $\mathbb{K}^n$ over any field $\mathbb{K}$

(c) $\operatorname{span}(\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{n-1}\})$, where $(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$ is a basis of a vector space $\mathbf{V}$ over a field $\mathbb{K}$

(d) $\operatorname{span}(\{\mathbf{v}_n, \mathbf{v}_{n-1}, \ldots, \mathbf{v}_1\})$, where $(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$ is a basis of a vector space $\mathbf{V}$ over a field $\mathbb{K}$

(e) $\{(x,y,z,t) \in \mathbb{R}^4 \mid x+y+z+t=0\}$

(f) $\{(x,y,z,t) \in \mathbb{R}^4 \mid 2x+y=z-t\}$

(g) $\operatorname{span}(\{(1,2,1,0),(1,3,1,1),(0,1,2,-1),(-3,1,2,2)\})$ over $\mathbb{R}$

(h) $2^{\{a,b,c\}}$ over $\mathbb{Z}_2$ (with the symmetric difference as vector addition).

Exercise E.4.4. Calculate the coordinates of the vector $\mathbf{v}$ with respect to the basis S:

(a) $\mathbf{v} = (1,2,3,4),\ S = \big((1,1,1,1),(1,1,1,0),(1,1,0,1),(1,0,1,1)\big)$

(b) $\mathbf{v} = \{a,b\},\ S = \big(\{a\},\{b\},\{c\}\big)$ in the vector space $2^{\{a,b,c\}}$ over $\mathbb{Z}_2$

(c) $\mathbf{v} = \mathbf{v}_1,\ S = (\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$

(d) $\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2 + \ldots + \mathbf{v}_n,\ S = (\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$.

Exercise E.4.5. Prove that if $(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$ is a basis of a vector space $\mathbf{V}$, then so is $(\mathbf{v}_1,\ \mathbf{v}_1+\mathbf{v}_2,\ \ldots,\ \mathbf{v}_1+\mathbf{v}_2+\ldots+\mathbf{v}_n)$.

Exercise E.4.6. Prove that $\dim \operatorname{span}(A \cup B) \leq \dim \operatorname{span}(A) + \dim \operatorname{span}(B)$ for subsets A, B of a vector space $\mathbf{V}$.

E.5 Matrices

Exercise E.5.1. For the following matrices A and B calculate the matrix $A \cdot B$:

(a) $A = \begin{pmatrix} 1 & 0 \\ 2 & 3 \\ 1 & 2 \\ -1 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 3 & -2 & 2 \\ 1 & 5 & 3 \end{pmatrix}$ over the field $\mathbb{R}$

(b) $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 4 & 0 \\ 2 & 5 & 1 \\ 3 & 6 & 2 \end{pmatrix}$ over the field $\mathbb{Z}_7$

(c) $A = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 3 & 2 & 3 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 1 \\ 1 & 0 \\ 2 & 0 \\ 1 & 1 \end{pmatrix}$ over the field $\mathbb{Z}_5$

Exercise E.5.2. Recall that a square matrix A is symmetric if and only if $A = A^T$. Is it true that multiplication of symmetric matrices is commutative?

Exercise E.5.3. Show for matrices A and B (of appropriate sizes) that $(A \cdot B)^T = B^T \cdot A^T$.

Exercise E.5.4. Use matrices to determine the dimension of the subspace W over the field $\mathbb{R}$:

(a) $W = \text{span}(\{(1, 2, 3), (1, -1, 2), (-1, 4, -1)\})$

(b) $W = \text{span}(\{(3, 1, 2, 1), (1, -1, 1, 2), (2, 3, 0, -3), (2, -1, 1, 2)\})$

E.6 Systems of linear equations

Exercise E.6.1. Find a basis and the dimension of the solution space over the field $\mathbb{R}$ of each of the following systems of linear equations:

$$\begin{cases} x + 2y - 2z + 2s - t = 0 \\ x + 2y - z + 3s - 2t = 0 \\ 2x + 4y - 7z + s + t = 0 \end{cases} \qquad \begin{cases} x + 2y - z + 3s - 4t = 0 \\ 2x + 4y - 2z - s + 5t = 0 \\ 2x + 4y - 2z + 4s - 2t = 0 \end{cases}$$

Exercise E.6.2. Discuss the solvability of the following systems of equations over the field $\mathbb{R}$ in terms of the parameters a and b:

$$\begin{cases} ax + by + az + t = 1 \\ ax + y + az + t = 1 \\ x + y + az + at = 1 \end{cases} \qquad \begin{cases} x + 2y + 3z + 4t = a \\ 2x + 3y + 4z + 5t = b \\ 3x + 4y + 5z + 6t = a \\ 4x + 5y + 6z + 7t = b \end{cases} \qquad \begin{cases} x + ay + az = 1 \\ ax + ay + z = 1 \\ ax + y + az = 1 \\ ax + ay + az = 1 \end{cases}$$

Exercise E.6.3. Find general solutions of the following systems of equations over the field $\mathbb{R}$:

(a) $\begin{cases} 2x + 2y + 3z + 4t = 2 \\ 2x + 3y + 4z + 6t = 0 \\ 3x + 4y + 5z + 6t = 1 \\ x + y + z + t = 1 \end{cases}$

(b) $\begin{cases} x + y + z - t + u = 1 \\ x - y + z + t + u = 1 \\ -x + y - z + t - u = 0 \end{cases}$

(c) $\begin{cases} x + y + z + t + u = 0 \\ x - 2y + 3z - 4t + 5u = 0 \end{cases}$

Exercise E.6.4. Find all solutions in $\mathbb{Z}_7$ of the following system of equations:

$$\begin{cases} 3x_1 + 4x_2 + 4x_3 + 6x_4 = 1 \\ 4x_1 + 5x_2 + 4x_3 + x_4 = 0 \\ 5x_1 + 6x_2 + 4x_3 + 3x_4 = 6 \end{cases} \quad \text{over the field } \mathbb{Z}_7.$$

Exercise E.6.5. Calculate the following determinants:

(a) $\det \begin{pmatrix} 1 & 2 & -2 & 1 \\ -3 & 1 & 3 & 2 \\ -1 & 2 & 4 & 2 \\ 5 & -1 & 2 & 2 \end{pmatrix}$, where the matrix belongs to $\mathbb{R}^4_4$

(b) $\det \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 2 & 1 \\ 3 & 2 & 3 & 2 \\ 4 & 3 & 4 & 3 \end{pmatrix}$, where the matrix belongs to $(\mathbb{Z}_5)^4_4$

(c) $\det(A)$, where $A \in \mathbb{K}^5_5$ has the entries $a_{ij} = i - j$.

Exercise E.6.6. The numbers 24026, 40262, 2624, 26240 and 62402 are divisible by 41. Prove that the following determinant is also divisible by 41:

$$\det \begin{pmatrix} 2 & 4 & 0 & 2 & 6 \\ 4 & 0 & 2 & 6 & 2 \\ 0 & 2 & 6 & 2 & 4 \\ 2 & 6 & 2 & 4 & 0 \\ 6 & 2 & 4 & 0 & 2 \end{pmatrix}.$$

Exercise E.6.7. Let $y \in \mathbb{R}$, $n \in \mathbb{N}$ and let $A \in \mathbb{R}^n_n$ be defined in the following way: $a_{ij} = y$ for $j \neq n - i + 1$ and $a_{ij} = 0$ for $j = n - i + 1$. Find the determinant of A for $n = 3$, and then find the general formula for the determinant of A for an arbitrary $n \in \mathbb{N}$. Justify your answer.

Exercise E.6.8. Show that for all matrices A and B of size 2×2, $\det(A \cdot B) = \det(A) \cdot \det(B)$.

Exercise E.6.9. Show, for invertible matrices A and B, that $(A \cdot B)^{-1} = B^{-1} \cdot A^{-1}$.

Exercise E.6.10. Calculate the inverse (if it exists) of each of the following matrices:

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \in (\mathbb{Z}_3)^4_4, \qquad B = \begin{pmatrix} 1 & 2 & 2 & 1 \\ 1 & 3 & 4 & 1 \\ 0 & 3 & 1 & 0 \\ 1 & 3 & 4 & 2 \end{pmatrix} \in \mathbb{R}^4_4, \qquad C = \begin{pmatrix} 1 & 2 & 2 & 1 \\ 1 & 3 & 4 & 1 \\ 0 & 3 & 1 & 0 \\ 1 & 3 & 4 & 2 \end{pmatrix} \in (\mathbb{Z}_5)^4_4.$$

E.7 Linear mappings. Matrices of linear mappings

Exercise E.7.1. Verify which of the following mappings are linear:

(a) $F\colon \mathbb{R}^3 \to \mathbb{R}^2,\ F(x,y,z) = (x,\ z)$

(b) $F\colon \mathbb{R}^2 \to \mathbb{R}^2,\ F(x,y) = (3x + y + 1,\ 2x - 3y)$

(c) $F\colon \mathbb{R}^2 \to \mathbb{R}^3,\ F(x,y) = (x + 2y,\ x - y,\ x)$

(d) $F\colon \mathbb{R} \to \mathbb{R},\ F(x) = |x|$

(e) $F\colon \mathbb{R}^3 \to \mathbb{R}^3,\ F(x,y,z) = (xy,\ x,\ z)$

Exercise E.7.2. Let $T\colon \mathbb{C} \to \mathbb{C},\ T(z) = \bar{z}$.

(a) Show that T is *not* linear if we consider $\mathbb{C}$ as a vector space over itself.

(b) Show that T is linear if we consider $\mathbb{C}$ as a vector space over $\mathbb{R}$.

Exercise E.7.3. Find $\ker F$, $\operatorname{im} F$, $\mathrm{r}(F)$ and the nullity of F for the following linear mappings $F\colon U \to V$:

(a) $U = V = \mathbb{R}^3,\ F(x,y,z) = (x + y + z,\ x + y,\ x)$

(b) $U = V = \mathbb{R}^3,\ F(x,y,z) = (x + y,\ x + y,\ x + 2y - z)$

(c) $U = \mathbb{R}^3, V = \mathbb{R}^2,\ F(x,y,z) = (x - y + z,\ x + y - z)$

Exercise E.7.4. Find the matrices of the linear mappings from Exercise E.7.3 in the following bases of the spaces U and V, respectively:

(a) the standard bases of unit vectors

(b) $\big((1,1,1),(1,1,0),(1,0,0)\big)$ and $\big((1,0,0),(1,1,0),(1,1,1)\big)$

(c) $\big((1,1,1),(1,0,0),(0,0,1)\big)$ and $\big((1,2),(3,3)\big)$

Exercise E.7.5. Find a formula and the matrix in the standard basis for the linear operator F such that:

(a) $F(1,1,1) = (1,0,1),\ F(1,1,0) = F(0,1,1) = (1,1,1)$

(b) $F(1,2,1) = (1,0,0),\ F(1,0,1) = (0,1,0),\ F(1,0,0) = (1,0,0)$

(c) $F(1,0,0,0) = F(0,1,0,0) = F(0,0,1,0) = F(0,0,0,1) = (1,1,1,1)$

Exercise E.7.6. Assuming that $A = M_S(F)$ is the matrix of a linear operator F in the basis S, find the matrix $B = M_R(F)$ of F in the basis R:

(a) $\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$, for $S = \big((1,1),(1,2)\big)$ and $R = \big((1,0),(0,1)\big)$

(b) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, for $S = \big((1,1,1),(0,1,2),(1,0,1)\big)$ and $R = \big((1,0,0),(0,1,0),(0,0,1)\big)$

(c) $\begin{pmatrix} 1 & 2 & 2 \\ 1 & 1 & 1 \\ 2 & 0 & 0 \end{pmatrix}$, for $S = \big((1,1,1),(0,1,2),(1,0,1)\big)$ and $R = \big((1,0,0),(0,1,0),(0,0,1)\big)$

(d) $\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$, for $S = \big((1,0,1,0),(0,0,1,1),(1,1,0,0),(1,0,1,1)\big)$ and $R = \big((1,1,1,0),(1,1,0,1),(1,0,1,1),(0,1,1,1)\big)$

Exercise E.7.7. For each pair of matrices A and B from Exercise E.7.6, find a matrix P such that $A = P^{-1} \cdot B \cdot P$. Verify your solution by matrix multiplication.

E.8 Eigenvalues and eigenvectors

Exercise E.8.1. Find the eigenvalues of $A = M_S(F)$:

(a) $A = \begin{pmatrix} 2+i & 1 \\ 2 & 2-i \end{pmatrix}$

(b) $A = \begin{pmatrix} 2 & -1 & 2 \\ 5 & -3 & 3 \\ -1 & 0 & -2 \end{pmatrix}$

Exercise E.8.2. Find all eigenvalues and a basis of each eigenspace for the following linear operators:

(a) $F\colon \mathbb{R}^2 \to \mathbb{R}^2,\ F(x,y) = (x,\ x+y)$

(b) $F\colon \mathbb{R}^3 \to \mathbb{R}^3,\ F(x,y,z) = (2x+y,\ y-z,\ 2y+4z)$

(c) $F\colon \mathbb{R}^3 \to \mathbb{R}^3,\ F(x,y,z) = (x-z,\ 2y,\ x+z)$

(d) $F\colon \mathbb{C}^2 \to \mathbb{C}^2,\ F(x,y) = (-y,\ x)$

(e) $F\colon \mathbb{C}^3 \to \mathbb{C}^3,\ F(x,y,z) = (x-z,\ 2y,\ x+z)$.

Exercise E.8.3. Assuming that each of the following matrices represents a linear operator F in a basis S, find all eigenvectors and a basis of each eigenspace:

(a) $A = M_S(F) = \begin{pmatrix} 3 & 1 & 1 \\ 2 & 4 & 2 \\ 1 & 3 & 3 \end{pmatrix}$

(b) $A = M_S(F) = \begin{pmatrix} 1 & 2 & 2 \\ 1 & 2 & -1 \\ -1 & 1 & 4 \end{pmatrix}$

(c) $A = M_S(F) = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Exercise E.8.4. For each operator F from Exercise E.8.3, determine whether there exists a basis T such that the matrix of the operator F in the basis T is diagonal.

Exercise E.8.5. For the matrix $A = \begin{pmatrix} 1 & 2 & -1 \\ 1 & -1 & 0 \\ 2 & -2 & 0 \end{pmatrix}$ calculate A^{100}.

Exercise E.8.6. For a linear mapping $F\colon \mathbb{R}^2 \to \mathbb{R}^2$ such that $F(1,2) = (2,4)$ and $F(-2,1) = (-2,1)$, calculate $F^{150}(1,0)$.